Representation Theory

Lecture Notes

Notes written by
Zhou Yimiao

Instructor
Shan Peng

Qiuzhen College, Tsinghua University

Contents

I Foundations of Finite Group Representations **1**

1 Introduction to Group Representations **2**
1.1 Groups and Symmetry . . . 2
1.2 Linear Representations . . . 3
1.3 Irreducibility and Indecomposability . . . 4
1.4 Fundamental Objectives . . . 5

2 Schur's Lemma, Maschke's Theorem, and Character Theory **6**
2.1 Schur's Lemma . . . 6
2.2 The Averaging Map and Maschke's Theorem . . . 6
2.3 Character Theory . . . 8
2.4 Orthogonality of Characters . . . 10

3 Decomposition of Representations and Compact Groups **12**
3.1 Decomposition of a Representation . . . 12
3.2 Irreducibility Criterion . . . 14
3.3 The Regular Representation . . . 14
3.4 Class Functions and Conjugacy Classes . . . 15
3.5 Representations of Compact Groups . . . 17
3.6 Linear Characters and Fourier Transforms . . . 18
3.7 The Group Algebra and its Center . . . 19

4 Group Algebras, Idempotents, and Orthogonality **21**
4.1 Convolution and the Group Algebra . . . 21
4.2 Central Idempotents . . . 22
4.3 Irreducible Characters as Idempotents . . . 23
4.4 Unitary Representations . . . 24
4.5 The Frobenius Determinant . . . 24
4.6 Algebras and Modules . . . 26

5 Induced Representations and Characters of Algebras **27**
5.1 The Converse of Maschke's Theorem . . . 27
5.2 Characters of Arbitrary Algebras . . . 28
5.3 Jordan–Hölder and Krull–Schmidt Theorems . . . 29
5.4 Induction, Restriction, and Frobenius Reciprocity . . . 31
5.5 Induced Characters and Automorphic Twists . . . 32

6 Mackey Formula and Applications **34**

6.1 The Mackey Formula . . . 34
6.2 Applications to Normal Subgroups . . . 35
6.3 The Ring of Virtual Characters . . . 36
6.4 Artin's Theorem on Induced Characters . . . 37
6.5 Representations of Supersolvable Groups and p-Groups . . . 37

II Algebras, Modules, and Categories 40

7 Representations of Algebras and Tensor Products 41
7.1 Tensor Products of Modules . . . 41
7.2 Representations and Semisimplicity . . . 43
7.3 The Jacobson Density Theorem . . . 45
7.4 The Jacobson Radical and Semisimplicity . . . 45

8 Categories, Functors, and Natural Transformations 48
8.1 Categories . . . 48
8.2 Functors . . . 49
8.3 Natural Transformations . . . 50
8.4 Adjoint Functors . . . 51
8.5 The Yoneda Lemma . . . 52

9 Quiver Representations 54
9.1 Quivers and Path Algebras . . . 54
9.2 Dynkin Quivers and Gabriel's Theorem . . . 56
9.3 Roots and the Weyl Group . . . 57
9.4 Reflection Functors and Coxeter Elements . . . 58
9.5 Detailed Example: D_4 . . . 60

III The Symmetric Group and Combinatorics 62

10 Okounkov-Vershik Approach to Symmetric Group Representations 63
10.1 Centralizers and Restriction of Representations . . . 63
10.2 The Branching Graph and Weights . . . 65
10.3 Combinatorial Weights . . . 66
10.4 Relation with Young Tableaux . . . 69

11 Symmetric Functions 71
11.1 The Ring of Symmetric Functions . . . 71
11.2 Bases of Symmetric Functions . . . 72
11.3 Schur Functions . . . 74
11.4 Orthogonality and Isometry . . . 75
11.5 The Frobenius Character Map . . . 76

12 Young Symmetrizers and Symmetric Group Representations 79
12.1 Tableaux and Stabilizers . . . 79
12.2 The Young Symmetrizer . . . 80
12.3 Irreducible Representations . . . 81

13 Schur-Weyl Duality 82

13.1 The Double Centralizer Theorem 83
13.2 The Schur Algebra 84
13.3 Schur-Weyl Duality 85

IV Continuous and Algebraic Representations 88

14 Introduction to Lie Algebras 89
14.1 Definitions and Examples 89
14.2 Subalgebras, Homomorphisms, and Representations 91
14.3 The Universal Enveloping Algebra 91

15 Representation Theory of $\mathfrak{sl}_2$ 93
15.1 The Highest Weight Vector 93
15.2 The Representation Chain 94
15.3 Classification Theorem 94

16 Representations of General Linear Groups and $\mathrm{SL}_2(\mathbb{F}_q)$ 96
16.1 Weyl Character Formula for $\mathrm{GL}(V)$ 96
16.2 Algebraic Representations of $\mathrm{GL}_N(\mathbb{C})$ 97
16.3 Structure of $\mathrm{SL}_2(\mathbb{F}_q)$ 98
16.4 Conjugacy Classes and Harish-Chandra Induction 99
16.4.1 Harish-Chandra Series 100

Bibliography 102

Part I

Foundations of Finite Group Representations

Chapter 1

Introduction to Group Representations

1.1 Groups and Symmetry

The language of groups is the standard mathematical language for symmetry. Historically, groups first appeared in Galois theory: one studies a polynomial equation by studying the permutations of its roots that preserve all algebraic relations visible over the base field. This point of view is still one of the guiding examples of representation theory.

Definition 1.1. A **group** is a set G equipped with an associative binary operation $G \times G \to G$, an identity element $1_G \in G$, and an inverse $g^{-1} \in G$ for every element $g \in G$.

Example 1.2 (Dihedral Group)**.** In these notes D_8 denotes the dihedral group of order 8, namely the group of rigid symmetries of a square. It can be generated by two reflections σ and τ whose reflecting axes meet at angle $\pi/4$, and it has the presentation

$$D_8 = \langle \sigma, \tau \mid \sigma^2 = \tau^2 = (\sigma\tau)^4 = 1 \rangle.$$

The same group appears in several natural ways:

- It is the Galois group of $X^4 - 2$ over $\mathbb{Q}$, acting by permutations of the four roots while preserving all rational algebraic relations among them.
- It is the group of Euclidean isometries of $\mathbb{R}^2$ that preserve a square centered at the origin. Since these isometries fix the origin, they are linear orthogonal transformations; hence this action gives a faithful representation $D_8 \hookrightarrow \mathrm{GL}_2(\mathbb{R})$.

Example 1.3 (Groups in Nature)**.** Groups occur throughout mathematics and physics.

- **Finite groups.** Basic examples include the symmetric groups $\mathfrak{S}_n$, cyclic groups $\mathbb{Z}/d\mathbb{Z}$, finite groups of Lie type such as $\mathrm{GL}_n(\mathbb{F}_q)$, and the 26 sporadic simple groups.
- **Lie groups and algebraic groups.** These are smooth manifolds or algebraic varieties endowed with compatible group structures, for example $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SO}_n(\mathbb{R})$, and $\mathrm{Sp}_{2n}(\mathbb{C})$. They arise naturally as continuous symmetries of geometric spaces and differential equations.

- **Groups in physics.** The unitary group

$$U(n) = \{A \in \mathrm{Mat}_n(\mathbb{C}) \mid A\overline{A}^T = I\}$$

governs finite-dimensional quantum evolution. For any associative unital ring A, the Heisenberg group

$$H_3(A) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; a, b, c \in A \right\}$$

is nilpotent of class at most 2. The Poincaré group is the group of isometries of Minkowski space-time; its subgroup fixing the origin is the Lorentz group.

Definition 1.4. More generally, a **representation** of a group G on a mathematical object X is an action

$$G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

satisfying $h \cdot (g \cdot x) = (hg) \cdot x$ and $1_G \cdot x = x$. Equivalently, it is a group homomorphism

$$\rho : G \to \mathrm{Aut}(X),$$

where $\mathrm{Aut}(X)$ denotes the automorphism group of X in the relevant category.

For instance, if X is a set then $\mathrm{Aut}(X)$ is the permutation group of X; if X is a vector space then $\mathrm{Aut}(X) = \mathrm{GL}(X)$; and if X is a topological space then $\mathrm{Aut}(X)$ is the group of homeomorphisms of X. In these notes, unless explicitly stated otherwise, the word "representation" will mean a linear representation.

1.2 Linear Representations

The main idea of representation theory is to study an abstract group by letting it act on a vector space. This replaces group elements by invertible matrices or linear operators, making linear algebra available.

Many naturally occurring actions can be linearized.

- If a finite group G acts on a finite set X, then G acts on the vector space

$$k[X] = \bigoplus_{x \in X} k e_x$$

by $g \cdot e_x = e_{g \cdot x}$.

- If G acts on an algebraic variety X, then it often acts linearly on cohomology groups such as $H^*(X, \mathbb{C})$.

- The rotation action of S^1 on itself gives a linear representation on the Hilbert space $L^2(S^1, \mathbb{C})$, leading to Fourier series.

Throughout this text, G is a group and k is a fixed ground field, usually $k = \mathbb{C}$ unless another field is specified.

Definition 1.5. A **linear representation** of G over k is a pair (V, ρ), where V is a k-vector space and

$$\rho : G \to \mathrm{GL}(V)$$

is a group homomorphism. Equivalently, it is an action $G \times V \to V$, written $(g, v) \mapsto g \cdot v$, such that each map $v \mapsto g \cdot v$ is k-linear and

1. $h \cdot (g \cdot v) = (hg) \cdot v$ for all $g, h \in G$ and $v \in V$;
2. $1_G \cdot v = v$ for all $v \in V$.

Definition 1.6. Let (V, ρ_V) and (W, ρ_W) be representations of G. A **morphism of representations**, or **intertwining operator**, is a k-linear map $\phi : V \to W$ such that, for every $g \in G$, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\phi} & W \\ {\scriptstyle \rho_V(g)}\downarrow & & \downarrow{\scriptstyle \rho_W(g)} \\ V & \xrightarrow[\phi]{} & W \end{array}$$

commutes. Equivalently,

$$\phi(g \cdot v) = g \cdot \phi(v) \qquad (g \in G,\ v \in V).$$

The vector space of all such morphisms is denoted $\mathrm{Hom}_G(V, W)$. If ϕ is a linear isomorphism, then V and W are said to be **isomorphic**, written $V \simeq W$.

Example 1.7 (Fundamental Linear Representations)**.**

1. **Trivial representation.** The one-dimensional space k is a representation with $\rho(g) = 1$ for all $g \in G$.
2. **Tautological representation.** If $G = \mathrm{GL}(V)$, then V is a representation by the identity homomorphism $\mathrm{GL}(V) \to \mathrm{GL}(V)$.
3. **Permutation representation.** If G acts on a finite set X, then the vector space $k[X] = \bigoplus_{x \in X} k e_x$ has a representation defined by $g \cdot e_x = e_{g \cdot x}$.
4. **Regular representation.** The group G acts on itself by left multiplication. The associated permutation representation $k[G]$ is called the left regular representation.
5. **Representations of $\mathbb{Z}$.** Since $\mathbb{Z}$ is generated by 1, a representation $\rho : \mathbb{Z} \to \mathrm{GL}(V)$ is the same thing as the choice of one invertible linear operator $T = \rho(1)$ on V.
6. **Representations of $\mathbb{Z}/d\mathbb{Z}$.** A representation of $\mathbb{Z}/d\mathbb{Z}$ is the same thing as an invertible linear operator $T \in \mathrm{GL}(V)$ satisfying $T^d = I$.

1.3 Irreducibility and Indecomposability

If V_1 and V_2 are representations of G, their **direct sum** $V_1 \oplus V_2$ is a representation by the diagonal action

$$g \cdot (v_1, v_2) = (g \cdot v_1, g \cdot v_2).$$

Definition 1.8. A **subrepresentation** of a representation V is a vector subspace $W \subset V$ that is stable under the action of G; that is, $g \cdot w \in W$ for all $g \in G$ and $w \in W$. If W is a subrepresentation, then the quotient vector space V/W inherits a representation by

$$g \cdot (v + W) = (g \cdot v) + W.$$

Definition 1.9. A nonzero representation V is **irreducible**, or **simple**, if its only subrepresentations are 0 and V. It is **indecomposable** if it cannot be written as a direct sum $V_1 \oplus V_2$ of two nonzero subrepresentations.

Every irreducible representation is indecomposable, but the converse is false in general.

Example 1.10. Let $G = \mathbb{Z}$ act on $V = k^2$ over an algebraically closed field k, and let the generator $1 \in \mathbb{Z}$ act by the unipotent matrix

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The line ke_1 is T-stable, so V is not irreducible. However, T has a single Jordan block. Thus there is no T-stable line complementary to ke_1, and V cannot be written as a direct sum of two nonzero subrepresentations. Hence V is indecomposable but not irreducible.

For finite groups, the behavior of representations depends strongly on the characteristic of the field.

- **Characteristic zero.** Over a field such as $\mathbb{C}$, every finite-dimensional representation of a finite group is a direct sum of irreducible representations. This is the semisimple theory.
- **Characteristic $p > 0$.** If $G = \mathbb{Z}/p\mathbb{Z}$ and $\mathrm{char}(k) = p$, then the unipotent matrix $T = I + N$ above satisfies $T^p = (I + N)^p = I + N^p = I$. Therefore it defines an indecomposable but reducible representation of the finite group $\mathbb{Z}/p\mathbb{Z}$. This phenomenon is one of the basic features of modular representation theory.

Remark 1.11 (The Permutation Functor)**.** The construction $X \mapsto k[X]$ is a functor from G-sets to k-linear representations. It preserves disjoint unions:

$$k[X \sqcup Y] \simeq k[X] \oplus k[Y].$$

However, it does not preserve irreducibility. A transitive G-set is indecomposable as a G-set, but its permutation representation is often reducible. For example, if $\mathfrak{S}_2$ acts on $X = \{1, 2\}$, then, when $\mathrm{char}(k) \neq 2$,

$$k[X] = k(e_1 + e_2) \oplus k(e_1 - e_2),$$

the direct sum of the trivial representation and the sign representation.

1.4 Fundamental Objectives

For a given group G, representation theory asks for both classification results and structural tools. The basic goals are to classify irreducible representations up to isomorphism; classify indecomposable representations when this is feasible; compute invariants such as dimensions and characters; understand spaces of morphisms $\mathrm{Hom}_G(V, W)$; and study the categorical and homological structure of $\mathrm{Rep}(G)$.

Chapter 2

Schur's Lemma, Maschke's Theorem, and Character Theory

2.1 Schur's Lemma

Schur's lemma is one of the basic rigidity statements in representation theory. It says that maps between irreducible representations are very rare: either they are zero, or they are isomorphisms. Over an algebraically closed field, the only endomorphisms of an irreducible finite-dimensional representation are scalar maps.

Theorem 2.1 (Schur's Lemma)**.** *Let V_1 and V_2 be irreducible representations of a group G over a field k.*

1. *If $V_1 \not\simeq V_2$, then $\mathrm{Hom}_G(V_1, V_2) = 0$.*
2. *If $V_1 = V_2 = V$, then $\mathrm{End}_G(V)$ is a division algebra.*
3. *If, in addition, V is finite-dimensional and k is algebraically closed, then every element of $\mathrm{End}_G(V)$ is a scalar operator. Thus $\mathrm{End}_G(V) \simeq k$.*

Proof. Let $f : V_1 \to V_2$ be a G-linear map. Then $\ker f$ is a subrepresentation of V_1, and $\mathrm{im}\, f$ is a subrepresentation of V_2. Since both representations are irreducible, either $f = 0$ or $\ker f = 0$ and $\mathrm{im}\, f = V_2$. Hence every nonzero map $f : V_1 \to V_2$ is an isomorphism. This proves the first assertion, and also shows that every nonzero element of $\mathrm{End}_G(V)$ is invertible.

Now assume that V is finite-dimensional and that k is algebraically closed. If $f \in \mathrm{End}_G(V)$, then f has an eigenvalue $\lambda \in k$. The operator $f - \lambda \,\mathrm{id}_V$ is still G-linear and has nonzero kernel. By irreducibility, this kernel is all of V, so $f = \lambda \,\mathrm{id}_V$. □

2.2 The Averaging Map and Maschke's Theorem

In this section G is finite and $\mathrm{char}(k)$ does not divide $|G|$. Equivalently, $|G|$ is invertible in k.

For any representation V of G, the fixed-point subspace is

$$V^G = \{v \in V \mid g \cdot v = v \text{ for all } g \in G\}.$$

The **averaging map**

$$\mathrm{Av} : V \to V^G, \qquad v \mapsto \frac{1}{|G|}\sum_{g\in G} g \cdot v$$

projects V onto V^G. Indeed, if $x \in V^G$, then $\mathrm{Av}(x) = x$, and for any $h \in G$,

$$h \cdot \mathrm{Av}(v) = \frac{1}{|G|}\sum_{g\in G}(hg) \cdot v = \mathrm{Av}(v).$$

Both hypotheses are essential: G must be finite so that the sum is finite, and $|G|$ must be invertible in k so that division by $|G|$ makes sense.

The same construction applies to spaces of linear maps. If V_1 and V_2 are representations, then G acts on $\mathrm{Hom}_k(V_1, V_2)$ by

$$(g \cdot f)(v) = g \cdot f(g^{-1} \cdot v).$$

The fixed points for this action are precisely the G-linear maps $\mathrm{Hom}_G(V_1, V_2)$. Therefore averaging an arbitrary k-linear map produces a G-linear one:

$$\mathrm{Av}(f) = \frac{1}{|G|}\sum_{g\in G} \rho_{V_2}(g) f \rho_{V_1}(g^{-1}).$$

Theorem 2.2 (Maschke's Theorem, 1899)**.** *Let G be finite and suppose that* $\mathrm{char}(k) \nmid |G|$. *Then every finite-dimensional representation of G over k is completely reducible; that is, it is a direct sum of irreducible representations.*

Proof. It is enough to show that every subrepresentation has a G-stable complement. Let $W \subset V$ be a subrepresentation. As a vector space, V has a k-linear projection $p : V \to W$ whose restriction to W is the identity. This projection need not be G-linear, so we average it:

$$p_G(x) = \frac{1}{|G|}\sum_{g\in G} g \cdot p(g^{-1} \cdot x).$$

By construction, p_G is G-linear. Moreover, if $x \in W$, then $g^{-1}\cdot x \in W$ and $p(g^{-1}\cdot x) = g^{-1}\cdot x$, so

$$p_G(x) = \frac{1}{|G|}\sum_{g\in G} g \cdot (g^{-1} \cdot x) = x.$$

Thus p_G is a G-linear projection from V onto W. Its kernel is a G-stable complement, and

$$V = W \oplus \ker p_G.$$

Repeating this argument, or arguing by induction on $\dim V$, gives a decomposition into irreducible representations. □

Remark 2.3. Maschke's theorem can fail when the characteristic of the field divides the group order. This is the beginning of modular representation theory.

Example 2.4. Let $G = \mathbb{Z}/p\mathbb{Z}$ and let k be a field of characteristic p. Define a representation on $V = k^2$ by sending the class of n to

$$\rho(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

This is well-defined because

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^p = \begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix} = I$$

in characteristic p. The line ke_1 is invariant, so the representation is not irreducible. There is no invariant complementary line, because the matrix has a single Jordan block. Hence this representation is indecomposable but not irreducible.

2.3 Character Theory

From now on in this chapter, G is a finite group, the ground field is $\mathbb{C}$, and every representation under discussion is finite-dimensional. Character theory replaces a complex representation by a complex-valued function on G.

Definition 2.5. The **character** of a representation (V, ρ_V) is the function $\chi_V : G \to \mathbb{C}$ defined by

$$\chi_V(g) = \operatorname{tr}(\rho_V(g)).$$

Equivalently, $\chi_V(g)$ is the sum of the eigenvalues of $\rho_V(g)$, counted with algebraic multiplicity.

For later use, the tensor product and dual carry the actions

$$g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w), \qquad (g \cdot \lambda)(v) = \lambda(g^{-1} \cdot v).$$

Proposition 2.6. *Let V be a complex representation of the finite group G, with character χ_V. Then:*

1. $\chi_V(1) = \dim V$.
2. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ *for all* $g \in G$.
3. $\chi_V(hgh^{-1}) = \chi_V(g)$ *for all* $g, h \in G$.

Proof. The first assertion follows because $\rho_V(1) = \mathrm{id}_V$. The third follows from invariance of trace under conjugation:

$$\rho_V(hgh^{-1}) = \rho_V(h)\rho_V(g)\rho_V(h)^{-1}.$$

For the second assertion, let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $\rho_V(g)$. Since G is finite, g has finite order, so each λ_i is a root of unity. Hence $\lambda_i^{-1} = \overline{\lambda_i}$, and

$$\chi_V(g^{-1}) = \sum_i \lambda_i^{-1} = \sum_i \overline{\lambda_i} = \overline{\chi_V(g)}.$$

□

Proposition 2.7. *Characters behave as follows under standard operations:*

1. $\chi_{V_1 \oplus V_2} = \chi_{V_1} + \chi_{V_2}$.
2. $\chi_{V_1 \otimes V_2} = \chi_{V_1}\chi_{V_2}$.

3. $\chi_{V^*} = \overline{\chi_V}$.

4. *For the natural representation on* $\mathrm{Hom}_k(V_1, V_2)$,

$$\chi_{\mathrm{Hom}_k(V_1,V_2)} = \overline{\chi_{V_1}}\chi_{V_2}.$$

Definition 2.8. A function $f : G \to \mathbb{C}$ is called a **class function**, or **central function**, if it is constant on conjugacy classes. Equivalently,

$$f(hgh^{-1}) = f(g) \qquad (g, h \in G).$$

The vector space of class functions is denoted $\mathrm{CF}(G)$.

Every character is a class function. Moreover, $\dim \mathrm{CF}(G)$ is the number of conjugacy classes of G, because a class function is determined by one complex number on each conjugacy class.

We shall use the following consequence of Schur's lemma.

Lemma 2.9 (Averaged Invariants)**.** *Let V_1 and V_2 be irreducible complex representations of a finite group G, and let $f : V_1 \to V_2$ be a linear map. Define*

$$f^\circ = \frac{1}{|G|} \sum_{g \in G} \rho_{V_2}(g) f \rho_{V_1}(g^{-1}).$$

Then:

1. *If $V_1 \not\cong V_2$, then $f^\circ = 0$.*

2. *If $V_1 = V_2 = V$, then*

$$f^\circ = \frac{\mathrm{tr}(f)}{\dim V} \mathrm{id}_V .$$

3. *If $\phi \in \mathrm{CF}(G)$ and V is irreducible, then*

$$u_\phi = \sum_{g \in G} \phi(g) \rho_V(g)$$

acts on V as a scalar operator, namely

$$u_\phi = \frac{1}{\dim V} \left(\sum_{g \in G} \phi(g) \chi_V(g) \right) \mathrm{id}_V .$$

Proof. The map f° is the average of f in $\mathrm{Hom}_k(V_1, V_2)$, so it is G-linear. The first assertion follows immediately from Schur's lemma. In the case $V_1 = V_2 = V$, Schur's lemma gives $f^\circ = \lambda \, \mathrm{id}_V$. Taking traces gives

$$\lambda \dim V = \mathrm{tr}(f^\circ) = \frac{1}{|G|} \sum_{g \in G} \mathrm{tr}(\rho_V(g) f \rho_V(g^{-1})) = \mathrm{tr}(f),$$

proving the second assertion.

For the third assertion, use that ϕ is a class function. For any $h \in G$,

$$\rho_V(h) u_\phi \rho_V(h)^{-1} = \sum_{g \in G} \phi(g) \rho_V(hgh^{-1}) = \sum_{x \in G} \phi(x) \rho_V(x) = u_\phi,$$

where $x = hgh^{-1}$. Thus u_ϕ is a G-endomorphism of the irreducible representation V, so Schur's lemma says that it is scalar. The scalar is determined by dividing the trace by $\dim V$. □

2.4 Orthogonality of Characters

Let $\mathrm{Fun}(G, \mathbb{C})$ be the vector space of all complex-valued functions on G. We use the Hermitian inner product

$$(\phi_1 \mid \phi_2) = \frac{1}{|G|} \sum_{g \in G} \overline{\phi_1(g)} \phi_2(g).$$

It is conjugate-linear in the first variable and linear in the second.

Theorem 2.10 (Frobenius, 1897)**.** *The irreducible characters of G form an orthonormal basis of* $\mathrm{CF}(G)$.

Proof. First we prove orthonormality. Let χ_{V_1} and χ_{V_2} be irreducible characters. Then

$$\begin{aligned}(\chi_{V_1} \mid \chi_{V_2}) &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_1}(g)} \chi_{V_2}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}_k(V_1, V_2)}(g).\end{aligned}$$

Put $U = \mathrm{Hom}_k(V_1, V_2)$. The operator

$$P = \frac{1}{|G|} \sum_{g \in G} \rho_U(g)$$

is the projection from U onto $U^G = \mathrm{Hom}_G(V_1, V_2)$. Therefore

$$(\chi_{V_1} \mid \chi_{V_2}) = \mathrm{tr}(P) = \dim \mathrm{Hom}_G(V_1, V_2).$$

Schur's lemma now gives

$$(\chi_{V_1} \mid \chi_{V_2}) = \begin{cases} 1, & V_1 \simeq V_2, \\ 0, & V_1 \not\simeq V_2. \end{cases}$$

It remains to prove completeness. Let $\phi \in \mathrm{CF}(G)$ be orthogonal to every irreducible character. For every irreducible representation V, Lemma 2.9 applied to $\overline{\phi}$ gives

$$\sum_{g \in G} \overline{\phi(g)} \rho_V(g) = 0,$$

because its scalar is

$$\frac{1}{\dim V} \sum_{g \in G} \overline{\phi(g)} \chi_V(g) = \frac{|G|}{\dim V} (\phi \mid \chi_V) = 0.$$

By Maschke's theorem, the same operator is zero on every finite-dimensional representation, since every such representation is a direct sum of irreducibles.

Apply this to the regular representation $\mathbb{C}[G]$, with basis $\{e_g\}_{g\in G}$ and left action. Evaluating at e_1 gives

$$0 = \sum_{g\in G} \overline{\phi(g)}\, g \cdot e_1 = \sum_{g\in G} \overline{\phi(g)}\, e_g.$$

Since the e_g form a basis, all coefficients vanish. Thus $\phi(g) = 0$ for all $g \in G$, and the irreducible characters span $\mathrm{CF}(G)$. $\square$

Chapter 3

Decomposition of Representations and Compact Groups

3.1 Decomposition of a Representation

Throughout the first part of this chapter, let G be a finite group and let all representations be finite-dimensional complex representations. The main point of character theory is that it turns the problem of decomposing a representation into a problem of linear algebra in the space of class functions.

Theorem 3.1. *Let V_1 and V_2 be irreducible representations of G. Then*

$$V_1 \simeq V_2 \quad \Longleftrightarrow \quad \chi_{V_1} = \chi_{V_2}.$$

Thus the character of an irreducible representation determines its isomorphism class.

Proof. If $V_1 \simeq V_2$, then the two representations are conjugate as matrix representations after choosing bases, so their traces are equal. Conversely, by the orthonormality of irreducible characters,

$$(\chi_{V_1} \mid \chi_{V_2}) = \begin{cases} 1, & V_1 \simeq V_2, \\ 0, & V_1 \not\simeq V_2. \end{cases}$$

If $\chi_{V_1} = \chi_{V_2}$, then $(\chi_{V_1} \mid \chi_{V_2}) = (\chi_{V_1} \mid \chi_{V_1}) = 1$, and therefore $V_1 \simeq V_2$. □

Theorem 3.2 (Uniqueness of decomposition)**.** *Let V be a finite-dimensional complex representation. Suppose*

$$V \simeq \bigoplus_i W_i^{\oplus m_i},$$

where the W_i are pairwise non-isomorphic irreducible representations. Then

$$\chi_V = \sum_i m_i \chi_{W_i}, \qquad m_i = (\chi_V \mid \chi_{W_i}).$$

In particular, the character of V determines the multiplicity of every irreducible representation in V, and hence determines V up to isomorphism.

Proof. Characters are additive on direct sums, so $\chi_V = \sum_i m_i \chi_{W_i}$. Taking the inner product with χ_{W_j} gives

$$(\chi_V \mid \chi_{W_j}) = \sum_i m_i (\chi_{W_i} \mid \chi_{W_j}) = m_j$$

by the orthonormality of irreducible characters. □

When a representation is written as a direct sum of irreducibles, the individual summands are usually not canonical. What is canonical is the part of the representation built from all copies of a fixed irreducible.

Definition 3.3. Let W_i be an irreducible representation. The **W_i-isotypic component** of V is the largest subrepresentation of V isomorphic to a direct sum of copies of W_i. If

$$V \simeq \bigoplus_i W_i^{\oplus m_i},$$

then the W_i-isotypic component is isomorphic to $W_i^{\oplus m_i}$. It is independent of the chosen decomposition of V into irreducible summands.

Proposition 3.4. *Let W_i be an irreducible representation of G. The projection from V onto its W_i-isotypic component is the G-equivariant operator*

$$p_i = \frac{\dim W_i}{|G|} \sum_{g \in G} \overline{\chi_{W_i}(g)}\, \rho_V(g).$$

The operators p_i are mutually orthogonal idempotents and

$$V = \bigoplus_i p_i V$$

is the canonical decomposition of V into isotypic components.

Proof. First restrict p_i to an irreducible subrepresentation $W \subset V$. By the averaged form of Schur's lemma (Lemma 2.9), this restriction is scalar:

$$p_i|_W = \frac{\dim W_i}{\dim W} (\chi_{W_i} \mid \chi_W)\ \mathrm{id}_W\,.$$

If $W \simeq W_i$, this scalar is 1. If $W \not\simeq W_i$, it is 0. Hence p_i acts as the identity on every irreducible summand of type W_i and kills every irreducible summand of any other type. The statement follows after decomposing V into irreducibles. □

Example 3.5. Let $G = \mathbb{Z}/d\mathbb{Z}$ and let $u = \rho_V(1)$. Since $u^d = 1$, the minimal polynomial of u divides $x^d - 1$, which has distinct roots over $\mathbb{C}$. Thus u is diagonalizable, and the isotypic components of V are exactly the eigenspaces of u. The character χ_k defined by $\chi_k(1) = \exp(2\pi i k/d)$ corresponds to the eigenspace with eigenvalue $\exp(2\pi i k/d)$.

3.2 Irreducibility Criterion

The character inner product gives a short test for irreducibility.

Proposition 3.6 (Irreducibility criterion)**.** *A complex representation V of a finite group is irreducible if and only if*

$$(\chi_V \mid \chi_V) = 1.$$

Proof. Write

$$V \simeq \bigoplus_i W_i^{\oplus m_i}$$

with the W_i pairwise non-isomorphic and irreducible. Then $\chi_V = \sum_i m_i \chi_{W_i}$, and therefore

$$(\chi_V \mid \chi_V) = \sum_i m_i^2.$$

The multiplicities m_i are non-negative integers. The sum of their squares is 1 exactly when one multiplicity is 1 and all the others are 0, which is precisely the condition that V is irreducible. □

3.3 The Regular Representation

Let $\mathbb{C}[G]$ denote the regular representation, with basis $\{e_h\}_{h \in G}$ and action

$$g \cdot e_h = e_{gh}.$$

If $g \neq 1$, the permutation $h \mapsto gh$ has no fixed points. Hence the character of the regular representation is

$$\chi_{\mathbb{C}[G]}(g) = \begin{cases} |G|, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

For any irreducible representation W,

$$(\chi_{\mathbb{C}[G]} \mid \chi_W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{\mathbb{C}[G]}(g)}\, \chi_W(g) = \chi_W(1) = \dim W.$$

Therefore every irreducible representation occurs in the regular representation with multiplicity equal to its dimension:

$$\mathbb{C}[G] \simeq \bigoplus_{W \in \mathrm{Irr}(G)} W^{\oplus \dim W}.$$

Taking dimensions gives the important identity

$$|G| = \sum_{W \in \mathrm{Irr}(G)} (\dim W)^2.$$

3.4 Class Functions and Conjugacy Classes

Theorem 3.7. *The number of isomorphism classes of irreducible complex representations of G is equal to the number of conjugacy classes of G:*

$$|\operatorname{Irr}(G)| = |\operatorname{Conj}(G)|.$$

Proof. Let $\mathrm{CF}(G)$ be the vector space of complex-valued class functions on G. The indicator functions of the conjugacy classes form one basis of $\mathrm{CF}(G)$, so

$$\dim \mathrm{CF}(G) = |\operatorname{Conj}(G)|.$$

On the other hand, Frobenius orthogonality says that the irreducible characters form an orthonormal basis of $\mathrm{CF}(G)$. Hence

$$\dim \mathrm{CF}(G) = |\operatorname{Irr}(G)|.$$

Comparing these two dimensions proves the claim. □

The character table

Let

$$\operatorname{Irr}(G) = \{W_1, \dots, W_h\}, \qquad \operatorname{Conj}(G) = \{C_1, \dots, C_h\}.$$

Choose one representative $c_j \in C_j$ for each conjugacy class and write

$$T_{ij} = \chi_{W_i}(c_j).$$

The matrix $T = (T_{ij})$ is the **character table** of G. Its rows are irreducible characters, and its columns are conjugacy classes.

If

$$K = \operatorname{diag}\left(\frac{|C_1|}{|G|}, \dots, \frac{|C_h|}{|G|}\right),$$

then row orthogonality is the matrix equation

$$TKT^* = I_h.$$

Since T is square and invertible, this is equivalent to

$$T^*T = K^{-1}.$$

Expanding the last equation gives column orthogonality.

Corollary 3.8 (Column orthogonality)**.** *For conjugacy classes C and C',*

$$\sum_{W \in \operatorname{Irr}(G)} \overline{\chi_W(C)}\, \chi_W(C') = \begin{cases} |G|/|C|, & C = C', \\ 0, & C \neq C'. \end{cases}$$

Example 3.9 (The character table of $\mathfrak{S}_3$)**.** The group $\mathfrak{S}_3$ has three conjugacy classes:

$$C_0 = \{1\}, \qquad C_1 = \{(12), (13), (23)\}, \qquad C_2 = \{(123), (132)\}.$$

Let π be the character of the permutation representation on $\mathbb{C}^3$. Since $\pi(g)$ is the number of fixed points of g and the permutation representation is the direct sum of its invariant diagonal line and the standard representation

$$\{(x_1, x_2, x_3) \in \mathbb{C}^3 \mid x_1 + x_2 + x_3 = 0\},$$

we have

$$\chi_{\mathrm{std}}(g) = \#\,\mathrm{Fix}_{\{1,2,3\}}(g) - 1.$$

Thus χ_{std} has values $(2, 0, -1)$ on (C_0, C_1, C_2), and

$$(\chi_{\mathrm{std}} \mid \chi_{\mathrm{std}}) = \frac{1}{6}\big(1 \cdot 2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2\big) = 1.$$

Hence the standard representation is irreducible. Together with the trivial and sign representations it gives three pairwise non-isomorphic irreducibles. Since $\mathfrak{S}_3$ has three conjugacy classes, this list is complete; equivalently, their squared dimensions sum to $1^2 + 1^2 + 2^2 = 6 = |\mathfrak{S}_3|$.

	C_0 (# 1)	C_1 (# 3)	C_2 (# 2)
triv	1	1	1
sgn	1	-1	1
V_{std}	2	0	-1

Example 3.10 (Representations of $\mathfrak{S}_4$)**.** The five conjugacy classes of $\mathfrak{S}_4$ have cycle types

$$1^4, \qquad 2\,1^2, \qquad 2^2, \qquad 3\,1, \qquad 4,$$

with respective sizes $1, 6, 3, 8, 6$. For the standard representation

$$V_{\mathrm{std}} = \{(x_1, x_2, x_3, x_4) \in \mathbb{C}^4 \mid x_1 + x_2 + x_3 + x_4 = 0\},$$

the fixed-point formula gives

$$\chi_{\mathrm{std}}(g) = \#\,\mathrm{Fix}_{\{1,2,3,4\}}(g) - 1.$$

Its values on the five classes above are $(3, 1, -1, 0, -1)$, so

$$(\chi_{\mathrm{std}} \mid \chi_{\mathrm{std}}) = \frac{1}{24}\big(9 + 6 + 3 + 0 + 6\big) = 1.$$

Thus V_{std} is irreducible. The sign character has values $(1, -1, 1, 1, -1)$, and hence $V_{\mathrm{std}} \otimes \mathrm{sgn}$ has character $(3, -1, -1, 0, 1)$. Since $|\chi_{\mathrm{sgn}}(g)| = 1$, this character has norm 1, while

$$(\chi_{\mathrm{std}} \mid \chi_{\mathrm{std}}\chi_{\mathrm{sgn}}) = \frac{1}{24}\big(9 - 6 + 3 + 0 - 6\big) = 0.$$

Thus the twist is irreducible and is not isomorphic to V_{std}.

A 2-dimensional representation is obtained from the quotient

$$\mathfrak{S}_4 \longrightarrow \mathfrak{S}_4/K \simeq \mathfrak{S}_3,$$

where

$$K = \{1, (12)(34), (13)(24), (14)(23)\}$$

is the Klein four subgroup. Pulling back the standard representation of $\mathfrak{S}_3$ along this quotient gives an irreducible 2-dimensional representation of $\mathfrak{S}_4$: an invariant subspace for the pullback would also be invariant under the quotient $\mathfrak{S}_3$.

We have now obtained five pairwise non-isomorphic irreducible representations: the trivial and sign representations, the two distinct 3-dimensional representations above, and this 2-dimensional pullback. Since $\mathfrak{S}_4$ has five conjugacy classes, the list is complete. The dimension identity gives the same check:

$$1^2 + 1^2 + 3^2 + 3^2 + 2^2 = 24 = |\mathfrak{S}_4|.$$

3.5 Representations of Compact Groups

Many constructions for finite groups have compact-group analogues. The finite average

$$\frac{1}{|G|}\sum_{g\in G}$$

is replaced by integration against Haar measure.

Definition 3.11. A **topological group** is a group G equipped with a Hausdorff topology such that multiplication $G\times G\to G$ and inversion $G\to G$ are continuous. If the underlying topological space is compact, then G is called a **compact group**.

Definition 3.12. A finite-dimensional complex representation of a topological group G is a finite-dimensional complex vector space V together with a continuous group homomorphism

$$\rho : G \to \mathrm{GL}(V).$$

Basic examples of compact groups include finite groups with the discrete topology, the circle group S^1, the special unitary groups $\mathrm{SU}(n)$, and the orthogonal groups $\mathrm{O}(n)$ and $\mathrm{SO}(n)$.

Theorem 3.13 (Haar measure)**.** *Let G be a compact group. There is a unique normalized regular Borel Haar measure dg on G. It is characterized by the following properties:*

1. *Linearity: $f \mapsto \int_G f(g)\,dg$ is linear on continuous functions.*
2. *Positivity: if $f \geq 0$ and f is not identically zero, then $\int_G f(g)\,dg > 0$.*
3. *Translation invariance:*

$$\int_G f(g)\,dg = \int_G f(hg)\,dg = \int_G f(gh)\,dg$$

   for all $h \in G$.
4. *Normalization: $\int_G 1\,dg = 1$.*

For a proof and further discussion of Haar measure, see [3].

Example 3.14. If G is finite with the discrete topology, then

$$\int_G f(g)\,dg = \frac{1}{|G|}\sum_{g\in G} f(g).$$

If $G = S^1 = \{e^{i\theta} : 0 \le \theta < 2\pi\}$, then

$$\int_{S^1} f(z)\, dg = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta.$$

Using Haar measure, we define the inner product of two continuous functions by

$$(\phi \mid \psi) = \int_G \overline{\phi(g)}\, \psi(g)\, dg.$$

The usual averaging argument shows that every finite-dimensional continuous representation of a compact group admits a G-invariant Hermitian inner product. Consequently, such representations are completely reducible.

Irreducible characters are still orthonormal with respect to this inner product. The precise compact-group replacement for the finite statement is the Peter-Weyl theorem.

Theorem 3.15 (Peter-Weyl theorem, informal form)**.** *Let G be a compact group. The matrix coefficients of irreducible finite-dimensional unitary representations of G span a dense subspace of $L^2(G)$. In particular, irreducible characters form an orthonormal basis of the closed subspace of $L^2(G)$ consisting of square-integrable class functions.*

For a proof of the Peter–Weyl theorem, see [3].

3.6 Linear Characters and Fourier Transforms

Definition 3.16. A **linear character** of G is a continuous group homomorphism

$$\chi : G \to \mathbb{C}^\times.$$

Equivalently, it is a one-dimensional complex representation of G. The set of linear characters is denoted by $\widehat{G}$.

The set $\widehat{G}$ is an abelian group under pointwise multiplication; the identity element is the trivial character.

Theorem 3.17. *Let G be a finite abelian group. Every irreducible complex representation of G is one-dimensional. Hence*

$$\mathrm{Irr}(G) = \widehat{G}.$$

Proof. Since G is abelian, every conjugacy class is a singleton. Therefore $|\operatorname{Conj}(G)| = |G|$, and Theorem 3.7 gives $|\operatorname{Irr}(G)| = |G|$. The dimension formula says

$$|G| = \sum_{W \in \mathrm{Irr}(G)} (\dim W)^2.$$

This is a sum of $|G|$ positive integer squares, so every term must be 1. Thus every irreducible representation is one-dimensional. □

Example 3.18. Let $G = \mathbb{Z}/d\mathbb{Z}$ and let $\zeta = \exp(2\pi i/d)$. The linear characters are

$$\chi_k(1) = \zeta^k, \qquad k = 0, \dots, d-1.$$

Hence $\widehat{G} \simeq \mathbb{Z}/d\mathbb{Z}$.

For the circle group S^1, the continuous linear characters are

$$\chi_n(e^{i\theta}) = e^{in\theta}, \qquad n \in \mathbb{Z}.$$

Their orthogonality is the standard orthogonality of Fourier modes:

$$(\chi_n \mid \chi_m) = \frac{1}{2\pi}\int_0^{2\pi} e^{-in\theta}e^{im\theta}\,d\theta = \delta_{n,m}.$$

The Fourier transform

Let G be a finite abelian group. Since the characters of G form an orthonormal basis of the space of functions on G, every function $\phi : G \to \mathbb{C}$ has a Fourier expansion. Define

$$\widehat{\phi}(\chi) = (\chi \mid \phi) = \frac{1}{|G|}\sum_{g\in G} \overline{\chi(g)}\,\phi(g), \qquad \chi \in \widehat{G}.$$

Then Fourier inversion is the identity

$$\phi = \sum_{\chi\in\widehat{G}} \widehat{\phi}(\chi)\chi.$$

3.7 The Group Algebra and its Center

The same decomposition theory also has an algebraic formulation.

Definition 3.19. Let G be a finite group and let k be a field. The **group algebra** $k[G]$ is the k-vector space with basis $\{g : g \in G\}$. Multiplication is defined by extending the group law bilinearly:

$$\left(\sum_{g\in G} a_g g\right)\left(\sum_{h\in G} b_h h\right) = \sum_{z\in G}\left(\sum_{g\in G} a_g b_{g^{-1}z}\right) z.$$

A representation of G over k is the same thing as a left $k[G]$-module: the group action extends linearly to $k[G]$, and conversely the action of the basis elements $g \in G$ recovers the group action.

Theorem 3.20 (Artin-Wedderburn for the complex group algebra)**.** *Let $\{W_i\}$ be a complete set of irreducible complex representations of G. The representation maps*

$$\rho_i : \mathbb{C}[G] \to \mathrm{End}_{\mathbb{C}}(W_i)$$

assemble to an algebra isomorphism

$$\prod_i \rho_i : \mathbb{C}[G] \xrightarrow{\sim} \prod_i \mathrm{End}_{\mathbb{C}}(W_i).$$

This is the group-algebra case of the Artin–Wedderburn theorem proved in Chapter 7; see Theorem 7.19.

Taking centers gives

$$Z(\mathbb{C}[G]) \simeq \prod_i \mathbb{C},$$

because the center of a full matrix algebra consists of scalar matrices. The central primitive idempotent corresponding to W_i is

$$e_i = \frac{\dim W_i}{|G|} \sum_{g \in G} \overline{\chi_{W_i}(g)}\, g \in Z(\mathbb{C}[G]).$$

For any representation V, the operator $\rho_V(e_i)$ is exactly the projection p_i from Proposition 3.4.

Chapter 4

Group Algebras, Idempotents, and Orthogonality

4.1 Convolution and the Group Algebra

Let G be a finite group and let k be a field. The group algebra $k[G]$ has basis $\{g : g \in G\}$, and every element can be written uniquely as

$$a = \sum_{g \in G} a_g g.$$

Thus there is a natural vector-space isomorphism

$$k[G] \xrightarrow{\sim} \mathrm{Fun}(G, k), \qquad a \longmapsto \widehat{a}, \qquad \widehat{a}(g) = a_g.$$

Under this identification, multiplication in $k[G]$ becomes **convolution**. If $f = \widehat{a}$ and $f' = \widehat{b}$, then

$$(f * f')(z) = \sum_{gh=z} f(g) f'(h) = \sum_{g \in G} f(g) f'(g^{-1}z).$$

The identity element for convolution is the delta function

$$\delta_1(g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

This formula can be expressed geometrically using the multiplication map $m : G \times G \to G$ and the projections $P_1, P_2 : G \times G \to G$:

$$\begin{array}{ccccc} & & G \times G & & \\ & \swarrow^{P_1} & \downarrow m & \searrow^{P_2} & \\ G & & G & & G. \end{array}$$

For each $z \in G$, the value $(f * f')(z)$ is the sum over the fiber $m^{-1}(z)$ of the product $P_1^* f \cdot P_2^* f'$.

Definition 4.1. Let A be an associative unital algebra. Its **center** is

$$Z(A) = \{a \in A : ab = ba \text{ for all } b \in A\}.$$

Proposition 4.2. *The center of the convolution algebra* $\mathrm{Fun}(G,k)$ *is the space of class functions on* G*:*

$$Z(\mathrm{Fun}(G,k),*) = \mathrm{CF}(G,k).$$

Proof. Let f be a class function and let φ be any function on G. Then

$$(f*\varphi)(z) = \sum_{g\in G} f(zg^{-1})\varphi(g), \qquad (\varphi * f)(z) = \sum_{g\in G} \varphi(g) f(g^{-1}z).$$

Since zg^{-1} and $g^{-1}z$ are conjugate, $f(zg^{-1}) = f(g^{-1}z)$. Hence $f*\varphi = \varphi * f$.

Conversely, suppose f belongs to the center. Taking $\varphi = \delta_g$ gives

$$f(zg^{-1}) = f(g^{-1}z) \qquad \text{for all } g, z \in G.$$

Replacing z by zg gives $f(z) = f(g^{-1}zg)$. Thus f is constant on conjugacy classes. □

4.2 Central Idempotents

Let A be a finite-dimensional unital algebra over a field k.

Definition 4.3. An element $e \in A$ is an **idempotent** if $e^2 = e$. Two idempotents e and e' are **orthogonal** if

$$ee' = e'e = 0.$$

An idempotent is **central** if it lies in $Z(A)$.

Definition 4.4. A nonzero central idempotent e is **primitive** if it cannot be written as

$$e = e' + e''$$

with e' and e'' nonzero orthogonal central idempotents. A **complete set of primitive central idempotents** is a collection $\{e_1, \dots, e_n\}$ such that

$$e_i e_j = 0 \quad (i \neq j), \qquad \sum_{i=1}^{n} e_i = 1,$$

and each e_i is primitive and central.

Theorem 4.5. *Every finite-dimensional unital algebra* A *has a complete set of primitive central idempotents. This set is unique up to reordering.*

Proof. First note that any family of nonzero mutually orthogonal central idempotents is linearly independent: if $\sum_i c_i e_i = 0$, then multiplying by e_j gives $c_j e_j = 0$, hence $c_j = 0$.

Since $Z(A)$ is finite-dimensional, there is a maximal decomposition

$$1 = e_1 + \cdots + e_n$$

into mutually orthogonal nonzero central idempotents. Each e_i must be primitive. If, say, $e_i = e_i' + e_i''$ with e_i' and e_i'' nonzero orthogonal central idempotents, then replacing e_i by e_i' and e_i'' would give a longer orthogonal decomposition of 1, contradicting maximality.

For uniqueness, let

$$1 = e_1 + \cdots + e_n = e_1' + \cdots + e_m'$$

be two complete decompositions. Fix i. Multiplying the second decomposition by e_i gives

$$e_i = \sum_{j=1}^{m} e_i e'_j.$$

The terms $e_i e'_j$ are mutually orthogonal central idempotents. Since e_i is primitive, exactly one of them is nonzero; call it $e_i e'_j$. Then $e_i = e_i e'_j$. Applying the same argument to the primitive idempotent e'_j gives $e'_j = e_i e'_j$. Hence $e_i = e'_j$. This matches the two decompositions term by term, up to reordering. □

4.3 Irreducible Characters as Idempotents

Now take $k = \mathbb{C}$. By the previous section, the primitive central idempotents of $\mathbb{C}[G]$ may be viewed as special class functions on G. Character theory gives them explicitly.

Theorem 4.6. *Let W_i be an irreducible complex representation of G with character χ_i. Define*

$$\varepsilon_i(g) = \frac{\dim W_i}{|G|}\chi_i(g^{-1}).$$

Under the identification $\mathbb{C}[G] \simeq \mathrm{Fun}(G, \mathbb{C})$, the functions ε_i are precisely the primitive central idempotents. Equivalently,

$$e_i = \frac{\dim W_i}{|G|} \sum_{g \in G} \chi_i(g^{-1})\, g$$

is the primitive central idempotent corresponding to W_i.

Proof. Since $\chi_i(g^{-1}) = \overline{\chi_i(g)}$, the formula is the same central idempotent already obtained from the isotypic projection formula. Acting on a representation V, the element

$$e_i = \frac{\dim W_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)}\, g$$

gives the projection onto the W_i-isotypic component of V. Hence e_i acts by the identity on W_i and by zero on every irreducible representation not isomorphic to W_i.

It follows that $e_i^2 = e_i$ and $e_i e_j = 0$ for $i \neq j$, because these identities can be checked on every irreducible representation and then on the regular representation. Also,

$$\sum_i e_i$$

acts as the identity on every irreducible representation, hence on the regular representation, so $\sum_i e_i = 1$ in $\mathbb{C}[G]$.

It remains to check primitivity. If $e_i = e' + e''$ is a sum of orthogonal central idempotents, then e' and e'' vanish on every irreducible other than W_i. On W_i they are scalar idempotents by Schur's lemma, and their sum is the identity, so exactly one acts as zero. A group-algebra element that acts as zero on every irreducible acts as zero on the regular representation and is therefore zero. Thus one of e', e'' is zero, and each e_i is primitive. Therefore the e_i form a complete set of primitive central idempotents. □

Remark 4.7. If all characters of G are real-valued, then $\chi_i(g^{-1}) = \chi_i(g)$ and the idempotent is simply the scalar multiple $(\dim W_i/|G|)\chi_i$ as a class function. In general, the inverse in $\chi_i(g^{-1})$ is essential.

4.4 Unitary Representations

Definition 4.8. A **unitary representation** of a group G is a complex representation V together with a positive definite Hermitian form $B(-,-)$ such that

$$B(\rho(g)v, \rho(g)w) = B(v,w) \qquad \text{for all } g \in G,\ v, w \in V.$$

Theorem 4.9. *Let G be a finite group. Every finite-dimensional complex representation of G admits a G-invariant positive definite Hermitian form. If the representation is irreducible, this form is unique up to multiplication by a positive real scalar.*

Proof. Start with any positive definite Hermitian form B' on V and average it:

$$B(v,w) = \frac{1}{|G|} \sum_{g \in G} B'(\rho(g)v, \rho(g)w).$$

This is still positive definite, and replacing g by gh in the sum shows that $B(\rho(h)v, \rho(h)w) = B(v,w)$ for every $h \in G$.

Now assume V is irreducible, and let B_1 and B_2 be two invariant positive definite Hermitian forms. There is a unique linear operator $A : V \to V$ such that

$$B_1(v,w) = B_2(v, Aw) \qquad \text{for all } v, w \in V.$$

Using the invariance of both forms,

$$B_2(v, Aw) = B_1(v,w) = B_1(\rho(g)^{-1}v, \rho(g)^{-1}w) = B_2(v, \rho(g)A\rho(g)^{-1}w).$$

Since B_2 is nondegenerate, $A = \rho(g)A\rho(g)^{-1}$ for all g. Thus A is G-equivariant. By Schur's lemma, $A = \lambda \,\mathrm{id}_V$. Positivity of B_1 and B_2 forces $\lambda \in \mathbb{R}_{>0}$. □

Corollary 4.10. *Every finite-dimensional unitary representation of a group is completely reducible. In particular, every finite-dimensional complex representation of a finite group is completely reducible.*

Proof. Let $W \subset V$ be a subrepresentation. If $v \in W^{\perp}$, then for $w \in W$ and $g \in G$,

$$B(gv, w) = B(v, g^{-1}w) = 0,$$

because $g^{-1}w \in W$. Hence $W^{\perp}$ is also a subrepresentation, and

$$V = W \oplus W^{\perp}.$$

Iterating this argument gives a decomposition into irreducible subrepresentations. □

4.5 The Frobenius Determinant

Historically, representation theory arose in part from the study of group determinants. Let

$$G = \{g_1, \ldots, g_n\}$$

and introduce commuting variables $\{x_g : g \in G\}$.

Definition 4.11. The **group matrix** of G is

$$X_G = (x_{g_i g_j^{-1}})_{1 \le i,j \le n}.$$

Its determinant is a homogeneous polynomial of degree n in the variables x_g, called the **Frobenius determinant** or **group determinant**.

Remark 4.12. Some sources use $x_{g_i^{-1} g_j}$ or $x_{g_i g_j}$ instead. These choices differ by reindexing rows, columns, or variables. The determinant may acquire an overall sign, but, after the corresponding relabeling of variables, its irreducible factors are unchanged up to nonzero scalar multiples.

Theorem 4.13 (Frobenius factorization)**.** *Let $W_1, \dots, W_r$ be the irreducible complex representations of G. For each j, define*

$$P_j(\underline{x}) = \det \left(\sum_{g \in G} x_g \, \rho_j(g) \right).$$

Then

$$\det X_G = \prod_{j=1}^{r} P_j(\underline{x})^{\dim W_j}.$$

Each P_j has degree $\dim W_j$. Moreover, the polynomials P_j are irreducible and pairwise non-proportional.

Proof. Consider the element

$$x = \sum_{g \in G} x_g g$$

in the group algebra over the polynomial ring $\mathbb{C}[x_g : g \in G]$. The matrix X_G is the matrix of left multiplication by x on the regular representation, with respect to the basis indexed by G.

Over $\mathbb{C}$, the regular representation decomposes as

$$\mathbb{C}[G] \simeq \bigoplus_{j=1}^{r} W_j^{\oplus \dim W_j}.$$

On the summand W_j, the element x acts by the matrix $\sum_g x_g \rho_j(g)$. Taking determinants on both sides of the regular representation decomposition gives the stated factorization. The assertion that the factors are irreducible and pairwise non-proportional is the classical irreducibility theorem for the group determinant; see [2]. □

Example 4.14. Let $G = \mathbb{Z}/n\mathbb{Z}$ and write $\zeta = \exp(2\pi i/n)$. Since all irreducible representations of G are one-dimensional, the group determinant of the cyclic group factors into linear forms:

$$\det X_G = \prod_{i=0}^{n-1} \left(\sum_{j=0}^{n-1} \zeta^{ij} x_j \right).$$

This is the usual factorization of a circulant determinant.

4.6 Algebras and Modules

The representation theory of groups is a special case of the representation theory of associative algebras.

Definition 4.15. Let A be an associative unital k-algebra. A **representation** of A is a vector space M together with a unital algebra homomorphism

$$\rho : A \to \mathrm{End}_k(M).$$

Equivalently, a representation of A is a left A-module: the action is

$$a \cdot m = \rho(a)m,$$

and the homomorphism and unit conditions give the associativity and unit laws

$$(ab) \cdot m = a \cdot (b \cdot m), \qquad 1_A \cdot m = m.$$

The standard vocabulary translates as follows:

- subrepresentations are submodules;
- irreducible representations are simple modules;
- indecomposable representations are indecomposable modules.

Example 4.16 (Fundamental algebras)**.** The following examples will appear repeatedly.

1. **Group algebras.** For $A = k[G]$, the category of left A-modules is equivalent to the category of k-linear representations of G.
2. **Polynomial rings.** Giving a module over $k[x]$ is the same as giving a vector space M together with one linear operator $T : M \to M$, where x acts by T.
3. **Tensor algebras.** If V is a vector space, then

   $$T^\bullet V = \bigoplus_{n \geq 0} V^{\otimes n}$$

   is the free associative algebra generated by V. A linear map $\phi : V \to \mathrm{End}_k(M)$ extends uniquely to an algebra homomorphism $T^\bullet V \to \mathrm{End}_k(M)$.
4. **Symmetric algebras.** The symmetric algebra is

   $$S^\bullet V = T^\bullet V / \langle v \otimes v' - v' \otimes v \rangle.$$

   It is the free commutative algebra generated by V; after choosing a basis of V, it becomes a polynomial ring.
5. **Exterior algebras.** The exterior algebra is

   $$\bigwedge^\bullet V = T^\bullet V / \langle v \otimes v : v \in V \rangle.$$

   It imposes the alternating relation $v \wedge v = 0$.

Chapter 5

Induced Representations and Characters of Algebras

Let G be a finite group and let k be an algebraically closed field, unless otherwise stated.

5.1 The Converse of Maschke's Theorem

Maschke's theorem says that if $\mathrm{char}(k)$ does not divide $|G|$, then every finite-dimensional representation of G over k is completely reducible. In the terminology of Chapter 7, this is equivalent to saying that the group algebra $k[G]$ is semisimple; see Theorem 7.19. The converse is also true.

Proposition 5.1. *Suppose $k[G]$ is semisimple. Then* $\mathrm{char}(k)$ *does not divide $|G|$.*

Proof. Let k denote the trivial $k[G]$-module. If $k[G]$ is semisimple, then the trivial module is a direct summand of the regular module $k[G]$. Thus there exist $k[G]$-module maps

$$i : k \to k[G], \qquad p : k[G] \to k$$

such that $p \circ i = \mathrm{id}_k$.

We now compute all such maps. A G-equivariant map $i : k \to k[G]$ is determined by $i(1)$, and this vector must be fixed by the left regular action of G. The fixed subspace of $k[G]$ is one-dimensional, spanned by

$$\Lambda = \sum_{g \in G} g.$$

Hence $i(1) = c\Lambda$ for some $c \in k$.

Similarly, a G-equivariant map $p : k[G] \to k$ is determined by the image of the basis elements $g \in G$, and equivariance forces all these images to be equal. Thus p is a scalar multiple of the augmentation map

$$\varepsilon : k[G] \to k, \qquad \varepsilon\left(\sum_{g \in G} a_g g\right) = \sum_{g \in G} a_g.$$

Therefore every composition $k \to k[G] \to k$ is a scalar multiple of

$$\varepsilon(\Lambda) = |G| \in k.$$

If $\mathrm{char}(k)$ divided $|G|$, this scalar would be zero, so no such composition could be the identity on k. This contradicts the fact that k is a direct summand of $k[G]$. Hence $|G| \neq 0$ in k. □

Example 5.2. Let $G = \mathbb{Z}/p\mathbb{Z}$ and let k have characteristic p. If k is algebraically closed, then the trivial representation is the only irreducible representation of G.

Indeed, let g be a generator of G. On any irreducible representation, the operator $\rho(g)$ has an eigenvector. The span of this eigenvector is stable under G, so irreducibility forces the representation to be one-dimensional. The scalar λ by which g acts satisfies $\lambda^p = 1$. In characteristic p,

$$x^p - 1 = (x-1)^p,$$

so $\lambda = 1$. Thus every irreducible representation is trivial. Since $\mathrm{char}(k)$ divides $|G|$, the preceding converse to Maschke's theorem also shows directly that $k[G]$ is not semisimple.

5.2 Characters of Arbitrary Algebras

Characters also make sense for representations of associative algebras. Let A be an associative unital k-algebra, let all modules be unital, and let V be a finite-dimensional A-module with structure map

$$\rho : A \to \mathrm{End}_k(V).$$

Definition 5.3. The **character** of V is the linear functional

$$\chi_V : A \to k, \qquad \chi_V(a) = \mathrm{tr}(\rho(a)).$$

Since $\mathrm{tr}(XY) = \mathrm{tr}(YX)$, we have

$$\chi_V(ab) = \chi_V(ba).$$

Thus χ_V vanishes on the commutator subspace

$$[A, A] = \mathrm{Span}\{ab - ba : a, b \in A\}.$$

The quotient vector space

$$A/[A, A]$$

is called the **cocenter** of A, and every character factors through the cocenter.

Theorem 5.4. *Let A be a k-algebra.*

1. *Let $W_1, \dots, W_r$ be pairwise non-isomorphic finite-dimensional irreducible A-modules. Then their characters $\chi_{W_1}, \dots, \chi_{W_r}$ are linearly independent.*
2. *If A is finite-dimensional and semisimple, then the irreducible characters form a basis of the dual cocenter:*

$$\{\chi_W : W \in \mathrm{Irr}(A)\} \text{ is a basis of } (A/[A,A])^*.$$

Proof. For (1), use the Jacobson density theorem (Theorem 7.14, proved in Chapter 7). The map

$$A \longrightarrow \bigoplus_{i=1}^{r} \mathrm{End}_k(W_i)$$

induced by the representations is surjective. Suppose $\sum_i c_i \chi_{W_i} = 0$. Fix j. By surjectivity, choose $a \in A$ whose image on W_j is the matrix unit E_{11} and whose image on W_i is zero for $i \neq j$. Evaluating the relation on a gives $c_j = 0$. Since j was arbitrary, all coefficients vanish.

For (2), first consider $M_n(k)$. Its commutator subspace is the space of trace-zero matrices, so

$$(M_n(k)/[M_n(k), M_n(k)])^*$$

is one-dimensional and is spanned by the trace. If A is finite-dimensional and semisimple, then Theorem 7.19 gives

$$A \simeq \bigoplus_{W \in \mathrm{Irr}(A)} \mathrm{End}_k(W).$$

Passing to cocenters gives one copy of k for each irreducible module, and the corresponding trace functionals are exactly the irreducible characters. Hence these characters form a basis of $(A/[A, A])^*$. □

5.3 Jordan–Hölder and Krull–Schmidt Theorems

Definition 5.5. Let V be a finite-dimensional representation of an algebra A. A **Jordan–Hölder filtration**, or **composition series**, of V is a chain of A-submodules

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$$

such that each quotient V_i/V_{i-1} is irreducible.

For an irreducible module W, define its multiplicity in such a filtration by

$$[V : W] = \#\{i : V_i/V_{i-1} \simeq W\}.$$

Theorem 5.6 (Jordan–Hölder)**.** *Every finite-dimensional A-module has a composition series, and the multiplicity $[V : W]$ is independent of the chosen composition series.*

Proof. Existence follows by repeatedly choosing a proper submodule of maximal dimension; the successive quotients are simple, and dimensions strictly decrease. For uniqueness, we argue by induction on $\dim V$. There is nothing to prove when $V = 0$. Given two composition series, let M and N be their respective penultimate terms. Thus M and N are maximal proper submodules of V. If $M = N$, the induction hypothesis applied to this common submodule finishes the proof.

Suppose $M \neq N$. Maximality gives $M + N = V$, and the second isomorphism theorem gives

$$M/(M \cap N) \simeq V/N, \qquad N/(M \cap N) \simeq V/M.$$

Both quotients are irreducible. Choose a composition series of $M \cap N$; one exists because dimensions strictly decrease along proper submodules. Appending first M and then V gives a composition series of V whose last two factors are V/N and V/M; appending first

N and then V gives the same two factors in the opposite order. By induction, the original series truncated at M has the same factors as the series obtained through $M \cap N$, and similarly for N. Hence the two original composition series have the same factors with the same multiplicities. □

When $\mathrm{char}(k) = 0$, there is also a short character-theoretic verification. The linear independence of irreducible characters proved above gives

$$\chi_V = \sum_{i=1}^{n} \chi_{V_i/V_{i-1}} = \sum_{W \in \mathrm{Irr}(A)} [V : W]\chi_W.$$

The coefficients in this expansion are uniquely determined by χ_V. In positive characteristic the same argument determines multiplicities only modulo $\mathrm{char}(k)$, so it cannot replace the preceding proof.

If $[V : W] \neq 0$, then W is called a **composition factor** of V. The common value

$$\ell(V) = \sum_{W} [V : W]$$

is the **length** of V.

Theorem 5.7 (Krull–Schmidt)**.** *Every finite-dimensional representation of an algebra decomposes as a finite direct sum of indecomposable representations. This decomposition is unique up to isomorphism and reordering of the indecomposable summands.*

Proof. Existence follows by induction on dimension: if a nonzero module is not indecomposable, split it into two nonzero submodules and apply induction to both.

For uniqueness, first let M be indecomposable and let $f \in \mathrm{End}_A(M)$. For all sufficiently large n, the kernels and images of the powers of f have stabilized. If $x = f^n y \in \ker(f^n)$, then $f^{2n} y = 0$, hence $y \in \ker(f^n)$ and $x = 0$. Rank–nullity now gives the Fitting decomposition

$$M = \ker(f^n) \oplus \mathrm{im}(f^n).$$

Indecomposability forces one summand to vanish. Thus f is either invertible or nilpotent. In the latter case $1 - f$ is invertible. Hence, for every f, either f or $1 - f$ is invertible, so the standard local-ring criterion shows that $\mathrm{End}_A(M)$ is local. Equivalently, its nonunits form the unique maximal ideal; in particular, a finite sum of nonunits cannot equal 1.

Now suppose

$$X = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$

with all summands indecomposable. Let ι_i, p_i and $\jmath_j, q_j$ be the inclusions and projections for the two decompositions. In the local ring $\mathrm{End}_A(M_1)$ we have

$$1_{M_1} = \sum_{j=1}^{s} p_1 \jmath_j q_j \iota_1.$$

Hence at least one summand, say $p_1 \jmath_j q_j \iota_1$, is invertible. It write this invertible endomorphism as a. Then $a^{-1} p_1 \jmath_j$ is a left inverse of $q_j \iota_1 : M_1 \to N_j$. Thus M_1 is isomorphic to a nonzero direct summand of N_j. Since N_j is indecomposable, $q_j \iota_1$ is an isomorphism, and

so is the restriction $p_1|_{N_j} : N_j \to M_1$. Relative to $X = M_1 \oplus \bigoplus_{i=2}^{r} M_i$, the submodule N_j is therefore a graph over M_1, whence

$$X = N_j \oplus \bigoplus_{i=2}^{r} M_i.$$

Passing to the quotient by N_j gives

$$\bigoplus_{i=2}^{r} M_i \simeq \bigoplus_{k \neq j} N_k.$$

Induction on the number of summands proves uniqueness up to isomorphism and reordering. □

5.4 Induction, Restriction, and Frobenius Reciprocity

Let $H \subset G$ be a subgroup. There are three basic functors relating representations of H and representations of G.

Definition 5.8. The **restriction** functor is

$$\mathrm{Res}_H^G : G\text{-mod} \to H\text{-mod}.$$

It keeps the same vector space and restricts the action from G to H.

The **induction** functor is

$$\mathrm{Ind}_H^G : H\text{-mod} \to G\text{-mod}, \qquad N \longmapsto k[G] \otimes_{k[H]} N,$$

where $k[G]$ is regarded as a $(k[G], k[H])$-bimodule.

The **coinduction** functor is

$$\mathrm{CoInd}_H^G : H\text{-mod} \to G\text{-mod}, \qquad N \longmapsto \mathrm{Hom}_{k[H]}(k[G], N),$$

where $k[G]$ is a left $k[H]$-module. The natural left G-action on the Hom space is right translation:

$$(g \cdot f)(x) = f(xg) \qquad (g, x \in G).$$

The restriction functor can be written in two useful ways:

$$\mathrm{Res}_H^G(M) \simeq k[G] \otimes_{k[G]} M, \qquad \mathrm{Res}_H^G(M) \simeq \mathrm{Hom}_{k[G]}(k[G], M).$$

In the first expression, $k[G]$ is viewed as a $(k[H], k[G])$-bimodule. In the second, it is viewed as a $(k[G], k[H])$-bimodule.

Theorem 5.9 (Frobenius reciprocity)**.** *Induction is left adjoint to restriction, and coinduction is right adjoint to restriction. Explicitly, for an H-module N and a G-module M, there are natural isomorphisms*

$$\mathrm{Hom}_G(\mathrm{Ind}_H^G N, M) \simeq \mathrm{Hom}_H(N, \mathrm{Res}_H^G M)$$

and

$$\mathrm{Hom}_G(M, \mathrm{CoInd}_H^G N) \simeq \mathrm{Hom}_H(\mathrm{Res}_H^G M, N).$$

Proof. These are the usual Hom-tensor adjunctions. For example,

$$\begin{aligned}\mathrm{Hom}_G(k[G] \otimes_{k[H]} N, M) &\simeq \mathrm{Hom}_H(N, \mathrm{Hom}_G(k[G], M)) \\ &\simeq \mathrm{Hom}_H(N, \mathrm{Res}^G_H M).\end{aligned}$$

The second adjunction follows similarly from the Hom version of the same formalism. □

Proposition 5.10. *For finite groups, induction and coinduction are naturally isomorphic:*

$$\mathrm{Ind}^G_H \simeq \mathrm{CoInd}^G_H .$$

Consequently, induction is both left and right adjoint to restriction.

Proof. Choose left coset representatives R for G/H, so

$$G = \coprod_{r \in R} rH.$$

As a vector space,

$$\mathrm{Ind}^G_H(N) = \bigoplus_{r \in R} r \otimes N.$$

If $g \in G$, then $gr = r'h$ for a unique $r' \in R$ and $h \in H$, and

$$g \cdot (r \otimes n) = r' \otimes hn.$$

A coinduced vector is an H-equivariant function $f : k[G] \to N$. Such a function is determined by its values on the representatives r^{-1}, where $r \in R$. The map

$$f \longmapsto \sum_{r \in R} r \otimes f(r^{-1})$$

gives an isomorphism from $\mathrm{CoInd}^G_H(N)$ to $\mathrm{Ind}^G_H(N)$. Checking the above formula for the G-action shows that this isomorphism is G-equivariant. Although the displayed formula uses a choice of representatives, the resulting isomorphism is natural in N. □

5.5 Induced Characters and Automorphic Twists

Now assume $k = \mathbb{C}$. Let

$$\mathrm{res} : \mathrm{CF}(G) \to \mathrm{CF}(H)$$

be restriction of class functions. The induced class function is

$$\mathrm{ind} : \mathrm{CF}(H) \to \mathrm{CF}(G), \qquad (\mathrm{ind}\, f)(s) = \sum_{\substack{r \in G/H \\ r^{-1}sr \in H}} f(r^{-1}sr),$$

where G/H denotes any set of left coset representatives. The value is independent of the chosen representatives.

Lemma 5.11. *For every finite-dimensional H-representation W,*

$$\chi_{\mathrm{Ind}^G_H W} = \mathrm{ind}(\chi_W).$$

Proof. Write

$$\mathrm{Ind}_H^G W = \bigoplus_{r \in G/H} r \otimes W$$

as a vector space. The element $s \in G$ sends the summand $r \otimes W$ to the summand $sr \otimes W$. A summand contributes to the trace only when $srH = rH$, equivalently when $r^{-1}sr \in H$. On that summand, the induced action is the action of $r^{-1}sr$ on W. Summing these traces gives the stated formula. □

Definition 5.12 (Automorphic twists)**.** For $s \in G$, write

$${}^sH = sHs^{-1}.$$

If Y is an H-module, define the ${}^sH$-module ${}^sY$ to have the same underlying vector space as Y, with action

$$(shs^{-1}) \cdot y = h \cdot y.$$

This gives an equivalence from H-modules to ${}^sH$-modules.

Lemma 5.13. *Induction is compatible with conjugation. For every $s \in G$ and every H-module Y, there is a natural isomorphism of G-modules*

$$\mathrm{Ind}_H^G(Y) \simeq \mathrm{Ind}_{{}^sH}^G({}^sY).$$

Proof. Define

$$k[G] \otimes_{k[H]} Y \longrightarrow k[G] \otimes_{k[{}^sH]} {}^sY, \qquad x \otimes y \longmapsto xs^{-1} \otimes y.$$

This is well-defined: for $h \in H$,

$$xh \otimes y \longmapsto xhs^{-1} \otimes y = xs^{-1}(shs^{-1}) \otimes y = xs^{-1} \otimes hy.$$

This is exactly the image of $x \otimes hy$. The inverse is $x \otimes y \mapsto xs \otimes y$, and the maps commute with the left action of G. □

Chapter 6

Mackey Formula and Applications

Throughout this chapter all groups are finite. Except for the last corollary, representations are finite-dimensional complex representations.

6.1 The Mackey Formula

Let H and L be subgroups of a finite group G. The group $L \times H$ acts on G by

$$(l, h) \cdot g = lgh^{-1}.$$

The orbits for this action are the double cosets LgH. We write $[L\backslash G/H]$ for a choice of representatives of these double cosets. For $r \in G$, set ${}^rH = rHr^{-1}$. If Y is an H-module, then ${}^rY$ denotes the corresponding ${}^rH$-module: the element $rhr^{-1} \in {}^rH$ acts on ${}^rY$ as h acts on Y.

Theorem 6.1 (Mackey Formula)**.** *Let $H, L \leq G$, and let Y be a representation of H. For any choice of representatives $[L\backslash G/H]$, there is an isomorphism of L-modules*

$$\operatorname{Res}^G_L \operatorname{Ind}^G_H(Y) \simeq \bigoplus_{r \in [L\backslash G/H]} \operatorname{Ind}^L_{L\cap {}^rH} \left(\operatorname{Res}^{{}^rH}_{L\cap {}^rH}({}^rY)\right).$$

Proof. By definition,

$$\operatorname{Ind}^G_H(Y) = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} Y.$$

As an (L, H)-biset, G is the disjoint union of its double cosets:

$$G = \coprod_{r \in [L\backslash G/H]} LrH.$$

Therefore, after restricting the induced representation to L,

$$\operatorname{Res}^G_L \operatorname{Ind}^G_H(Y) = \bigoplus_{r \in [L\backslash G/H]} \mathbb{C}[LrH] \otimes_{\mathbb{C}[H]} Y.$$

It remains to identify a single double-coset summand. For each r, define

$$\Phi_r : \mathbb{C}[L] \otimes_{\mathbb{C}[L\cap {}^rH]} {}^rY \longrightarrow \mathbb{C}[LrH] \otimes_{\mathbb{C}[H]} Y,$$
$$l \otimes y \longmapsto lr \otimes y.$$

This map is well-defined. Indeed, if $a \in L \cap {}^{r}H$, then $a = rhr^{-1}$ for some $h \in H$, and

$$\Phi_r(la \otimes y) = lar \otimes y = lrh \otimes y = lr \otimes hy = \Phi_r(l \otimes ay).$$

The map is plainly L-equivariant and surjective. Its inverse is obtained by writing an element of LrH as lrh and sending

$$lrh \otimes y \longmapsto l \otimes hy.$$

The ambiguity in the expression lrh is exactly the balancing by $L \cap {}^{r}H$, so the inverse is well-defined. Hence

$$\mathbb{C}[LrH] \otimes_{\mathbb{C}[H]} Y \simeq \operatorname{Ind}_{L \cap {}^{r}H}^{L} \operatorname{Res}_{L \cap {}^{r}H}^{{}^{r}H}({}^{r}Y),$$

and summing over all double cosets proves the formula. □

Corollary 6.2 (Mackey's Intertwining Number Theorem)**.** *For $Y \in \operatorname{Rep}(H)$ and $Z \in \operatorname{Rep}(L)$, there is a natural isomorphism*

$$\operatorname{Hom}_G(\operatorname{Ind}_H^G Y, \operatorname{Ind}_L^G Z) \simeq \bigoplus_{r \in [L\backslash G/H]} \operatorname{Hom}_{L \cap {}^{r}H}\left(\operatorname{Res}_{L \cap {}^{r}H}^{{}^{r}H}({}^{r}Y), \operatorname{Res}_{L \cap {}^{r}H}^{L}(Z)\right).$$

Proof. For finite groups, induction is also right adjoint to restriction. Thus

$$\operatorname{Hom}_G(\operatorname{Ind}_H^G Y, \operatorname{Ind}_L^G Z) \simeq \operatorname{Hom}_L(\operatorname{Res}_L^G \operatorname{Ind}_H^G Y, Z).$$

Applying the Mackey formula to the first argument and using additivity of Hom gives the stated direct sum. □

6.2 Applications to Normal Subgroups

Proposition 6.3. *Let W be a representation of H, and put $V = \operatorname{Ind}_H^G W$. Then V is irreducible if and only if W is irreducible and, for every representative s of a non-identity double coset in $H\backslash G/H$,*

$$\operatorname{Hom}_{H \cap {}^{s}H}\left(\operatorname{Res}_{H \cap {}^{s}H}^{{}^{s}H}({}^{s}W), \operatorname{Res}_{H \cap {}^{s}H}^{H}(W)\right) = 0.$$

Equivalently, the two restricted modules in this Hom-space have no common irreducible constituent.

Proof. Since we are over $\mathbb{C}$, the category of G-representations is semisimple; hence V is irreducible if and only if $\dim \operatorname{End}_G(V) = 1$. Applying the intertwining theorem with $L = H$ and $Z = W$ gives

$$\operatorname{End}_G(V) \simeq \bigoplus_{s \in [H\backslash G/H]} \operatorname{Hom}_{H \cap {}^{s}H}\left(\operatorname{Res}_{H \cap {}^{s}H}^{{}^{s}H}({}^{s}W), \operatorname{Res}_{H \cap {}^{s}H}^{H}(W)\right).$$

The identity double coset contributes $\operatorname{Hom}_H(W, W)$. This summand is one-dimensional exactly when W is irreducible. All other summands must vanish, and this is precisely the condition stated above. □

Corollary 6.4. *Let $H \triangleleft G$ be a normal subgroup. Then $V = \operatorname{Ind}_H^G W$ is irreducible if and only if W is irreducible and*

$$W \not\simeq {}^xW \qquad \text{for every non-identity coset } x \in G/H.$$

Proof. If H is normal, then ${}^xH = H$ and $H \cap {}^xH = H$ for all $x \in G$. The previous proposition reduces the condition to $\operatorname{Hom}_H({}^xW, W) = 0$ for every $x \notin H$. Since W and ${}^xW$ are irreducible, Schur's lemma says that this Hom-space vanishes exactly when the two representations are not isomorphic. □

6.3 The Ring of Virtual Characters

Let G be a finite group. The genuine characters form the additive monoid

$$R^+(G) = \{\chi_V \mid V \in \operatorname{Rep}(G)\} = \bigoplus_{W \in \operatorname{Irr}(G)} \mathbb{Z}_{\geq 0}\, \chi_W.$$

The **ring of virtual characters**, or representation ring, is the Grothendieck group

$$R(G) = \bigoplus_{W \in \operatorname{Irr}(G)} \mathbb{Z}\, \chi_W.$$

Tensor product of representations makes $R(G)$ a commutative ring:

$$\chi_{V \otimes V'} = \chi_V \chi_{V'}.$$

Since irreducible characters form a basis of the space of complex class functions, there is a natural isomorphism

$$R(G) \otimes_{\mathbb{Z}} \mathbb{C} \simeq \operatorname{CF}(G, \mathbb{C}).$$

We use the standard bilinear character pairing

$$\langle \alpha, \beta \rangle_G = \frac{1}{|G|} \sum_{g \in G} \alpha(g^{-1})\beta(g),$$

which agrees with the usual Hermitian inner product on genuine characters.

Lemma 6.5 (Projection Formula)**.** *Let $H \leq G$. Let* $\operatorname{res} : R(G) \to R(H)$ *and* $\operatorname{ind} : R(H) \to R(G)$ *be the homomorphisms induced by restriction and induction. For $w \in R(H)$ and $v \in R(G)$,*

$$\operatorname{ind}\big(w \cdot \operatorname{res}(v)\big) = \operatorname{ind}(w) \cdot v.$$

Consequently, the image of $\operatorname{ind} : R(H) \to R(G)$ *is an ideal of $R(G)$.*

Proof. It is enough to prove the identity for genuine representations. Let W be an H-module and let V be a G-module. Define

$$\begin{aligned} \Psi : \operatorname{Ind}_H^G(W \otimes \operatorname{Res}_H^G V) &\longrightarrow \operatorname{Ind}_H^G(W) \otimes V, \\ g \otimes (w \otimes v) &\longmapsto (g \otimes w) \otimes gv. \end{aligned}$$

Its inverse is

$$(g \otimes w) \otimes v \longmapsto g \otimes (w \otimes g^{-1}v).$$

These maps respect the balancing over $\mathbb{C}[H]$ and are G-equivariant. Therefore

$$\mathrm{Ind}_H^G(W \otimes \mathrm{Res}_H^G V) \simeq \mathrm{Ind}_H^G(W) \otimes V.$$

Taking characters gives the projection formula. The ideal statement follows because multiplying an induced virtual character by any $v \in R(G)$ again gives an induced virtual character. □

6.4 Artin's Theorem on Induced Characters

Artin's theorem says that, after allowing rational coefficients, every complex character of G is built from characters induced from cyclic subgroups.

Theorem 6.6 (Artin)**.** *Let V be a representation of G. There exist a positive integer d_V, cyclic subgroups $C_i \leq G$, linear characters $\chi_i \in \widehat{C_i}$, and integers n_i such that*

$$d_V \chi_V = \sum_i n_i \, \mathrm{ind}_{C_i}^G(\chi_i).$$

Equivalently, the map

$$\bigoplus_{C \leq G \text{ cyclic}} R(C) \xrightarrow{\text{ind}} R(G)$$

has finite cokernel.

Proof. First we show that the induced characters

$$\mathrm{ind}_C^G(\chi), \qquad C \leq G \text{ cyclic}, \quad \chi \in \widehat{C},$$

span $\mathrm{CF}(G, \mathbb{C})$ over $\mathbb{C}$. If not, there is a nonzero class function ϕ orthogonal to all of them. By Frobenius reciprocity,

$$0 = \langle \mathrm{ind}_C^G(\chi), \phi \rangle_G = \langle \chi, \mathrm{res}_C^G \phi \rangle_C$$

for every cyclic subgroup C and every $\chi \in \widehat{C}$. Since C is cyclic, its linear characters form a basis of $\mathrm{CF}(C, \mathbb{C})$. Hence $\mathrm{res}_C^G \phi = 0$ for every cyclic C. Taking $C = \langle g \rangle$ shows $\phi(g) = 0$ for all $g \in G$, a contradiction.

Choose a finite subset of these induced characters that forms a basis of $\mathrm{CF}(G, \mathbb{C})$. Expressed in the irreducible character basis, the corresponding change-of-basis matrix has integer entries, because induction sends genuine characters to genuine characters. Its inverse has rational entries. Thus each irreducible character, and therefore each genuine character χ_V, is a rational linear combination of characters induced from cyclic subgroups. Clearing denominators gives the integer d_V. The same full-rank lattice statement is equivalent to finite cokernel of the displayed induction map. □

6.5 Representations of Supersolvable Groups and p-Groups

We use the following standard classes of finite groups.

- A group is **solvable** if it has a filtration

  $$\{1\} = G_0 \subset G_1 \subset \cdots \subset G_n = G$$

  such that $G_{i-1} \triangleleft G_i$ and each quotient G_i/G_{i-1} is abelian.

- A group is **supersolvable** if it has such a filtration with each $G_i \triangleleft G$ and each quotient G_i/G_{i-1} cyclic.
- A group is **nilpotent** if it has a central filtration: each $G_i \triangleleft G$ and

$$G_i/G_{i-1} \leq Z(G/G_{i-1}).$$

Definition 6.7. A **p-group** is a finite group whose order is a power of a prime p.

Lemma 6.8. *Let G be a p-group acting on a finite set X. If*

$$X^G = \{x \in X \mid gx = x \text{ for all } g \in G\},$$

then

$$|X| \equiv |X^G| \pmod{p}.$$

Proof. Decompose X into G-orbits. Every orbit has cardinality $[G : \mathrm{Stab}_G(x)]$, hence its size is a power of p. The orbits of size 1 are exactly the fixed points, and every other orbit has size divisible by p. Summing the orbit sizes gives the congruence. □

Theorem 6.9. *Every finite p-group is nilpotent, and therefore supersolvable.*

Proof. We first show that $Z(G)$ is nontrivial. Let G act on itself by conjugation. The fixed points are precisely $Z(G)$, so the previous lemma gives

$$|Z(G)| \equiv |G| \equiv 0 \pmod{p}.$$

Since $1 \in Z(G)$, this forces $|Z(G)| \geq p$.

Now argue by induction on $|G|$. Choose a central subgroup $C \leq Z(G)$ of order p. By induction, G/C is nilpotent. Lifting a central series for G/C and inserting C at the bottom gives a central series for G, so G is nilpotent. Refining the same argument with subgroups of order p gives a normal series with cyclic quotients; hence G is supersolvable. □

Theorem 6.10. *Let G be a finite supersolvable group. Every irreducible complex representation V of G is induced from a one-dimensional representation of a subgroup: there exist $H \leq G$ and a linear character $\chi \in \widehat{H}$ such that*

$$V \simeq \mathrm{Ind}_H^G(\mathbb{C}_\chi).$$

Proof. We use strong induction on $|G|$. Let $q : G \to G/\ker\rho$ be the quotient map. If $\ker\rho \neq 1$, then $G/\ker\rho$ is a smaller supersolvable group, so the induction hypothesis gives $V \simeq \mathrm{Ind}_{\overline{H}}^{G/\ker\rho}(\mathbb{C}_{\overline{\chi}})$, viewed as a representation of $G/\ker\rho$. For $H = q^{-1}(\overline{H})$ and $\chi = \overline{\chi} \circ q|_H$ there is a natural isomorphism

$$\mathrm{Inf}_{G/\ker\rho}^G \mathrm{Ind}_{\overline{H}}^{G/\ker\rho}(\mathbb{C}_{\overline{\chi}}) \simeq \mathrm{Ind}_H^G(\mathbb{C}_\chi).$$

Thus we may assume that ρ is faithful. If G is abelian, then every irreducible complex representation is one-dimensional, so there is nothing to prove.

Assume G is nonabelian. Since G is supersolvable, the quotient $G/Z(G)$ has a nontrivial normal cyclic subgroup. Let A be its inverse image in G. Then $A \triangleleft G$, the quotient $A/Z(G)$ is cyclic, and therefore A is abelian. Moreover, A is not contained in $Z(G)$.

Restrict V to A. Because A is abelian, we have a decomposition into A-isotypic components

$$V = \bigoplus_{\lambda} V_\lambda,$$

where λ runs over the characters of A that occur in V. More than one character occurs: if A acted by a single character on all of V, then $\rho(A)$ would consist of scalar operators, forcing A to be central in G by faithfulness.

The group G permutes the components V_λ, and irreducibility of V implies that this action is transitive on the set of occurring characters. Choose one character λ, and let $H = \mathrm{Stab}_G(\lambda)$. Then H is a proper subgroup of G, the space $W = V_\lambda$ is an irreducible H-module, and the natural map

$$\mathrm{Ind}_H^G W \longrightarrow V$$

is an isomorphism. Indeed, the distinct translates of W are exactly the distinct A-isotypic components of V, indexed by G/H. If $0 \neq U \subseteq W$ is H-stable, then the direct sum of its distinct G-translates is a nonzero G-submodule of V, hence equals V. Its λ-isotypic component is U, whereas that of V is W, so $U = W$. Thus W is irreducible. The map above sends the coset summands of $\mathrm{Ind}_H^G W$ isomorphically onto the distinct isotypic components; it is therefore an isomorphism.

The subgroup H is again supersolvable. By the induction hypothesis, $W \simeq \mathrm{Ind}_L^H(\mathbb{C}_\chi)$ for some $L \leq H$ and some linear character χ of L. Transitivity of induction gives

$$V \simeq \mathrm{Ind}_H^G W \simeq \mathrm{Ind}_H^G \mathrm{Ind}_L^H(\mathbb{C}_\chi) \simeq \mathrm{Ind}_L^G(\mathbb{C}_\chi).$$

□

Corollary 6.11. *Let G be a p-group, and let k be a field of characteristic p. The only irreducible k-representation of G is the trivial representation.*

Proof. Let V be an irreducible k-representation of G, and choose $0 \neq v \in V$. Let X be the $\mathbb{F}_p$-span of the finite orbit Gv. Then X is a finite set, stable under G, and its cardinality is a power of p. By the fixed-point lemma,

$$|X^G| \equiv |X| \equiv 0 \pmod{p}.$$

Since $0 \in X^G$, this congruence implies that X^G contains a nonzero vector x. The line kx is a trivial subrepresentation of V. By irreducibility, $V = kx$, so V is the trivial representation. □

Part II

Algebras, Modules, and Categories

Chapter 7

Representations of Algebras and Tensor Products

Throughout this chapter, k is an algebraically closed field, and all algebras are associative unital k-algebras. Unless otherwise stated, modules are unital left modules, and A-mod denotes the category of all such modules. Finite-dimensionality is stated explicitly when it is required.

7.1 Tensor Products of Modules

Definition 7.1. Let M be a right A-module and let N be a left A-module. The **tensor product of M and N over A** is the quotient

$$M \otimes_A N = (M \otimes_k N)/\langle ma \otimes n - m \otimes an \rangle,$$

where $a \in A$, $m \in M$, and $n \in N$.

Theorem 7.2 (Universal Property)**.** *Let X be a k-vector space. If a bilinear map $f : M \times N \to X$ satisfies the balanced relation*

$$f(ma, n) = f(m, an) \qquad (a \in A),$$

then there is a unique k-linear map

$$\widehat{f} : M \otimes_A N \to X$$

such that

$$f(m, n) = \widehat{f}(m \otimes n).$$

Proof. The usual universal property of $M \otimes_k N$ gives a linear map $M \otimes_k N \to X$. The balanced relation says exactly that this map vanishes on the subspace generated by $ma \otimes n - m \otimes an$, so it descends uniquely to the quotient $M \otimes_A N$. □

Tensor product over an algebra is additive in each variable. For example,

$$(M_1 \oplus M_2) \otimes_A N \simeq (M_1 \otimes_A N) \oplus (M_2 \otimes_A N).$$

Definition 7.3. A vector space M is an (A, B)**-bimodule** if it is a left A-module and a right B-module, and the two actions commute:

$$(am)b = a(mb) \qquad (a \in A,\ b \in B,\ m \in M).$$

Proposition 7.4. *Bimodules induce natural module structures on tensor products.*

1. *If M is an (A, B)-bimodule and N is a left B-module, then $M \otimes_B N$ is naturally a left A-module by*
$$a \cdot (m \otimes n) = (am) \otimes n.$$
2. *Consequently, tensoring with M defines a functor*
$$M \otimes_B - : B\text{-}mod \longrightarrow A\text{-}mod.$$

Proof. The displayed formula respects the balancing over B because

$$a \cdot (mb \otimes n) = (am)b \otimes n = am \otimes bn = a \cdot (m \otimes bn).$$

Thus the action is well-defined on $M \otimes_B N$. A B-linear map $N \to N'$ induces $M \otimes_B N \to M \otimes_B N'$, giving the functor. □

Theorem 7.5 (Hom–Tensor Adjunction)**.** *Let M be an (A, B)-bimodule. For every left B-module X and every left A-module Y, there is a natural isomorphism*

$$\mathrm{Hom}_A(M \otimes_B X, Y) \simeq \mathrm{Hom}_B\big(X, \mathrm{Hom}_A(M, Y)\big).$$

Here $\mathrm{Hom}_A(M, Y)$ is a left B-module by

$$(b \cdot f)(m) = f(mb) \qquad (b \in B,\ f \in \mathrm{Hom}_A(M, Y),\ m \in M).$$

Proof. Given an A-linear map $\beta : M \otimes_B X \to Y$, define

$$\widetilde{\beta}(x)(m) = \beta(m \otimes x).$$

This gives an A-linear map $M \to Y$ for each x. Moreover,

$$\widetilde{\beta}(bx)(m) = \beta(m \otimes bx) = \beta(mb \otimes x) = \widetilde{\beta}(x)(mb) = (b \cdot \widetilde{\beta}(x))(m),$$

so $x \mapsto \widetilde{\beta}(x)$ is B-linear.

Conversely, if $\alpha : X \to \mathrm{Hom}_A(M, Y)$ is B-linear, define

$$\beta_\alpha(m \otimes x) = \alpha(x)(m).$$

Indeed,

$$\alpha(x)(mb) = (b \cdot \alpha(x))(m) = \alpha(bx)(m),$$

so the B-linearity of α makes this formula balanced, and A-linearity follows from the A-linearity of each map $\alpha(x)$. These two constructions are inverse to one another. □

Remark 7.6. For a group algebra $k[G]$, the tensor product over k of two $k[G]$-modules is again a $k[G]$-module by the diagonal action

$$g \cdot (v \otimes w) = gv \otimes gw.$$

This uses the Hopf-algebra structure of $k[G]$, encoded by $\Delta(g) = g \otimes g$. A general algebra A has no such diagonal map. Thus, if M and N are left A-modules, the vector-space tensor product $M \otimes_k N$ is naturally an $(A \otimes_k A)$-module, but not naturally an A-module.

7.2 Representations and Semisimplicity

Lemma 7.7 (Schur's Lemma for Algebras)**.** *Let V_1 and V_2 be simple A-modules, and let $\phi : V_1 \to V_2$ be A-linear.*

1. *If $\phi \neq 0$, then ϕ is an isomorphism.*
2. *If V is finite-dimensional and k is algebraically closed, then*

$$\mathrm{End}_A(V) \simeq k.$$

Equivalently, every A-endomorphism of V is scalar.

Proof. The kernel and image of ϕ are submodules. Since V_1 and V_2 are simple, a nonzero map has zero kernel and full image. For the second statement, an endomorphism T of V has an eigenvalue $\lambda \in k$. Then $T - \lambda \,\mathrm{id}_V$ is not invertible, hence cannot be a nonzero endomorphism of the simple module V. Therefore $T = \lambda \,\mathrm{id}_V$. □

Definition 7.8. A finite-dimensional representation of A is **semisimple**, or **completely reducible**, if it is a direct sum of simple modules.

Proposition 7.9. *Let V be a semisimple A-module. Then the evaluation map*

$$\bigoplus_{X \in \mathrm{Irr}(A)} X \otimes_k \mathrm{Hom}_A(X, V) \longrightarrow V, \qquad x \otimes f \longmapsto f(x),$$

is a canonical A-module isomorphism.

Proof. Choose a decomposition

$$V \simeq \bigoplus_i X_i^{\oplus n_i}$$

with the X_i pairwise non-isomorphic simple modules. By Schur's lemma,

$$\mathrm{Hom}_A(X_i, V) \simeq k^{n_i},$$

and $\mathrm{Hom}_A(X, V) = 0$ for simple modules X not isomorphic to one of the X_i. Therefore the summand $X_i \otimes_k \mathrm{Hom}_A(X_i, V)$ is naturally the X_i-isotypic component $X_i^{\oplus n_i}$. Summing over all simple isomorphism classes recovers V. □

Remark 7.10. If V is not semisimple, the image of the same evaluation map is the **socle** of V: the largest semisimple submodule of V.

Lemma 7.11. *Let M be a finite direct sum of simple A-modules. Every submodule $N \subseteq M$ is a direct summand of M and is itself a finite direct sum of simple modules.*

Proof. We argue by induction on the number of simple summands of M. The claim is clear when $M = 0$. Write $M = S \oplus M'$ with S simple. If $S \subseteq N$, then

$$N = S \oplus (N \cap M'),$$

and the induction hypothesis applied inside M' proves the result.

Suppose instead that $S \cap N = 0$, and let $p : M \to M'$ be the projection. Then $p|_N$ is injective, so N is isomorphic to the submodule $N' = p(N) \subseteq M'$. By induction,

$M' = N' \oplus C$ and N' is a finite direct sum of simple modules. Relative to $S \oplus N'$, the submodule N is the graph of an A-linear map $N' \to S$. Hence

$$S \oplus N' = N \oplus S,$$

and therefore $M = N \oplus S \oplus C$. This also shows that N is a finite direct sum of simple modules. □

Proposition 7.12. *Let $V_1, \dots, V_m$ be pairwise non-isomorphic finite-dimensional simple A-modules, and let*

$$V = \bigoplus_{i=1}^{m} V_i^{\oplus n_i}.$$

If $W \subseteq V$ is a submodule, then:

1. *$W \simeq \bigoplus_{i=1}^{m} V_i^{\oplus r_i}$ for some $0 \le r_i \le n_i$.*
2. *Under the decomposition into isotypic components, the inclusion $W \hookrightarrow V$ is a direct sum of maps*

   $$V_i^{\oplus r_i} \hookrightarrow V_i^{\oplus n_i}.$$

   After identifying the multiplicity spaces with k^{r_i} and k^{n_i}, each such map is given, in the column-vector convention, by a full-rank matrix in $\mathrm{Mat}_{n_i \times r_i}(k)$.

Proof. By Lemma 7.11, W is a finite direct sum of simple modules and is a direct summand of V. If S is a simple summand of W, then some coordinate projection $S \to V_i$ is nonzero, so Schur's lemma gives $S \simeq V_i$. Thus

$$W \simeq \bigoplus_{i=1}^{m} V_i^{\oplus r_i},$$

for some $r_i \ge 0$. The inclusion $W \hookrightarrow V$ induces injections

$$\mathrm{Hom}_A(V_i, W) \hookrightarrow \mathrm{Hom}_A(V_i, V).$$

Schur's lemma identifies these two spaces with k^{r_i} and k^{n_i}, respectively, so $r_i \le n_i$. The inclusion cannot have a component from the V_i-isotypic part of W to the V_j-isotypic part of V when $i \neq j$, since $\mathrm{Hom}_A(V_i, V_j) = 0$. Finally, Schur's lemma identifies

$$\mathrm{Hom}_A\big(V_i^{\oplus r_i}, V_i^{\oplus n_i}\big) \simeq \mathrm{Mat}_{n_i \times r_i}(k).$$

Because the inclusion is injective, the corresponding matrix has full column rank. This proves both assertions. □

Corollary 7.13. *Let V be an irreducible finite-dimensional representation of A, and let $v_1, \dots, v_n \in V$ be linearly independent. For any vectors $w_1, \dots, w_n \in V$, there exists $a \in A$ such that*

$$a v_i = w_i \qquad (1 \le i \le n).$$

Proof. Consider the A-linear map

$$\Phi : A \longrightarrow V^{\oplus n}, \qquad a \longmapsto (av_1, \dots, av_n).$$

If the desired element a did not exist for some target tuple $(w_1, \ldots, w_n)$, then $\operatorname{im}\Phi$ would be a proper submodule of $V^{\oplus n}$. By the previous proposition,

$$\operatorname{im}\Phi \simeq V^{\oplus r} \qquad \text{for some } r < n.$$

The inclusion into $V^{\oplus n}$ is given by a full-rank $n \times r$ matrix on multiplicity spaces. Since $\Phi(1) = (v_1, \ldots, v_n)$, the vectors v_i would all be linear combinations of r vectors in V. This contradicts the linear independence of $v_1, \ldots, v_n$. □

7.3 The Jacobson Density Theorem

Theorem 7.14 (Jacobson Density Theorem)**.** *Let A be a k-algebra.*

1. *If $V \in \operatorname{Irr}(A)$ is finite-dimensional, then the action map*

$$\rho : A \longrightarrow \operatorname{End}_k(V)$$

   is surjective.

2. *If $V_1, \ldots, V_r$ are pairwise non-isomorphic finite-dimensional irreducible representations of A, then the simultaneous action map*

$$\rho = \bigoplus_{i=1}^{r} \rho_i : A \longrightarrow \bigoplus_{i=1}^{r} \operatorname{End}_k(V_i)$$

   is surjective.

Proof. For (1), let $C \in \operatorname{End}_k(V)$ and choose a basis $v_1, \ldots, v_n$ of V. By the preceding corollary, there is $a \in A$ such that $av_i = C(v_i)$ for every i. Hence $\rho(a) = C$.

For (2), view the codomain as a left A-module by left composition: a acts on $\operatorname{End}_k(V_i)$ through $\rho_i(a)$. As a left A-module,

$$\operatorname{End}_k(V_i) \simeq V_i^{\oplus \dim V_i}.$$

Thus the codomain is semisimple, with distinct isotypic components indexed by the non-isomorphic V_i. The image of ρ is a submodule. Its projection to the ith component is the image of ρ_i, which is all of $\operatorname{End}_k(V_i)$ by part (1). Therefore the image contains every isotypic component of the codomain, so it is the whole codomain. □

7.4 The Jacobson Radical and Semisimplicity

Definition 7.15. Let A be a finite-dimensional algebra. The **Jacobson radical** of A, denoted $\operatorname{rad}(A)$, is

$$\operatorname{rad}(A) = \{a \in A \mid aV = 0 \text{ for every } V \in \operatorname{Irr}(A)\}.$$

Lemma 7.16. *The radical has the following properties.*

1. $\operatorname{rad}(A)$ *is a two-sided ideal.*
2. $\operatorname{rad}(A)$ *is nilpotent:* $\operatorname{rad}(A)^N = 0$ *for all sufficiently large N.*

3. Every nilpotent two-sided ideal $I \subseteq A$ is contained in $\operatorname{rad}(A)$.

Proof. For (1), suppose $a \in \operatorname{rad}(A)$ and $b, c \in A$. For every simple module V and every $v \in V$,

$$(cab)v = c\big(a(bv)\big) = 0,$$

so cab annihilates every simple module. Thus $\operatorname{rad}(A)$ is a two-sided ideal.

For (3), let I be a nilpotent two-sided ideal, say $I^N = 0$. If V is simple, then IV is a submodule of V, because I is a left ideal. Hence $IV = 0$ or $IV = V$. The second possibility is impossible: if $IV = V$, then $I^m V = V$ for all $m \geq 1$, contradicting $I^N = 0$. Thus $IV = 0$ for every simple V, so $I \subseteq \operatorname{rad}(A)$.

For (2), take a composition series of the left regular module A:

$$0 = A_0 \subset A_1 \subset \cdots \subset A_n = A.$$

Each quotient A_i/A_{i-1} is simple, so $\operatorname{rad}(A)$ annihilates it. Hence

$$\operatorname{rad}(A) A_i \subseteq A_{i-1}.$$

Iterating gives $\operatorname{rad}(A)^n A = 0$. Since $1 \in A$, this implies $\operatorname{rad}(A)^n = 0$. □

Theorem 7.17. *Let A be a finite-dimensional algebra. Then A has only finitely many isomorphism classes of irreducible representations. If $V_1, \ldots, V_r$ is the complete list, then*

$$A/\operatorname{rad}(A) \simeq \bigoplus_{i=1}^{r} \operatorname{End}_k(V_i).$$

Proof. First, every irreducible representation V of A is finite-dimensional: if $0 \neq v \in V$, then the map $A \to V$, $a \mapsto av$, is surjective, so $\dim V \leq \dim A$.

If $V_1, \ldots, V_r$ are pairwise non-isomorphic irreducible modules, the density theorem gives a surjection

$$A \twoheadrightarrow \bigoplus_{i=1}^{r} \operatorname{End}_k(V_i).$$

Therefore

$$\sum_{i=1}^{r} (\dim V_i)^2 \leq \dim A.$$

Since each summand is at least 1, only finitely many irreducible isomorphism classes can occur.

Now take $V_1, \ldots, V_r$ to be the complete list. The kernel of the simultaneous action map is exactly the set of elements annihilating every irreducible module, namely $\operatorname{rad}(A)$. The first isomorphism theorem gives the displayed isomorphism. □

Definition 7.18. A finite-dimensional algebra A is called **semisimple** if

$$\operatorname{rad}(A) = 0.$$

Theorem 7.19 (Artin–Wedderburn Classification)**.** *For a finite-dimensional algebra A over an algebraically closed field k, the following conditions are equivalent.*

1. *A is semisimple:* $\operatorname{rad}(A) = 0$.
2. *The dimension formula holds:*
$$\dim A = \sum_{V \in \operatorname{Irr}(A)} (\dim V)^2.$$
3. *A is isomorphic to a finite direct product of matrix algebras:*
$$A \simeq \bigoplus_i \operatorname{Mat}_{n_i}(k).$$
4. *Every finite-dimensional representation of A is completely reducible.*
5. *The left regular representation of A is completely reducible.*

Proof. From the preceding theorem,
$$\dim A - \dim \operatorname{rad}(A) = \sum_{V \in \operatorname{Irr}(A)} (\dim V)^2.$$
Thus (1) and (2) are equivalent.

If (1) holds, then
$$A \simeq A/\operatorname{rad}(A) \simeq \bigoplus_i \operatorname{End}_k(V_i) \simeq \bigoplus_i \operatorname{Mat}_{\dim V_i}(k),$$
so (3) holds.

If (3) holds, then modules over each matrix algebra $\operatorname{Mat}_n(k)$ are direct sums of the standard simple module k^n, and modules over a finite product are direct sums over the different factors. Hence every finite-dimensional A-module is completely reducible, proving (4). Clearly (4) implies (5).

Finally assume (5). As a left A-module, write
$$A \simeq \bigoplus_i V_i^{\oplus n_i}$$
with the V_i simple. By definition, $\operatorname{rad}(A)$ annihilates every simple A-module, hence every summand of the left regular module A. Thus $\operatorname{rad}(A)A = 0$, and evaluating at $1 \in A$ gives $\operatorname{rad}(A) = 0$. Therefore (5) implies (1), closing the chain of equivalences. □

Chapter 8

Categories, Functors, and Natural Transformations

8.1 Categories

Definition 8.1. A **category** $\mathcal{C}$ consists of the following data:

1. a collection of **objects**, denoted $\mathrm{Ob}(\mathcal{C})$;
2. for every pair of objects A, B, a set of **morphisms**
$$\mathrm{Hom}_{\mathcal{C}}(A, B);$$
3. for every triple of objects A, B, C, a composition map
$$\begin{aligned}\mathrm{Hom}_{\mathcal{C}}(A, B) \times \mathrm{Hom}_{\mathcal{C}}(B, C) &\longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, C),\\ (f, g) &\longmapsto g \circ f.\end{aligned}$$

This data is required to satisfy:

1. **Associativity:** if
$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D,$$
then
$$h \circ (g \circ f) = (h \circ g) \circ f.$$
2. **Identities:** for each object A, there is a morphism $1_A \in \mathrm{Hom}_{\mathcal{C}}(A, A)$ such that
$$f \circ 1_A = f, \qquad 1_A \circ g = g$$
whenever these compositions are defined.

We write $f : A \to B$ to mean $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. A morphism $f : A \to B$ is an **isomorphism** if there exists $g : B \to A$ such that
$$g \circ f = 1_A, \qquad f \circ g = 1_B.$$

Example 8.2. Here are the standard first examples.

1. **Concrete categories.** These are categories whose objects are sets with extra structure and whose morphisms preserve that structure. Examples include:
$$\mathbf{Set}, \quad \mathbf{Grp}, \quad \mathbf{Ring}, \quad \mathbf{Alg}_k, \quad \mathbf{Vect}_k, \quad R\text{-}\mathbf{Mod}, \quad \mathbf{Man}.$$
2. **Monoids as categories.** A category with exactly one object $\bullet$ is the same thing as a monoid: the elements are $\mathrm{End}(\bullet)$, and multiplication is composition.
3. **Posets as categories.** If $(P, \leq)$ is a partially ordered set, make a category whose objects are the elements of P and where
$$\mathrm{Hom}(x, y) = \begin{cases} \{*\}, & x \leq y, \\ \emptyset, & x \not\leq y. \end{cases}$$
Reflexivity gives identities, and transitivity gives composition.

Definition 8.3. A category is called **small** if $\mathrm{Ob}(\mathcal{C})$ is a set rather than a proper class.

Given a category $\mathcal{C}$, its **opposite category** $\mathcal{C}^{\mathrm{op}}$ has the same objects and reversed morphisms:
$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A).$$
Composition is also reversed. Thus $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$. This is analogous to the opposite algebra A^{op}.

If $\mathcal{C}$ and $\mathcal{D}$ are categories, the **product category** $\mathcal{C} \times \mathcal{D}$ has objects (C, D) and morphisms
$$\mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((C, D), (C', D')) = \mathrm{Hom}_{\mathcal{C}}(C, C') \times \mathrm{Hom}_{\mathcal{D}}(D, D').$$

A **subcategory** $\mathcal{D}$ of $\mathcal{C}$ consists of some objects of $\mathcal{C}$ and some morphisms between them, closed under composition and containing the identity morphisms of its objects. It is **full** if, for all objects A, B of $\mathcal{D}$,
$$\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B).$$

8.2 Functors

Definition 8.4. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **covariant functor** $F : \mathcal{C} \to \mathcal{D}$ assigns:

1. to each object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;
2. to each morphism $f : A \to B$, a morphism
$$F(f) : F(A) \to F(B).$$

These assignments satisfy
$$F(1_A) = 1_{F(A)}, \qquad F(g \circ f) = F(g) \circ F(f).$$

A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ reverses arrows. Equivalently, it is a covariant functor
$$\mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{D}.$$

Example 8.5. Important examples of functors include the following.

1. The **identity functor** $\mathbb{1}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ sends every object and morphism to itself.
2. **Forgetful functors** forget part of the structure. For example, **Grp** $\to$ **Set** sends a group to its underlying set. If $R \subset R'$ is a ring extension, restriction of scalars gives a functor
$$R'\text{-}\mathbf{Mod} \longrightarrow R\text{-}\mathbf{Mod}.$$
3. For a fixed object $A \in \mathcal{C}$, the functor
$$h_A = \mathrm{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \to \mathbf{Set}$$
is covariant. A morphism $f : B \to C$ acts by postcomposition:
$$\mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, C), \qquad g \longmapsto f \circ g.$$
The functor $\mathrm{Hom}_{\mathcal{C}}(-, A)$ is contravariant; a morphism $f : B \to C$ acts by precomposition.
4. The Hom construction is a bifunctor
$$\mathrm{Hom}_R(-, -) : (R\text{-}\mathbf{Mod})^{\mathrm{op}} \times R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}.$$
For a commutative ring R, the tensor product is a covariant bifunctor
$$- \otimes_R - : R\text{-}\mathbf{Mod} \times R\text{-}\mathbf{Mod} \longrightarrow R\text{-}\mathbf{Mod}.$$

Definition 8.6. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. For objects A, B of $\mathcal{C}$, there is an induced map
$$\mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)).$$
The functor F is:

- **faithful** if this map is injective for all A, B;
- **full** if this map is surjective for all A, B;
- **fully faithful** if this map is bijective for all A, B;
- **essentially surjective** if every object of $\mathcal{D}$ is isomorphic to some $F(C)$.

8.3 Natural Transformations

Functors are maps between categories. Natural transformations are maps between functors.

Definition 8.7. Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A **natural transformation** $\alpha : F \to G$ is a family of morphisms
$$\alpha_A : F(A) \to G(A), \qquad A \in \mathrm{Ob}(\mathcal{C}),$$
such that for every morphism $f : A \to B$ in $\mathcal{C}$,
$$G(f) \circ \alpha_A = \alpha_B \circ F(f).$$

Equivalently, the square

$$\begin{array}{ccc} F(A) & \xrightarrow{\alpha_A} & G(A) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(B) & \xrightarrow[\alpha_B]{} & G(B) \end{array}$$

commutes.

Two functors F and G are **naturally isomorphic** if there are natural transformations

$$\alpha : F \to G, \qquad \beta : G \to F,$$

which are inverse to each other componentwise.

Example 8.8. If $f : M \to M'$ is a morphism of R-modules, precomposition gives a natural transformation between covariant representable functors

$$\alpha(f) : \mathrm{Hom}_R(M', -) \longrightarrow \mathrm{Hom}_R(M, -).$$

At a target module N, the component is

$$\alpha(f)_N : \mathrm{Hom}_R(M', N) \longrightarrow \mathrm{Hom}_R(M, N), \qquad g \longmapsto g \circ f.$$

Definition 8.9. A functor $F : \mathcal{C} \to \mathcal{D}$ is an **equivalence of categories** if there is a functor $G : \mathcal{D} \to \mathcal{C}$ such that

$$G \circ F \simeq \nVdash_{\mathcal{C}}, \qquad F \circ G \simeq \nVdash_{\mathcal{D}}.$$

The functor G is called a **quasi-inverse** of F.

Remark 8.10. One rarely asks for categories to be strictly isomorphic, since equality of objects is usually too rigid. Equivalence of categories is the more useful notion. A standard criterion says that F is an equivalence if and only if F is fully faithful and essentially surjective.

Example 8.11. Let $\mathbf{Vect}_k$ be the category of vector spaces over k. The double-dual functor $D^2(V) = V^{**}$ is covariant, and there is a canonical natural transformation

$$\iota : \nVdash_{\mathbf{Vect}_k} \to D^2, \qquad \iota_V(v)(\phi) = \phi(v).$$

The map ι_V is always injective, and it is an isomorphism exactly when V is finite-dimensional. Thus, on the full subcategory $\mathbf{Vect}_k^{\mathrm{f.d.}}$ of finite-dimensional vector spaces, ι is a natural isomorphism. The duality functor

$$D : \mathbf{Vect}_k^{\mathrm{f.d.}} \longrightarrow (\mathbf{Vect}_k^{\mathrm{f.d.}})^{\mathrm{op}}$$

is therefore an equivalence of categories.

8.4 Adjoint Functors

Adjoint functors formalize the idea that one construction is universal with respect to another.

Definition 8.12. Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say that F is **left adjoint** to G, and that G is **right adjoint** to F, if there are natural bijections

$$\alpha_{M,N} : \mathrm{Hom}_{\mathcal{D}}(F(M), N) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(M, G(N))$$

for all $M \in \mathcal{C}$ and $N \in \mathcal{D}$. We write

$$F \dashv G.$$

If a functor has a left adjoint, that left adjoint is unique up to natural isomorphism. More precisely, after the adjunction bijections (equivalently, the units and counits) are fixed, there is a unique natural isomorphism compatible with those data. The same statement holds for right adjoints.

Example 8.13. 1. The forgetful functor

$$\mathbf{Alg}_k \longrightarrow \mathbf{Set}$$

has a left adjoint: the free algebra functor $X \mapsto k\langle X \rangle$.

2. If M is an (A, B)-bimodule, then, for the categories of all unital modules,

$$M \otimes_B - : B\text{-mod} \to A\text{-mod}$$

is left adjoint to

$$\mathrm{Hom}_A(M, -) : A\text{-mod} \to B\text{-mod}.$$

This is the Hom–Tensor adjunction from the previous chapter.

Unit and counit. Suppose $F \dashv G$. The adjunction determines two natural transformations:

$$\eta : \mathbb{1}_{\mathcal{C}} \to GF, \qquad \varepsilon : FG \to \mathbb{1}_{\mathcal{D}}.$$

The **unit** $\eta_M : M \to GF(M)$ corresponds under the adjunction to the identity map of $F(M)$. The **counit** $\varepsilon_N : FG(N) \to N$ corresponds to the identity map of $G(N)$.

Conversely, a pair (η, ε) defines an adjunction precisely when the triangle identities hold:

$$(G\varepsilon) \circ (\eta G) = \mathbb{1}_G, \qquad (\varepsilon F) \circ (F\eta) = \mathbb{1}_F.$$

8.5 The Yoneda Lemma

The Yoneda lemma says that an object can be recovered from the functor of maps out of it. It is one of the basic reasons representable functors are so useful.

Definition 8.14. Let $\mathcal{C}$ be a category. A functor $F : \mathcal{C} \to \mathbf{Set}$ is **representable** if it is naturally isomorphic to

$$h_A = \mathrm{Hom}_{\mathcal{C}}(A, -)$$

for some object A of $\mathcal{C}$.

Theorem 8.15 (Yoneda's Lemma)**.** *Let A be an object of $\mathcal{C}$ and let $F : \mathcal{C} \to \mathbf{Set}$ be a functor. Then there is a canonical bijection*

$$\mathrm{Nat}(h_A, F) \xrightarrow{\sim} F(A), \qquad \eta \longmapsto \eta_A(1_A).$$

Proof. Given $\eta : h_A \to F$, evaluation at the identity gives

$$\eta_A(1_A) \in F(A).$$

Conversely, for $x \in F(A)$ define a family of maps

$$\eta^x_C : \mathrm{Hom}_{\mathcal{C}}(A, C) \longrightarrow F(C), \qquad g \longmapsto F(g)(x).$$

If $u : C \to D$, then

$$F(u)\big(\eta^x_C(g)\big) = F(u \circ g)(x) = \eta^x_D(u \circ g),$$

so η^x is natural. Moreover, $\eta^x_A(1_A) = x$. In the other direction, naturality of η for a morphism $g : A \to C$ gives

$$\eta_C(g) = F(g)\big(\eta_A(1_A)\big).$$

Thus the two constructions are inverse. □

Corollary 8.16. *For objects A, B of $\mathcal{C}$, there is a canonical bijection*

$$\mathrm{Nat}(h_A, h_B) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(B, A), \qquad \varphi \longmapsto \varphi_A(1_A).$$

Proof. Apply Yoneda's lemma to $F = h_B$ and use $h_B(A) = \mathrm{Hom}_{\mathcal{C}}(B, A)$. □

Corollary 8.17. *The assignment*

$$A \longmapsto h_A = \mathrm{Hom}_{\mathcal{C}}(A, -)$$

defines a functor

$$\mathcal{C}^{\mathrm{op}} \longrightarrow \mathrm{Fun}(\mathcal{C}, \mathbf{Set}).$$

By Yoneda's lemma, this functor is fully faithful. Hence $\mathcal{C}^{\mathrm{op}}$ is equivalent to the full subcategory of representable functors on $\mathcal{C}$.

Chapter 9

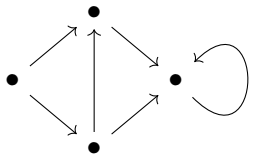
Quiver Representations

9.1 Quivers and Path Algebras

Definition 9.1. A **quiver** Q is a directed graph. We write I for its set of vertices and H for its set of arrows. If $h \in H$ is an arrow $i \to j$, then i is its **source** and j is its **target**. Following the notation used in the lectures, we write

$$h' = i, \qquad h'' = j.$$

For example, a quiver may have several arrows and loops:



Definition 9.2. A **representation** of Q over $\mathbb{C}$ consists of a vector space V_i for every vertex $i \in I$ and a linear map

$$x_h \colon V_{h'} \longrightarrow V_{h''}$$

for every arrow $h \in H$. We denote such a representation by (V_i, x_h). Its **dimension vector** is

$$\dim V = (\dim V_i)_{i \in I}.$$

Unless explicitly stated otherwise, representations in these notes are finite-dimensional, meaning that

$$\sum_{i \in I} \dim_{\mathbb{C}} V_i < \infty.$$

Definition 9.3. A **morphism** $\varphi \colon (V_i, x_h) \to (V_i', x_h')$ of quiver representations is a collection of linear maps $\varphi_i \colon V_i \to V_i'$ such that, for every arrow h, the square

$$\begin{array}{ccc} V_{h'} & \xrightarrow{\varphi_{h'}} & V'_{h'} \\ {\scriptstyle x_h}\downarrow & & \downarrow{\scriptstyle x'_h} \\ V_{h''} & \xrightarrow{\varphi_{h''}} & V'_{h''} \end{array}$$

commutes. The resulting category is denoted $\mathrm{Rep}(Q)$.

Definition 9.4. The **path algebra** P_Q is the $\mathbb{C}$-vector space with basis given by all oriented paths in Q, including the trivial path e_i of length zero at each vertex i. Multiplication is concatenation: if the target of p_1 is the source of p_2, then

$$p_1 p_2 = \text{the path obtained by following } p_1 \text{ and then } p_2;$$

otherwise $p_1 p_2 = 0$.

Remark 9.5. If I is finite, then $\sum_{i \in I} e_i$ is the identity of P_Q. The algebra is generated by the idempotents e_i and the length-one paths a_h for $h \in H$, with relations

$$e_i^2 = e_i, \qquad e_i e_j = 0 \quad (i \neq j),$$

and

$$e_i a_h = \delta_{i,h'} a_h, \qquad a_h e_i = \delta_{i,h''} a_h.$$

These relations encode the source and target of each arrow.

Definition 9.6. A right P_Q-module M is called **locally finite** if, for every $m \in M$, only finitely many terms me_i are nonzero and

$$m = \sum_{i \in I} m e_i.$$

For finite-dimensional modules over a finite quiver this condition is automatic.

Proposition 9.7. *There is an equivalence of categories*

$$\mathrm{Rep}(Q) \simeq \mathrm{mod}^{\mathrm{lf}}\text{–}P_Q,$$

where the right-hand side denotes finite-dimensional locally finite right P_Q-modules.

Proof. Let $V = (V_i, x_h)$ be a representation. Put $M = \bigoplus_{i \in I} V_i$. The idempotent e_i acts on the right as the projection $M \to V_i$. If $h : i \to j$, define right multiplication by the arrow a_h to be x_h on the summand V_i and zero on the other summands:

$$v a_h = \begin{cases} x_h(v), & v \in V_i, \\ 0, & v \in V_k,\ k \neq i. \end{cases}$$

For a longer path, the action is the corresponding composition of arrow maps. The incidence relations above show that this gives a right P_Q-module.

Conversely, let M be a locally finite right P_Q-module. Set $V_i = Me_i$. Orthogonality of the idempotents gives $M = \bigoplus_i Me_i$. For an arrow $h : i \to j$, right multiplication by a_h maps Me_i into Me_j, because

$$(me_i)a_h = m(e_i a_h) = m a_h, \qquad (m a_h) e_j = m(a_h e_j) = m a_h.$$

Thus $x_h(v) = v a_h$ defines a map $V_i \to V_j$. These two constructions are inverse to one another, also on morphisms. $\square$

9.2 Dynkin Quivers and Gabriel's Theorem

Definition 9.8. A **subrepresentation** of (V_i, x_h) is a family of subspaces $W_i \subseteq V_i$ such that

$$x_h(W_{h'}) \subseteq W_{h''}$$

for every arrow h.

Let Γ be a finite connected undirected graph without loops, with vertex set I. Its adjacency matrix is $R_\Gamma = (r_{ij})_{i,j\in I}$, where r_{ij} is the number of edges joining i and j. The associated symmetric Cartan matrix is

$$A_\Gamma = 2\,\mathrm{Id} - R_\Gamma.$$

Definition 9.9. The graph Γ is a **Dynkin diagram** if A_Γ is positive definite. A quiver is a **Dynkin quiver** if its underlying undirected graph is a Dynkin diagram.

We use the following classical classification without proof; see [10].

Theorem 9.10 (Dynkin Classification)**.** *A connected graph is a Dynkin diagram if and only if it is one of the ADE diagrams*

$$A_n\ (n \geq 1), \qquad D_n\ (n \geq 4), \qquad E_6,\ E_7,\ E_8.$$

Here A_n is a chain, D_n is the chain with one extra leaf attached to the next-to-last vertex, and E_6, E_7, E_8 are the three exceptional simply-laced diagrams.

Definition 9.11. A quiver is **of finite type** if it has only finitely many isomorphism classes of indecomposable finite-dimensional representations.

Remark 9.12. Gabriel's theorem is the first point where the linear algebra of quiver representations meets the root systems familiar from Lie theory. A representation of a quiver is a system of vector spaces and matrices modulo changes of bases. The theorem says that, in finite type, the indecomposable objects are controlled by the positive roots of the corresponding ADE root system.

Gabriel's finite-type criterion is another classical input; see [10].

Theorem 9.13 (Gabriel's Theorem, Part I)**.** *A connected finite quiver Q is of finite type if and only if Q is a Dynkin quiver. Equivalently, a finite quiver is of finite type precisely when each of its connected components is a Dynkin quiver.*

Example 9.14. We record the basic examples.

1. For any vertex j, the representation S_j with $V_j = \mathbb{C}$, all other $V_i = 0$, and all arrow maps zero is simple.

2. For the one-vertex quiver A_1, the only indecomposable representation is S_1.

3. For A_2 with orientation $\bullet_1 \to \bullet_2$, a representation is just one linear map $\varphi\colon V_1 \to V_2$. After choosing bases adapted to $\ker\varphi$ and $\operatorname{im}\varphi$, it decomposes as a direct sum of copies of

   $$S_1 = (\mathbb{C} \to 0), \qquad S_2 = (0 \to \mathbb{C}), \qquad M = (\mathbb{C} \xrightarrow{\mathrm{id}} \mathbb{C}).$$

   The representation M fits into the nonsplit exact sequence

   $$0 \longrightarrow S_2 \longrightarrow M \longrightarrow S_1 \longrightarrow 0.$$

4. For A_3 with orientation $\bullet \to \bullet \to \bullet$, the six indecomposable representations are the interval representations:

$$(\mathbb{C} \to 0 \to 0), \qquad (0 \to \mathbb{C} \to 0), \qquad (0 \to 0 \to \mathbb{C}),$$
$$(\mathbb{C} \xrightarrow{1} \mathbb{C} \to 0), \qquad (0 \to \mathbb{C} \xrightarrow{1} \mathbb{C}), \qquad (\mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}).$$

   For the orientation $\bullet \xrightarrow{\varphi} \bullet \xleftarrow{\psi} \bullet$, one again obtains six indecomposables. The classification is obtained by putting the two subspaces $\operatorname{im}\varphi$ and $\operatorname{im}\psi$ of the middle space in normal form. The indecomposable supported on all three vertices is

$$\mathbb{C} \xrightarrow{1} \mathbb{C} \xleftarrow{1} \mathbb{C}.$$

9.3 Roots and the Weyl Group

Let Γ be a Dynkin diagram with vertices $1, \dots, N$ and Cartan matrix A_Γ. Define the symmetric bilinear form on $\mathbb{R}^N$ by

$$B(x, y) = x^T A_\Gamma y.$$

Since Γ is Dynkin, this form is positive definite. For $x = (x_i) \in \mathbb{Z}^N$,

$$B(x, x) = 2\sum_i x_i^2 - 2\sum_{i<j} r_{ij} x_i x_j \in 2\mathbb{Z}.$$

Definition 9.15. A **root** is an integer vector $\alpha \in \mathbb{Z}^N$ satisfying

$$B(\alpha, \alpha) = 2.$$

The standard basis vectors α_i are called the **simple roots**. The set of roots is denoted Φ_Γ.

Positive definiteness implies that Φ_Γ is finite.

Lemma 9.16. *Let $\alpha = \sum_i k_i \alpha_i$ be a root. Then either all $k_i \geq 0$, or all $k_i \leq 0$.*

Proof. Suppose that both signs occur. Write

$$\beta = \sum_{k_i > 0} k_i \alpha_i, \qquad \gamma = \sum_{k_i < 0} k_i \alpha_i,$$

so that $\alpha = \beta + \gamma$ and both β, γ are nonzero integer vectors. Since B is positive definite and even on the lattice, $B(\beta, \beta) \geq 2$ and $B(\gamma, \gamma) \geq 2$.

The supports of β and γ are disjoint. Hence the diagonal terms do not contribute to $B(\beta, \gamma)$, and

$$B(\beta, \gamma) = \sum_{k_i > 0,\ k_j < 0} k_i k_j (-r_{ij}) \geq 0,$$

because $k_i k_j < 0$ and $-r_{ij} \leq 0$. Therefore

$$B(\alpha, \alpha) = B(\beta, \beta) + 2B(\beta, \gamma) + B(\gamma, \gamma) \geq 4,$$

contradicting $B(\alpha, \alpha) = 2$. □

Thus roots split into positive and negative roots:

$$\Phi_\Gamma = \Phi_\Gamma^+ \sqcup \Phi_\Gamma^-,$$

where Φ_Γ^+ consists of roots whose simple-root coefficients are all nonnegative.

Definition 9.17. For a root α, define the reflection

$$s_\alpha(v) = v - B(v,\alpha)\alpha.$$

This is the usual reflection through the hyperplane orthogonal to α, since $B(\alpha,\alpha) = 2$. The simple reflections are $s_i = s_{\alpha_i}$, and the **Weyl group** W is the subgroup of $\mathrm{GL}(\mathbb{R}^N)$ generated by $s_1, \dots, s_N$.

The reflections preserve B and the lattice $\mathbb{Z}^N$, hence they permute the finite root set. This action is faithful: the simple roots belong to the root set and span $\mathbb{R}^N$. Therefore W is finite. For type A_n, the roots may be written as $\varepsilon_i - \varepsilon_j$ with $i \neq j$, and the Weyl group is the symmetric group $\mathfrak{S}_{n+1}$.

9.4 Reflection Functors and Coxeter Elements

Let $V = (V_i, x_h)$ be a representation. Its dimension vector will be written

$$d(V) = \sum_{i=1}^{N} (\dim V_i)\alpha_i.$$

Reflection functors are the categorical counterpart of simple reflections on the root lattice.

Definition 9.18. A vertex i is a **sink** if every arrow incident to i points toward i. It is a **source** if every arrow incident to i points away from i. Let $\overline{Q}_i$ be the quiver obtained from Q by reversing all arrows incident to i.

Definition 9.19. Let i be a sink. Define

$$F_i^+ \colon \mathrm{Rep}(Q) \longrightarrow \mathrm{Rep}(\overline{Q}_i)$$

as follows. For $k \neq i$, set $(F_i^+ V)_k = V_k$. At the vertex i, set

$$(F_i^+ V)_i = \ker\left(\bigoplus_{j \to i} V_j \xrightarrow{\varphi} V_i \right),$$

where φ is the sum of the incoming arrow maps. In $\overline{Q}_i$, the arrows incident to i now go out of i; their maps are the restrictions of the natural projections $\bigoplus_{j\to i} V_j \to V_j$ to this kernel.

Dually, if i is a source, define

$$F_i^- \colon \mathrm{Rep}(Q) \longrightarrow \mathrm{Rep}(\overline{Q}_i)$$

by keeping all spaces away from i and setting

$$(F_i^- V)_i = \mathrm{coker}\left(V_i \xrightarrow{\psi} \bigoplus_{i \to j} V_j \right),$$

where ψ is the direct sum of the outgoing arrow maps. The reversed arrows into i are induced by the quotient map to this cokernel.

For the proof of the second part of Gabriel's theorem, we use the following root-theoretic lemma. It is the precise form of the sink-adapted reduction for finite ADE root systems; see [10].

Lemma 9.20 (Sink-adapted root reduction)**.** *Let $Q_0 = Q$ be a Dynkin quiver. A sequence $i_1, \dots, i_t$ is* sink-adapted *if, recursively, i_r is a sink of Q_{r-1} and $Q_r = \overline{(Q_{r-1})}_{i_r}$. For every positive root α there is a sink-adapted sequence such that, with*

$$\alpha_0 = \alpha, \qquad \alpha_r = s_{i_r}\alpha_{r-1},$$

all intermediate roots $\alpha_0, \dots, \alpha_{t-1}$ are positive and α_t is simple.

Moreover, choose an ordering of the vertices by successively removing a sink and repeat the resulting sink sequence cyclically. Starting from any positive root, its successive images remain positive only for finitely many steps; at the first step that would leave Φ_Γ^+, the current root is the simple root at the sink being reflected.

Proposition 9.21. *Let $V \in \mathrm{Rep}(Q)$ be indecomposable.*

1. *If i is a sink, then either $V \simeq S_i$, or the total incoming map $\varphi\colon \bigoplus_{j\to i} V_j \to V_i$ is surjective. In the surjective case,*

$$d(F_i^+V) = s_i(d(V)), \qquad F_i^-F_i^+(V) \simeq V,$$

and F_i^+V is indecomposable.

2. *If i is a source, then either $V \simeq S_i$, or the total outgoing map $\psi\colon V_i \to \bigoplus_{i\to j} V_j$ is injective. In the injective case,*

$$d(F_i^-V) = s_i(d(V)), \qquad F_i^+F_i^-(V) \simeq V,$$

and F_i^-V is indecomposable.

Proof. We prove the sink case; the source case is dual. Let

$$\varphi\colon \bigoplus_{j\to i} V_j \longrightarrow V_i$$

be the total incoming map. Choose a complement W to $\operatorname{im}\varphi$ in V_i. The representation with the same spaces as V away from i and with $\operatorname{im}\varphi$ at i is a subrepresentation. The complement W gives a direct summand isomorphic to $\dim W$ copies of S_i. Since V is indecomposable, either $V \simeq S_i$ or $W = 0$, which is exactly surjectivity of φ.

Assume now that φ is surjective. The exact sequence

$$0 \longrightarrow (F_i^+V)_i \longrightarrow \bigoplus_{j\to i} V_j \xrightarrow{\varphi} V_i \longrightarrow 0$$

gives

$$\dim(F_i^+V)_i = \sum_{j\to i} \dim V_j - \dim V_i.$$

All other dimensions are unchanged. This is precisely the formula for the simple reflection s_i applied to $d(V)$. The same exact sequence also shows that applying the opposite reflection functor reconstructs V.

To justify indecomposability, note that the total outgoing map from the new source i in F_i^+V is the inclusion

$$\ker\varphi \hookrightarrow \bigoplus_{j\to i} V_j,$$

so it is injective. Consequently F_i^+V has no direct summand isomorphic to S_i, and no nonzero direct summand is killed by F_i^-. Since reflection functors are additive, a nontrivial decomposition of F_i^+V would therefore give a nontrivial decomposition of $F_i^-F_i^+V \simeq V$, a contradiction. □

Definition 9.22. For an ordering of the vertices $1,\dots,N$, the product

$$C = s_1 s_2 \cdots s_N$$

is called a **Coxeter element**.

Theorem 9.23 (Gabriel's Theorem, Part II)**.** *For a Dynkin quiver Q, the dimension-vector map induces a bijection*

$$\left\{ \begin{array}{c} \textit{indecomposable representations of } Q \\ \textit{up to isomorphism} \end{array} \right\} \longleftrightarrow \Phi_\Gamma^+.$$

Proof sketch. Choose an ordering of the vertices by repeatedly removing a sink: label the first removed sink by N, the next by $N-1$, and so on. Then the sequence of reflection functors

$$F_N^+,\ F_{N-1}^+,\dots,F_1^+$$

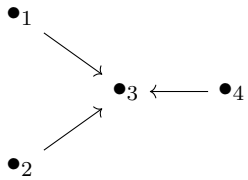
may be applied cyclically whenever the current representation is not the simple representation at the sink being reflected.

Let V be indecomposable. As long as no simple representation is encountered, Proposition 9.21 says that the dimension vector is transformed by the corresponding simple reflection. By Lemma 9.20, the fixed cyclic sink sequence cannot keep the dimension vector positive forever. At the first reflection that would make it negative, the current dimension vector is the simple root at that sink, so the current representation is S_i. Reversing the reflection sequence expresses $d(V)$ as a positive root.

Conversely, for a positive root choose the sink-adapted sequence supplied by Lemma 9.20. Start with its terminal simple representation and reverse the sequence. Proposition 9.21 shows at each nonexceptional step that the result remains indecomposable and that its dimension vector is transformed by the inverse simple reflection. Thus every positive root occurs. Finally, two indecomposables with the same dimension vector follow the same fixed cyclic sink sequence to the same simple representation; reversing the sequence recovers isomorphic representations. This proves uniqueness. □

9.5 Detailed Example: D_4

Consider the D_4 quiver in which the vertices 1, 2, and 4 point toward the central sink 3:



Write a vector as (a, b, c, d), where c is the coordinate at the central vertex. The quadratic form is

$$B(x, x) = (a - c)^2 + (b - c)^2 + (d - c)^2 + a^2 + b^2 + d^2 - c^2.$$

Solving $B(x, x) = 2$ gives the 12 positive roots of type D_4. The largest positive root is

$$\theta = \alpha_1 + \alpha_2 + 2\alpha_3 + \alpha_4.$$

The reflection identity

$$s_3 s_2 s_1 s_4(\alpha_3) = \theta$$

suggests constructing the indecomposable representation of dimension vector θ by reversing the corresponding reflection functors:

$$V_\theta \simeq \big(F_3^- \circ F_2^- \circ F_1^- \circ F_4^-\big)(S_3).$$

The construction can be read as follows.

1. Start with S_3: only $V_3 = \mathbb{C}$ is nonzero.
2. Applying F_4^- creates a one-dimensional space at vertex 4; the relevant arrow is now oriented out of vertex 3.
3. Applying F_1^- and F_2^- similarly creates one-dimensional spaces at vertices 1 and 2.
4. Reflecting at vertex 3 restores the original orientation and replaces V_3 by

$$\operatorname{coker}\left(\mathbb{C} \xrightarrow{(1,1,1)} \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}\right) \simeq \mathbb{C}^2.$$

We have therefore obtained an indecomposable representation for the original orientation, with dimension vector

$$(1, 1, 2, 1).$$

This vector is exactly the maximal positive root θ.

Since $s_4(\theta) = \theta$, one may apply an additional reflection F_4^- without changing the dimension vector; it only changes the orientation of the edge between vertices 3 and 4.

Part III

The Symmetric Group and Combinatorics

Chapter 10

Okounkov-Vershik Approach to Symmetric Group Representations

For a finite set I, write

$$\mathfrak{S}_I = \mathrm{Perm}(I)$$

for its permutation group. For $I = \{1, \ldots, n\}$, we use the usual notation $\mathfrak{S}_n$. We also use the convention $\mathfrak{S}_0 = \{1\}$.

By the character theory of finite groups, the number of irreducible complex representations of $\mathfrak{S}_n$ equals the number of conjugacy classes. These conjugacy classes are determined by cycle type, hence by partitions of n:

$$P_n = \left\{ \lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d > 0) \;\middle|\; \sum_{i=1}^{d} \lambda_i = n \right\}.$$

The goal is to construct the irreducible representations and to identify the natural bijection

$$P_n \longleftrightarrow \mathrm{Irr}(\mathfrak{S}_n).$$

There are three standard ways to do this:

1. by Young symmetrizers;
2. by Schur-Weyl duality;
3. by the inductive method of Okounkov and Vershik.

This chapter follows the third approach. The basic structure is the chain

$$\mathfrak{S}_0 \subset \mathfrak{S}_1 \subset \mathfrak{S}_2 \subset \cdots \subset \mathfrak{S}_n,$$

and the basic operations are restriction and induction along this chain.

10.1 Centralizers and Restriction of Representations

Definition 10.1. Let A be a finite-dimensional semisimple $\mathbb{C}$-algebra, and let $B \subseteq A$ be a semisimple subalgebra. The **centralizer** of B in A is

$$Z_B(A) = \{a \in A \mid ab = ba \text{ for all } b \in B\}.$$

If U is a B-module and V is an A-module, then $\mathrm{Hom}_B(U,V)$ is naturally a left module over $Z_B(A)$: for $a \in Z_B(A)$ and $f \in \mathrm{Hom}_B(U,V)$,

$$(af)(u) = af(u).$$

This is still B-linear, because

$$(af)(bu) = abf(u) = baf(u) = b(af)(u).$$

Thus we get a representation

$$\rho_{U,V} \colon Z_B(A) \longrightarrow \mathrm{End}(\mathrm{Hom}_B(U,V)).$$

Lemma 10.2. *There is a natural algebra isomorphism*

$$Z_B(A) \simeq \bigoplus_{\substack{U \in \mathrm{Irr}(B) \\ V \in \mathrm{Irr}(A)}} \mathrm{End}(\mathrm{Hom}_B(U,V)).$$

Proof. By Artin-Wedderburn,

$$A \simeq \bigoplus_{V \in \mathrm{Irr}(A)} \mathrm{End}(V).$$

Taking elements that commute with the action of B gives

$$Z_B(A) \simeq \bigoplus_{V \in \mathrm{Irr}(A)} \mathrm{End}_B(V).$$

Now decompose the restriction of V to B:

$$V \simeq \bigoplus_{U \in \mathrm{Irr}(B)} U \otimes \mathrm{Hom}_B(U,V).$$

The factor B acts only on U, so its commutant on this summand is $\mathrm{End}(\mathrm{Hom}_B(U,V))$. Summing over all U and V gives the claimed isomorphism. □

Corollary 10.3. *The algebra $Z_B(A)$ is commutative if and only if*

$$\dim \mathrm{Hom}_B(U,V) \le 1$$

for all $U \in \mathrm{Irr}(B)$ and $V \in \mathrm{Irr}(A)$.

Proof. A direct sum of matrix algebras is commutative exactly when every matrix algebra appearing in it is one-dimensional. □

We now specialize to symmetric groups. For $0 \le m \le n$, set

$$Z_m(n) = Z_{\mathbb{C}\mathfrak{S}_m}(\mathbb{C}\mathfrak{S}_n).$$

Definition 10.4. The **Jucys-Murphy elements** in $\mathbb{C}\mathfrak{S}_n$ are

$$L_k = \sum_{i=1}^{k-1} (i,k), \qquad 1 \le k \le n.$$

In particular, $L_1 = 0$.

We use the following standard centralizer theorem without proof; see [8].

Theorem 10.5 (Centralizer generation theorem)**.** *The centralizer $Z_m(n)$ is generated by*

$$Z_m(m), \qquad \mathbb{C}\mathfrak{S}_{\{m+1,\dots,n\}}, \qquad L_{m+1}, \dots, L_n.$$

Corollary 10.6. *The algebra $Z_{n-1}(n)$ is commutative.*

Proof. The theorem says that $Z_{n-1}(n)$ is generated by $Z_{n-1}(n-1)$ and L_n. Here $Z_{n-1}(n-1)$ is the center of $\mathbb{C}\mathfrak{S}_{n-1}$. A central element of $\mathbb{C}\mathfrak{S}_{n-1}$ commutes with

$$L_n = \sum_{i=1}^{n-1} (i, n)$$

because this sum is invariant under conjugation by $\mathfrak{S}_{n-1}$. Hence the generators commute. □

Corollary 10.7. *For every $V \in \mathrm{Irr}(\mathfrak{S}_n)$, the restriction*

$$\mathrm{Res}^{\mathfrak{S}_n}_{\mathfrak{S}_{n-1}} V$$

is multiplicity-free.

Proof. Apply the previous commutativity criterion with $A = \mathbb{C}\mathfrak{S}_n$ and $B = \mathbb{C}\mathfrak{S}_{n-1}$. □

10.2 The Branching Graph and Weights

Definition 10.8. The **branching graph** $\mathcal{G}_{\mathrm{rep}}$ has vertices

$$\coprod_{n \geq 0} \mathrm{Irr}(\mathfrak{S}_n).$$

There is an arrow $U \to V$, with $U \in \mathrm{Irr}(\mathfrak{S}_{n-1})$ and $V \in \mathrm{Irr}(\mathfrak{S}_n)$, precisely when U occurs in $\mathrm{Res}^{\mathfrak{S}_n}_{\mathfrak{S}_{n-1}} V$.

The beginning of the graph is

$$\begin{aligned} n = 0 &\ : \ \mathbf{1}, \\ n = 1 &\ : \ \mathbf{1}, \\ n = 2 &\ : \ \mathbf{1},\ \mathrm{sgn}, \\ n = 3 &\ : \ \mathbf{1},\ \mathrm{std},\ \mathrm{sgn}\,. \end{aligned}$$

The arrows are the ones determined by restriction. For instance, the standard representation of $\mathfrak{S}_3$ restricts to $\mathbf{1} \oplus \mathrm{sgn}$ for $\mathfrak{S}_2$.

Fix $V_n \in \mathrm{Irr}(\mathfrak{S}_n)$. Let $\mathrm{Path}(V_n)$ be the set of directed paths from the trivial representation at level 0 to V_n, and set

$$\mathrm{Path}_n = \coprod_{V_n \in \mathrm{Irr}(\mathfrak{S}_n)} \mathrm{Path}(V_n).$$

Since restriction is multiplicity-free at every step, repeated restriction gives a decomposition of V_n into one-dimensional lines indexed by these paths. Choosing a nonzero vector on each line gives the **Gelfand-Tsetlin basis**

$$\{v_p \mid p \in \mathrm{Path}(V_n)\},$$

well-defined up to rescaling each basis vector.

Proposition 10.9. *The vectors v_p are simultaneous eigenvectors for $L_1, \dots, L_n$.*

Proof. The element L_k lies in $Z_{k-1}(k)$. Along a fixed path, the chosen edge from level $k-1$ to level k selects an irreducible $\mathfrak{S}_{k-1}$-summand inside the level-k representation. Since L_k commutes with $\mathfrak{S}_{k-1}$, Schur's lemma says that it acts by a scalar on this summand, and hence on the Gelfand–Tsetlin line contained in it. This holds for every k. □

For a path p, record the eigenvalues by

$$\mathrm{wt}(p) = (w_1, \dots, w_n) \in \mathbb{C}^n.$$

Since $L_1 = 0$, always $w_1 = 0$.

Example 10.10. Consider the standard representation of $\mathfrak{S}_n$, realized as the subspace

$$\left\{ (x_1, \dots, x_n) \in \mathbb{C}^n \mid \sum_i x_i = 0 \right\}.$$

For $2 \le m \le n$, the path that first reaches the standard representation at level m has a Gelfand-Tsetlin vector proportional to

$$v_{p_m} = (\underbrace{1, \dots, 1}_{m-1}, 1-m, 0, \dots, 0).$$

The Jucys-Murphy eigenvalues are

$$\mathrm{wt}(p_m) = (0, 1, 2, \dots, m-2, -1, m-1, m, \dots, n-2).$$

10.3 Combinatorial Weights

The simultaneous eigenspace decomposition for the Jucys-Murphy elements is the Gelfand-Tsetlin decomposition. Let A be the algebra of all elements of $\mathbb{C}\mathfrak{S}_n$ that are diagonal in every Gelfand-Tsetlin basis.

Proposition 10.11. *The algebra A is generated by the Jucys-Murphy elements*

$$L_1, \dots, L_n.$$

Proof. The inclusion $\mathbb{C}[L_1, \dots, L_n] \subseteq A$ follows from the preceding proposition.

Conversely, the Gelfand-Tsetlin algebra is generated by the centers

$$Z(\mathbb{C}\mathfrak{S}_1), \dots, Z(\mathbb{C}\mathfrak{S}_n).$$

Indeed, for $U \in \mathrm{Irr}(\mathfrak{S}_k)$ let $e_U^{(k)} \in Z(\mathbb{C}\mathfrak{S}_k)$ be the corresponding primitive central idempotent. If

$$p = (V_0 \to V_1 \to \cdots \to V_n)$$

is a path, then the embedded central idempotents commute, and

$$e_p = \prod_{k=1}^{n} e_{V_k}^{(k)}$$

acts as the projector onto the Gelfand–Tsetlin line indexed by p. This follows successively from the multiplicity-free restriction at each level. The projectors e_p span the full algebra of operators diagonal in the Gelfand–Tsetlin bases, so the centers generate that algebra.

By the Jucys center theorem, each $Z(\mathbb{C}\mathfrak{S}_k)$ is generated by symmetric polynomials in $L_1, \ldots, L_k$; see [8, 9]. Hence every e_p, and therefore every element of the Gelfand–Tsetlin algebra, belongs to $\mathbb{C}[L_1, \ldots, L_n]$. □

Corollary 10.12. *The weight map*

$$\mathrm{wt}\colon \mathrm{Path}_n \longrightarrow \mathbb{C}^n$$

is injective.

Proof. Different paths define different Gelfand-Tsetlin lines. The projector onto each such line is a polynomial in $L_1, \ldots, L_n$, so two distinct paths cannot have the same joint eigenvalues. □

Let Wt_n be the image of the weight map. Two weights are equivalent if their paths end at the same irreducible representation. Then

$$\mathrm{Irr}(\mathfrak{S}_n) \simeq \mathrm{Wt}_n / \sim.$$

Theorem 10.13 (Key Properties of Weights)**.** *Let $p \in \mathrm{Path}_n$ have weight $w = (w_1, \ldots, w_n)$. For every $i < n$:*

1. *$w_i \neq w_{i+1}$.*
2. *If $w_{i+1} = w_i \pm 1$, then no other path differs from p only at level i.*
3. *If $w_{i+1} \neq w_i \pm 1$, then there is exactly one other path that differs from p only at level i, and its weight is*
$$s_i w = (w_1, \ldots, w_{i+1}, w_i, \ldots, w_n).$$
4. *If $w_{i+1} = w_i \pm 1$ and $i < n-1$, then $w_{i+2} \neq w_i$.*

Sketch of proof. Focus on the two consecutive levels i and $i+1$. The space spanned by all paths that agree with p except possibly at level i carries the rank-two degenerate affine Hecke algebra generated by the adjacent transposition $s_i = (i, i+1)$ and the operators L_i, L_{i+1}. The relations include

$$s_i^2 = 1, \qquad [L_i, L_{i+1}] = 0,$$

and the usual Jucys-Murphy relations

$$s_i L_i - L_{i+1} s_i = -1, \qquad s_i L_{i+1} - L_i s_i = 1.$$

On this local block, L_i and L_{i+1} act semisimply with one-dimensional joint eigenspaces, because they are diagonal in the Gelfand–Tsetlin basis and the weight map is injective. Thus the block is *calibrated*. The calibrated rank-two classification says that equal consecutive eigenvalues are impossible; eigenvalues differing by 1 or -1 give the one-dimensional case; otherwise the block is two-dimensional and its second weight is obtained by swapping the two eigenvalues. These conclusions give parts 1–3. The corresponding calibrated rank-three calculation, using the braid relation for adjacent transpositions, excludes the pattern $w_i, w_i \pm 1, w_i$ and gives part 4. We use these local classification results without proof; see [8, 9]. □

Definition 10.14. An adjacent transposition s_i is **admissible** for a tuple $w = (w_1, \dots, w_n)$ if

$$w_{i+1} \neq w_i \pm 1.$$

In that case it swaps w_i and w_{i+1}. Let $\sim_c$ be the equivalence relation generated by admissible swaps.

A tuple $w \in \mathbb{C}^n$ is a **combinatorial weight** if every tuple $w' = (w'_1, \dots, w'_n)$ that is c-equivalent to w satisfies

$$w'_1 = 0, \qquad w'_i \neq w'_{i+1} \quad (1 \leq i < n),$$

and

$$w'_{i+1} = w'_i \pm 1 \Rightarrow w'_{i+2} \neq w'_i \quad (1 \leq i < n-1).$$

The set of combinatorial weights is denoted cWt_n.

The theorem says that every representation-theoretic weight is a combinatorial weight:

$$\mathrm{Wt}_n \subseteq \mathrm{cWt}_n\,.$$

The following standard content-vector lemma will be used in the counting argument; see [9].

Lemma 10.15 (Content-vector normal form)**.** *Each $\sim_c$-class in cWt_n has a unique row-reading representative of the form*

$$(0, 1, \dots, \lambda_1 - 1,\ -1, 0, \dots, \lambda_2 - 2,\\ -2, -1, \dots, \lambda_3 - 3, \dots),$$

for a partition $\lambda \in P_n$. Moreover, the entire class consists of the content vectors of the standard Young tableaux of shape λ.

Sketch of proof. An induction removes the largest label, uses the local conditions to identify a unique removable outer corner, and then reverses the process by adding that corner. Adjacent admissible swaps are exactly swaps of incomparable consecutive labels in a standard tableau; the graph of linear extensions of a fixed Young diagram is connected. These observations give the asserted normal form and identify its entire equivalence class; full details appear in [9]. □

The lemma gives

$$|\,\mathrm{cWt}_n\,/{\sim_c}| = |P_n|.$$

The Key Properties theorem shows that Wt_n is a union of complete $\sim_c$-classes: every admissible swap of a representation-theoretic weight is again the weight of a path with the same endpoint. Sending a path to its endpoint therefore gives a surjection

$$\mathrm{Wt}_n / \sim_c \longrightarrow \mathrm{Irr}(\mathfrak{S}_n).$$

Consequently,

$$|\,\mathrm{Irr}(\mathfrak{S}_n)| \le |\,\mathrm{Wt}_n / \sim_c| \le |\,\mathrm{cWt}_n / \sim_c| = |P_n| = |\,\mathrm{Irr}(\mathfrak{S}_n)|.$$

All inequalities are equalities. Since both weight sets are unions of $\sim_c$-classes, it follows that $\mathrm{Wt}_n = \mathrm{cWt}_n$; thus the Okounkov-Vershik weights recover the partition classification.

10.4 Relation with Young Tableaux

Definition 10.16. For $\lambda \in P_n$, the **Young diagram** Y_λ has λ_i boxes in row i, left-justified. A **standard Young tableau** of shape λ is a filling of Y_λ by $\{1, \dots, n\}$ such that entries increase from left to right along rows and from top to bottom along columns.

Write $\mathrm{SYT}(\lambda)$ for the set of standard Young tableaux of shape λ, and set

$$\mathrm{SYT}_n = \coprod_{\lambda \in P_n} \mathrm{SYT}(\lambda).$$

The **content** of a box b in row r and column c is

$$\mathrm{ct}(b) = c - r.$$

For a standard tableau T, define its content vector by

$$\mathrm{ct}_T = \big(\mathrm{ct}_T(1), \dots, \mathrm{ct}_T(n)\big),$$

where $\mathrm{ct}_T(k)$ is the content of the box containing k.

Proposition 10.17. *The content map gives a bijection*

$$\mathrm{SYT}_n \longrightarrow \mathrm{cWt}_n\,.$$

Proof. The Content-vector Normal Form Lemma says precisely that the $\sim_c$-class indexed by λ consists of the content vectors of $\mathrm{SYT}(\lambda)$. It also gives the inverse explicitly: starting from a content vector, add at step k the uniquely determined box of content w_k. Hence the content map is both injective and surjective. □

Corollary 10.18. *The branching graph of representations is canonically isomorphic to the branching graph of Young diagrams. Therefore*

$$\mathrm{Irr}(\mathfrak{S}_n) \longleftrightarrow P_n.$$

If $\lambda \in P_n$, the irreducible representation V_λ has a Gelfand-Tsetlin basis

$$\{v_T \mid T \in \mathrm{SYT}(\lambda)\},$$

and the Jucys-Murphy elements act by

$$L_k v_T = \mathrm{ct}_T(k) v_T.$$

These eigenvalue formulas are unchanged if the individual basis vectors are rescaled. To write the adjacent transpositions explicitly, choose the standard seminormal normalization and set

$$r_i(T) = \mathrm{ct}_T(i+1) - \mathrm{ct}_T(i).$$

If $s_i T$ is not standard, then $r_i(T) = \pm 1$ and

$$s_i v_T = r_i(T)^{-1} v_T.$$

If $s_i T$ is standard, the two tableaux form an unordered pair. Choose one member and call it T, and write $T' = s_i T$ for the other. On this two-dimensional block, the standard seminormal normalization may be chosen so that

$$\begin{aligned} s_i v_T &= r_i(T)^{-1} v_T + v_{T'}, \\ s_i v_{T'} &= \bigl(1 - r_i(T)^{-2}\bigr) v_T - r_i(T)^{-1} v_{T'}. \end{aligned}$$

Thus, after this normalization, the matrix of s_i is determined by the content difference. In an arbitrarily rescaled Gelfand–Tsetlin basis, the same matrices are conjugated by a diagonal change of basis; see [9].

Chapter 11

Symmetric Functions

11.1 The Ring of Symmetric Functions

For $N \geq 1$, let

$$\Lambda_N = \mathbb{Z}[x_1, \ldots, x_N]^{\mathfrak{S}_N}$$

be the ring of symmetric polynomials in N variables. It is graded by degree:

$$\Lambda_N = \bigoplus_{k \geq 0} \Lambda_N^k,$$

where Λ_N^k is the abelian group of homogeneous symmetric polynomials of degree k.

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_N)$, write

$$x^\alpha = x_1^{\alpha_1} \cdots x_N^{\alpha_N}.$$

If λ is a partition with $\ell(\lambda) \leq N$, the **monomial symmetric polynomial** m_λ is the sum of all distinct monomials obtained by permuting the entries of λ:

$$m_\lambda = \sum_{\alpha \in \mathfrak{S}_N \lambda} x^\alpha.$$

For instance,

$$m_{(1)} = x_1 + \cdots + x_N.$$

For fixed k, the set

$$\{m_\lambda \mid |\lambda| = k,\ \ell(\lambda) \leq N\}$$

is a $\mathbb{Z}$-basis of Λ_N^k.

If $N' \geq N$, evaluation at zero in the last variables gives a homomorphism

$$\rho_{N',N} \colon \Lambda_{N'} \longrightarrow \Lambda_N, \qquad x_i \longmapsto \begin{cases} x_i, & i \leq N, \\ 0, & i > N. \end{cases}$$

On monomial symmetric polynomials,

$$\rho_{N',N}(m_\lambda) = \begin{cases} m_\lambda, & \ell(\lambda) \leq N, \\ 0, & \ell(\lambda) > N. \end{cases}$$

Thus, for fixed degree k, the map

$$\rho^k_{N',N} \colon \Lambda^k_{N'} \longrightarrow \Lambda^k_N$$

is an isomorphism as soon as $N' \geq N \geq k$.

Define

$$\Lambda^k = \varprojlim_N \Lambda^k_N.$$

An element of Λ^k is a compatible sequence $(f_N)_{N\geq 1}$ with $\rho_{N',N}(f_{N'}) = f_N$. For $N \geq k$, the projection

$$\rho^k_N \colon \Lambda^k \longrightarrow \Lambda^k_N$$

is an isomorphism. This lets us regard m_λ as a stable symmetric function whenever $|\lambda| = k$.

Definition 11.1. The **ring of symmetric functions** is

$$\Lambda = \bigoplus_{k\geq 0} \Lambda^k.$$

Informally, one may think of its elements as symmetric polynomials in infinitely many variables, homogeneous of bounded degree.

11.2 Bases of Symmetric Functions

Let P denote the set of all partitions. For $\lambda = (\lambda_1, \lambda_2, \ldots)$, write

$$|\lambda| = \sum_i \lambda_i, \qquad \ell(\lambda) = |\{i \mid \lambda_i > 0\}|.$$

All infinite sums, products, and generating-function identities below are understood in the graded completion (and, when t occurs, in the t-adic completion). In every fixed homogeneous degree only finitely many terms contribute.

Definition 11.2 (Elementary Symmetric Functions)**.** For $r \geq 0$, the r-th **elementary symmetric function** is

$$e_r = m_{(1^r)} = \sum_{i_1 < \cdots < i_r} x_{i_1} \cdots x_{i_r},$$

with $e_0 = 1$. Its generating function is

$$E(t) = \sum_{r\geq 0} e_r t^r = \prod_{i\geq 1} (1 + x_i t).$$

The fundamental theorem of symmetric polynomials gives

$$\Lambda = \mathbb{Z}[e_1, e_2, \ldots].$$

Consequently, for a partition λ,

$$e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$$

and the elements e_λ form a $\mathbb{Z}$-basis of Λ.

Definition 11.3 (Complete Symmetric Functions)**.** For $r \geq 0$, the r-th **complete symmetric function** is the sum of all monomials of total degree r:

$$h_r = \sum_{|\lambda|=r} m_\lambda, \qquad h_0 = 1.$$

Its generating function is

$$H(t) = \sum_{r\geq 0} h_r t^r = \prod_{i\geq 1} (1 - x_i t)^{-1}.$$

The generating functions satisfy

$$H(t)E(-t) = 1.$$

Comparing coefficients gives, for $n \geq 1$,

$$\sum_{r=0}^{n} (-1)^r e_r h_{n-r} = 0.$$

Since the e_r freely generate Λ, there is a unique algebra endomorphism

$$\omega \colon \Lambda \longrightarrow \Lambda$$

with $\omega(e_r) = h_r$. The relation above implies

$$\omega(h_r) = e_r,$$

so ω is an involution. In particular, the products $h_\lambda = h_{\lambda_1} h_{\lambda_2} \cdots$ also form a $\mathbb{Z}$-basis of Λ.

Definition 11.4 (Power Sums)**.** For $r \geq 1$, the r-th **power sum** is

$$p_r = m_{(r)} = \sum_i x_i^r.$$

The power sums are encoded by

$$P(t) = \sum_{r\geq 1} p_r t^{r-1} = \sum_{i\geq 1} \frac{x_i}{1 - x_i t} = \frac{H'(t)}{H(t)}.$$

Equivalently,

$$P(-t) = \frac{E'(t)}{E(t)}.$$

Applying ω to $H'(t)/H(t) = P(t)$ gives $E'(t)/E(t) = \omega(P(t)) = P(-t)$. Comparing coefficients yields

$$\omega(p_r) = (-1)^{r-1} p_r.$$

The differential equation $H'(t) = P(t)H(t)$ gives

$$n h_n = \sum_{r=1}^{n} p_r h_{n-r}.$$

Over $\mathbb{Q}$ this recursion expresses the h_n in terms of the p_r and conversely. Hence, with

$$\Lambda_{\mathbb{Q}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{Q},$$

the products

$$p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots$$

form a $\mathbb{Q}$-basis of $\Lambda_{\mathbb{Q}}$.

If λ has n_i parts equal to i, define

$$z_\lambda = \prod_{i \geq 1} i^{n_i} n_i!.$$

This is the order of the centralizer in $\mathfrak{S}_{|\lambda|}$ of a permutation of cycle type λ.

Proposition 11.5. *The generating functions have the expansions*

$$H(t) = \sum_{\lambda \in P} z_\lambda^{-1} p_\lambda t^{|\lambda|},$$

and

$$E(t) = \sum_{\lambda \in P} \varepsilon_\lambda z_\lambda^{-1} p_\lambda t^{|\lambda|}, \qquad \varepsilon_\lambda = (-1)^{|\lambda| - \ell(\lambda)}.$$

Proof. Integrating $H'(t)/H(t) = P(t)$ gives

$$H(t) = \exp\left(\sum_{r \geq 1} \frac{p_r}{r} t^r \right).$$

Expanding the exponential,

$$H(t) = \prod_{r \geq 1} \sum_{n_r \geq 0} \frac{1}{n_r!} \left(\frac{p_r}{r} t^r \right)^{n_r}.$$

Choosing the nonnegative integers n_r is the same as choosing a partition λ with n_r parts of size r, and the coefficient is exactly z_λ^{-1}. Applying the involution ω, which satisfies $\omega(p_r) = (-1)^{r-1} p_r$, gives the formula for $E(t)$. $\square$

11.3 Schur Functions

A polynomial $f \in \mathbb{Z}[x_1, \ldots, x_N]$ is **alternating** if

$$w(f) = \operatorname{sgn}(w) f$$

for every $w \in \mathfrak{S}_N$. For a multi-index α, define

$$a_\alpha(x_1, \ldots, x_N) = \sum_{w \in \mathfrak{S}_N} \operatorname{sgn}(w)\, w(x^\alpha).$$

This is zero if two entries of α are equal.

Let λ be a partition with $\ell(\lambda) \leq N$, padded by zeros to length N, and set

$$\delta = (N-1, N-2, \ldots, 1, 0).$$

Then $\lambda + \delta$ has strictly decreasing entries, and

$$a_{\lambda+\delta} = \det\left(x_i^{\lambda_j + N - j} \right)_{1 \leq i,j \leq N}.$$

The special case

$$a_\delta = \prod_{i<j}(x_i - x_j)$$

is the Vandermonde determinant. Every alternating polynomial is divisible by a_δ.

Definition 11.6. The **Schur polynomial** attached to λ is

$$s_\lambda(x_1, \dots, x_N) = \frac{a_{\lambda+\delta}}{a_\delta}.$$

These polynomials are symmetric and homogeneous. Passing to the stable limit gives the **Schur functions** $s_\lambda \in \Lambda$.

The alternating polynomials $a_{\lambda+\delta}$ form a $\mathbb{Z}$-basis of the alternating polynomials, and multiplication by a_δ identifies symmetric polynomials with alternating polynomials. Therefore the Schur functions s_λ form a $\mathbb{Z}$-basis of Λ.

The Jacobi-Trudi identities express Schur functions in the complete and elementary bases:

$$s_\lambda = \det\big(h_{\lambda_i - i + j}\big)_{1 \le i,j \le \ell(\lambda)},$$

and

$$s_\lambda = \det\big(e_{\lambda'_i - i + j}\big)_{1 \le i,j \le \lambda_1},$$

where λ' is the transpose partition and we set $h_0 = e_0 = 1$, $h_r = e_r = 0$ for $r < 0$.

11.4 Orthogonality and Isometry

Let $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ be two sets of variables.

Theorem 11.7 (Cauchy identities)**.** *The Cauchy kernel has the following three expansions:*

$$\prod_{i,j}(1 - x_i y_j)^{-1} = \sum_{\lambda \in P} z_\lambda^{-1} p_\lambda(x) p_\lambda(y),$$

$$\prod_{i,j}(1 - x_i y_j)^{-1} = \sum_{\lambda \in P} h_\lambda(x) m_\lambda(y),$$

and

$$\prod_{i,j}(1 - x_i y_j)^{-1} = \sum_{\lambda \in P} s_\lambda(x) s_\lambda(y).$$

Sketch of proof. Taking the formal logarithm of the kernel gives $\sum_{r \ge 1} p_r(x)p_r(y)/r$, and expanding the exponential gives the first identity with the factors z_λ^{-1}. Expanding $\prod_j H_x(y_j)$ and grouping monomials of the same exponent partition gives the second identity. The third is the classical Schur Cauchy identity. In finitely many variables it follows from the bialternant formula and Cauchy–Binet, and passage to the stable limit gives the displayed identity; see [7]. □

Define a scalar product on Λ by declaring the complete and monomial bases to be dual:

$$\langle h_\lambda, m_\mu \rangle = \delta_{\lambda\mu}.$$

Lemma 11.8. *Fix a degree k. Let $\{u_\lambda\}_{\lambda\in P_k}$ and $\{v_\lambda\}_{\lambda\in P_k}$ be two bases of $\Lambda^k_{\mathbb{Q}}$. They are dual bases for this scalar product if and only if*

$$\sum_{\lambda\in P_k} u_\lambda(x)v_\lambda(y) = \left[\prod_{i,j}(1-x_iy_j)^{-1}\right]_{(k,k)},$$

where the subscript means the component of degree k in each set of variables.

Proof. Write

$$u_\lambda = \sum_\rho a_{\lambda\rho}h_\rho, \qquad v_\mu = \sum_\sigma b_{\mu\sigma}m_\sigma.$$

Then

$$\langle u_\lambda, v_\mu\rangle = \sum_\rho a_{\lambda\rho}b_{\mu\rho}.$$

On the other hand,

$$\sum_\lambda u_\lambda(x)v_\lambda(y) = \sum_{\rho,\sigma}\left(\sum_\lambda a_{\lambda\rho}b_{\lambda\sigma}\right)h_\rho(x)m_\sigma(y).$$

The Cauchy expansion in the h- and m-bases is

$$\sum_{\rho\in P_k} h_\rho(x)m_\rho(y).$$

Thus the displayed kernel identity is equivalent to the duality of the two bases. □

Corollary 11.9. *The scalar product satisfies*

$$\langle p_\lambda, p_\mu\rangle = z_\lambda\delta_{\lambda\mu}, \qquad \langle s_\lambda, s_\mu\rangle = \delta_{\lambda\mu}.$$

In particular, the Schur functions form an orthonormal basis, and the involution ω is an isometry:

$$\langle \omega u, \omega v\rangle = \langle u, v\rangle.$$

Proof. Compare the three Cauchy expansions above with the lemma. The power-sum expansion gives the first formula, and the Schur expansion gives the second. Since $\omega(p_r) = (-1)^{r-1}p_r$, the power-sum formula also shows that ω preserves the scalar product. □

11.5 The Frobenius Character Map

Let

$$R^n = R(\mathfrak{S}_n)\otimes_{\mathbb{Z}}\mathbb{Q}$$

be the rational vector space spanned by the irreducible characters of $\mathfrak{S}_n$. It carries the standard character inner product

$$\langle f, g\rangle_{\mathfrak{S}_n} = \frac{1}{n!}\sum_{w\in\mathfrak{S}_n} f(w^{-1})g(w).$$

Set

$$R = \bigoplus_{n \geq 0} R^n.$$

Multiplication in R is defined by induction:

$$f \circ g = \mathrm{Ind}_{\mathfrak{S}_n \times \mathfrak{S}_m}^{\mathfrak{S}_{n+m}}(f \boxtimes g), \qquad f \in R^n,\ g \in R^m.$$

For $w \in \mathfrak{S}_n$, let $\sigma(w) \in P_n$ be its cycle type.

Definition 11.10. The **Frobenius character map** is the linear map

$$\mathrm{Ch} \colon R^n \longrightarrow \Lambda_{\mathbb{Q}}^n$$

defined by

$$\mathrm{Ch}(f) = \frac{1}{n!} \sum_{w \in \mathfrak{S}_n} f(w)\, p_{\sigma(w)}.$$

Equivalently,

$$\mathrm{Ch}(f) = \sum_{\sigma \in P_n} f(\sigma)\, z_\sigma^{-1} p_\sigma,$$

where $f(\sigma)$ denotes the value of f on the conjugacy class of cycle type σ.

Proposition 11.11. *The Frobenius map*

$$\mathrm{Ch} \colon R \longrightarrow \Lambda_{\mathbb{Q}}$$

is an isometric ring isomorphism.

Proof. The isometry follows from the power-sum scalar product:

$$\begin{aligned} \langle \mathrm{Ch}(f), \mathrm{Ch}(g) \rangle &= \sum_{\sigma \in P_n} f(\sigma) g(\sigma) z_\sigma^{-1} \\ &= \langle f, g \rangle_{\mathfrak{S}_n}. \end{aligned}$$

Thus the map is injective. Moreover,

$$\dim_{\mathbb{Q}} R^n = |\operatorname{Irr}(\mathfrak{S}_n)| = |P_n| = \dim_{\mathbb{Q}} \Lambda_{\mathbb{Q}}^n.$$

Hence it is a degreewise vector-space isomorphism.

It remains to check multiplication. Fix $\nu \vdash n+m$. Splitting the multiset of cycles of type ν into submultisets of total sizes n and m gives

$$\frac{(f \circ g)(\nu)}{z_\nu} = \sum_{\substack{\lambda \cup \mu = \nu \\ |\lambda| = n,\ |\mu| = m}} \frac{f(\lambda) g(\mu)}{z_\lambda z_\mu}.$$

Indeed, for fixed λ and μ the number of invariant choices of the n letters is $z_\nu / (z_\lambda z_\mu)$. This is exactly the coefficient identity obtained by multiplying $p_\lambda p_\mu = p_{\lambda \cup \mu}$. Hence $\mathrm{Ch}(f \circ g) = \mathrm{Ch}(f)\mathrm{Ch}(g)$. □

Theorem 11.12. *Let χ^λ be the irreducible character of $\mathfrak{S}_n$ corresponding to the partition λ. Then*

$$\mathrm{Ch}(\chi^\lambda) = s_\lambda.$$

Proof. For a partition $\lambda \vdash n$, let

$$\mathfrak{S}_\lambda = \mathfrak{S}_{\lambda_1} \times \mathfrak{S}_{\lambda_2} \times \cdots \subseteq \mathfrak{S}_n$$

be the Young subgroup, and define the permutation representation

$$U_\lambda = \operatorname{Ind}_{\mathfrak{S}_\lambda}^{\mathfrak{S}_n}(\mathbf{1}).$$

Since $\operatorname{Ch}(\mathbf{1}_{\mathfrak{S}_r}) = h_r$ and Ch respects induction products,

$$\operatorname{Ch}(\chi_{U_\lambda}) = h_\lambda.$$

Recall that a **semistandard Young tableau** has weakly increasing rows and strictly increasing columns. Its weight records the multiplicity of each entry, and the **Kostka number** $K_{\mu\lambda}$ is the number of semistandard Young tableaux of shape μ and weight λ.

Young's rule and the corresponding Kostka expansion of complete symmetric functions are the standard identities

$$\chi_{U_\lambda} = \sum_{\mu \vdash n} K_{\mu\lambda} \chi^\mu,$$

and

$$h_\lambda = \sum_{\mu \vdash n} K_{\mu\lambda} s_\mu.$$

We use these identities without proof; see [9, 7]. The Kostka matrix is unitriangular in dominance order, hence invertible. Comparing the last two displayed expansions and using the linearity of Ch therefore gives

$$\operatorname{Ch}(\chi^\lambda) = s_\lambda.$$

□

Chapter 12

Young Symmetrizers and Symmetric Group Representations

The previous chapters classified the irreducible representations of $\mathfrak{S}_n$ using the Okounkov-Vershik method and connected them to Schur functions through the Frobenius character map. We now describe the older, more explicit construction due to Young. Its basic objects are idempotents, up to scalar, in the group algebra $\mathbb{C}\mathfrak{S}_n$.

12.1 Tableaux and Stabilizers

Let $\lambda \vdash n$ be a partition,

$$\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0), \qquad \sum_i \lambda_i = n.$$

Its Young diagram has λ_i boxes in row i, left-justified. For a box $b = (i, j)$, its **hook length** is

$$h(b) = \lambda_i - j + \lambda'_j - i + 1,$$

the number of boxes weakly to the right of b in its row together with those strictly below it in its column.

Definition 12.1. A **Young tableau** t of shape λ is a bijective filling of the boxes of the Young diagram by the numbers $1, \ldots, n$.

At this point no monotonicity condition is imposed. Thus a tableau is not necessarily standard. For example, if $\lambda = (4, 2, 1, 1)$, a tableau is any placement of the numbers $1, \ldots, 8$ in the eight boxes. For a fixed shape $\lambda \vdash n$, there are $n!$ tableaux, and $\mathfrak{S}_n$ acts transitively on them by permuting the entries.

Fix a tableau t. There are two natural subgroups of $\mathfrak{S}_n$:

$$P_t = \{\text{permutations preserving each row of } t\},$$

and

$$Q_t = \{\text{permutations preserving each column of } t\}.$$

Thus

$$P_t \simeq \mathfrak{S}_{\lambda_1} \times \cdots \times \mathfrak{S}_{\lambda_\ell},$$

whereas Q_t is the corresponding product for the column lengths, or equivalently for the transpose partition λ'. Moreover,

$$P_t \cap Q_t = \{1\},$$

because a permutation that preserves every row and every column fixes each box.

12.2 The Young Symmetrizer

Definition 12.2. In $\mathbb{C}\mathfrak{S}_n$, define the **row symmetrizer**

$$a_t = \sum_{p \in P_t} p$$

and the **column antisymmetrizer**

$$b_t = \sum_{q \in Q_t} \operatorname{sgn}(q) q.$$

The **Young symmetrizer** attached to t is

$$c_t = a_t b_t = \sum_{p \in P_t,\ q \in Q_t} \operatorname{sgn}(q) pq.$$

For $p \in P_t$ and $q \in Q_t$,

$$pa_t = a_t p = a_t, \qquad qb_t = b_t q = \operatorname{sgn}(q) b_t.$$

Hence c_t satisfies the row-symmetry and column-antisymmetry relations

$$pc_t = c_t, \qquad c_t q = \operatorname{sgn}(q) c_t.$$

Equivalently,

$$pc_t q = \operatorname{sgn}(q) c_t \qquad (p \in P_t,\ q \in Q_t).$$

Remark 12.3. The transformation property above characterizes the row symmetrization and column antisymmetrization imposed by t, but it does not by itself make c_t unique among all elements of the whole group algebra. For the construction, the crucial fact is instead the following standard quasi-idempotence theorem.

We use the classical Young symmetrizer theorem and the hook-length formula without proof; see [9, 4].

Theorem 12.4 (Young Symmetrizer Theorem)**.** *There is a nonzero scalar $\gamma_t \in \mathbb{C}$ such that*

$$c_t^2 = \gamma_t c_t.$$

In fact,

$$\gamma_t = \prod_{b \in \lambda} h(b) = \frac{n!}{f^\lambda},$$

where $h(b)$ is the hook length of the box b, and f^λ is the number of standard Young tableaux of shape λ.

Thus

$$e_t = \gamma_t^{-1} c_t$$

is an idempotent in $\mathbb{C}\mathfrak{S}_n$.

12.3 Irreducible Representations

The idempotent e_t, or equivalently the quasi-idempotent c_t, defines a left ideal

$$V_\lambda(t) = \mathbb{C}\mathfrak{S}_n c_t.$$

This is stable under left multiplication by $\mathfrak{S}_n$, hence is a representation of $\mathfrak{S}_n$.

We also use Young's irreducibility theorem without proof; see [9, 4].

Theorem 12.5 (Young's Irreducibility Theorem)**.** *For every partition $\lambda \vdash n$ and every tableau t of shape λ, the left ideal*

$$V_\lambda(t) = \mathbb{C}\mathfrak{S}_n c_t$$

is an irreducible representation of $\mathfrak{S}_n$. Its isomorphism class depends only on λ. Every irreducible representation of $\mathfrak{S}_n$ is obtained in this way for exactly one partition $\lambda \vdash n$.

Let $t' = w(t)$ for $w \in \mathfrak{S}_n$. Then

$$P_{t'} = wP_tw^{-1}, \qquad Q_{t'} = wQ_tw^{-1},$$

and therefore

$$c_{t'} = wc_tw^{-1}.$$

Right multiplication by w gives an isomorphism of left $\mathfrak{S}_n$-modules

$$\mathbb{C}\mathfrak{S}_n c_{t'} \xrightarrow{\sim} \mathbb{C}\mathfrak{S}_n c_t.$$

Thus the module is denoted simply by V_λ.

Example 12.6 (The Trivial Representation)**.** For the partition $\lambda = (n)$, the diagram has one row. Hence

$$P_t = \mathfrak{S}_n, \qquad Q_t = \{1\}.$$

The Young symmetrizer is

$$c_t = \sum_{w \in \mathfrak{S}_n} w.$$

The ideal $\mathbb{C}\mathfrak{S}_n c_t$ is one-dimensional, and every element of $\mathfrak{S}_n$ acts trivially on it.

Example 12.7 (The Sign Representation)**.** For the partition $\lambda = (1^n)$, the diagram has one column. Hence

$$P_t = \{1\}, \qquad Q_t = \mathfrak{S}_n.$$

The Young symmetrizer is

$$c_t = \sum_{w \in \mathfrak{S}_n} \operatorname{sgn}(w) w.$$

The ideal $\mathbb{C}\mathfrak{S}_n c_t$ is one-dimensional, and $\mathfrak{S}_n$ acts on it by the sign character.

Relation with the Okounkov–Vershik Construction

Young symmetrizers realize the irreducible $\mathfrak{S}_n$-modules as explicit left ideals in the regular representation $\mathbb{C}\mathfrak{S}_n$. The Okounkov-Vershik construction gives the Gelfand-Tsetlin basis and the branching rule efficiently, while Young symmetrizers connect directly to tensor spaces and therefore to Schur-Weyl duality. This is the perspective used in the next chapter.

Chapter 13

Schur-Weyl Duality

Let V be a finite-dimensional complex vector space and let $n \geq 1$. Recall that

$$\mathfrak{g} = \mathfrak{gl}(V) = \mathrm{End}(V).$$

Here $\mathfrak{gl}(V)$ is a Lie algebra with bracket $[x, y] = xy - yx$. By a representation of a Lie algebra $\mathfrak{a}$ on a vector space E we mean a linear map $\rho\colon \mathfrak{a} \to \mathrm{End}(E)$ satisfying

$$\rho([x, y]) = [\rho(x), \rho(y)].$$

These notions are developed systematically in the next chapter. The vector space V is the standard representation of $\mathfrak{g}$. On the tensor power $V^{\otimes n}$, there are two natural actions.

First, $\mathfrak{g}$ acts by the Leibniz rule:

$$x \cdot (v_1 \otimes \cdots \otimes v_n) = \sum_{i=1}^{n} v_1 \otimes \cdots \otimes xv_i \otimes \cdots \otimes v_n.$$

Write $\Delta(x)$ for this endomorphism, and write $x^{(i)}$ for the operator that applies x in the i-th tensor factor and the identity in all other factors. Then $\Delta(x) = \sum_i x^{(i)}$. Operators in distinct tensor factors commute, while $[x^{(i)}, y^{(i)}] = [x, y]^{(i)}$, so

$$[\Delta(x), \Delta(y)] = \sum_{i=1}^{n} [x, y]^{(i)} = \Delta([x, y]).$$

Thus the displayed formula indeed defines a representation of $\mathfrak{g}$. Second, $\mathfrak{S}_n$ acts by permuting tensor factors:

$$w \cdot (v_1 \otimes \cdots \otimes v_n) = v_{w^{-1}(1)} \otimes \cdots \otimes v_{w^{-1}(n)}.$$

These actions commute. Indeed, applying x to all tensor factors and then permuting the factors gives the same result as permuting first and then applying x. Schur-Weyl duality is the statement that, after passing to the generated endomorphism algebras, these two commuting actions are each other's full centralizer.

13.1 The Double Centralizer Theorem

Theorem 13.1 (Double Centralizer Theorem)**.** *Let E be a finite-dimensional vector space over $\mathbb{C}$. Let $A \subseteq \mathrm{End}(E)$ be a unital semisimple subalgebra, and set*

$$B = \mathrm{End}_A(E) = \{\varphi \in \mathrm{End}(E) \mid \varphi a = a\varphi \text{ for all } a \in A\}.$$

Then:

1. $\mathrm{End}_B(E) = A$.
2. *B is semisimple.*
3. *There is a decomposition*
$$E \simeq \bigoplus_{i \in I} V_i \otimes W_i$$
as an $A \otimes B$-module, where V_i runs over the irreducible A-modules appearing in E, and
$$W_i = \mathrm{Hom}_A(V_i, E).$$
The W_i are precisely the irreducible B-modules appearing in E.

Thus the irreducible A-modules and B-modules that occur in E are paired with one another.

Proof. Since A is semisimple, E decomposes into A-isotypic components:

$$E \simeq \bigoplus_{V \in \mathrm{Irr}(A)} V \otimes \mathrm{Hom}_A(V, E),$$

where $v \otimes f \mapsto f(v)$. If V does not occur in E, then $\mathrm{Hom}_A(V, E) = 0$, so we restrict the sum to the modules that do occur and write them as V_i. Put $W_i = \mathrm{Hom}_A(V_i, E)$.

On the summand $V_i \otimes W_i$, the algebra A acts on the first factor. Hence an endomorphism commuting with A must preserve the isotypic components and act only on the multiplicity spaces:

$$B = \mathrm{End}_A(E) \simeq \bigoplus_{i \in I} \mathrm{End}(W_i).$$

This proves that B is semisimple and that the W_i are the irreducible B-modules appearing in E.

Taking the centralizer again gives

$$\mathrm{End}_B(E) \simeq \bigoplus_{i \in I} \mathrm{End}(V_i).$$

But this is exactly the image of A on the sum of the A-isotypic components appearing in E. Since $A \subseteq \mathrm{End}(E)$ was already acting faithfully as a subalgebra, $\mathrm{End}_B(E) = A$. $\square$

13.2 The Schur Algebra

Now let $E = V^{\otimes n}$. The symmetric group action gives an algebra homomorphism

$$\mathbb{C}\mathfrak{S}_n \longrightarrow \mathrm{End}(V^{\otimes n}).$$

Let A be its image. Since $\mathbb{C}\mathfrak{S}_n$ is semisimple, so is its quotient A.

Define

$$B = \mathrm{End}_A(V^{\otimes n}) = \mathrm{End}_{\mathbb{C}\mathfrak{S}_n}(V^{\otimes n}).$$

This algebra is called the **Schur algebra** $S(\dim V, n)$ (with this choice of parameters). For a Lie algebra $\mathfrak{g}$, its universal enveloping algebra is

$$\mathcal{U}(\mathfrak{g}) = T^\bullet(\mathfrak{g})/\langle x \otimes y - y \otimes x - [x,y] \mid x, y \in \mathfrak{g}\rangle;$$

its universal property identifies Lie representations of $\mathfrak{g}$ with unital algebra representations of $\mathcal{U}(\mathfrak{g})$. The construction is treated in detail in the next chapter. The $\mathfrak{g}$-action commutes with $\mathfrak{S}_n$, so it lands in B and extends to an algebra homomorphism

$$\rho\colon \mathcal{U}(\mathfrak{g}) \longrightarrow B.$$

Theorem 13.2. *The homomorphism*

$$\rho\colon \mathcal{U}(\mathfrak{g}) \longrightarrow \mathrm{End}_{\mathbb{C}\mathfrak{S}_n}(V^{\otimes n})$$

is surjective.

We first isolate the elementary tensor fact used in the proof.

Lemma 13.3. *Let U be a finite-dimensional complex vector space, and let $n \geq 1$.*

1. *The symmetric tensor space $S^nU = (U^{\otimes n})^{\mathfrak{S}_n}$ is linearly spanned by pure powers $u^{\otimes n}$, with $u \in U$.*
2. *If R is a finite-dimensional unital associative algebra, then $(R^{\otimes n})^{\mathfrak{S}_n}$ is generated as an algebra by the elements*

$$\Delta(a) = \sum_{i=1}^{n} 1 \otimes \cdots \otimes a \otimes \cdots \otimes 1, \qquad a \in R.$$

Proof. For (1), use polarization. The coefficient of $t_1 \cdots t_n$ in

$$(t_1u_1 + \cdots + t_nu_n)^{\otimes n}$$

is the symmetrization of $u_1 \otimes \cdots \otimes u_n$. Since the left side is a linear combination of pure powers, pure powers span S^nU.

For (2), it is enough by (1), applied to the vector space R, to show that $a^{\otimes n}$ lies in the algebra generated by the $\Delta(b)$. Fix $a \in R$. In the commutative polynomial ring $\mathbb{C}[x_1, \ldots, x_n]$, the elementary symmetric polynomial $x_1 \cdots x_n$ is a polynomial in the power sums

$$p_k = x_1^k + \cdots + x_n^k, \qquad 1 \leq k \leq n.$$

Substitute

$$x_i \longmapsto 1 \otimes \cdots \otimes a \otimes \cdots \otimes 1$$

with a in the i-th tensor factor. These substituted elements commute with one another, and p_k maps to $\Delta(a^k)$. Hence $a^{\otimes n}$ is a polynomial in $\Delta(a), \Delta(a^2), \ldots, \Delta(a^n)$. Since each a^k is an element of R, it belongs to the algebra generated by all $\Delta(b)$. □

Proof of Theorem 13.2. Under ρ, an element $x \in \mathfrak{g} = \mathrm{End}(V)$ acts as

$$\Delta(x) = \sum_{i=1}^{n} 1 \otimes \cdots \otimes x \otimes \cdots \otimes 1$$

on $V^{\otimes n}$. Therefore $\rho(\mathcal{U}(\mathfrak{g}))$ is the algebra generated by these operators.

Using the canonical identification

$$\mathrm{End}(V^{\otimes n}) \simeq \mathrm{End}(V)^{\otimes n},$$

the action of $\mathfrak{S}_n$ permutes the tensor factors. Hence

$$\mathrm{End}_{\mathbb{C}\mathfrak{S}_n}(V^{\otimes n}) = \big(\mathrm{End}(V)^{\otimes n}\big)^{\mathfrak{S}_n}.$$

By Lemma 13.3(2), this invariant algebra is generated by the elements $\Delta(x)$ with $x \in \mathrm{End}(V)$. Thus ρ is surjective. □

13.3 Schur-Weyl Duality

Theorem 13.4 (Schur-Weyl Duality)**.** *Let $A \subseteq \mathrm{End}(V^{\otimes n})$ be the image of $\mathbb{C}\mathfrak{S}_n$, and let $B \subseteq \mathrm{End}(V^{\otimes n})$ be the image of $\mathcal{U}(\mathfrak{gl}(V))$. Then:*

1. *A and B are mutual centralizers.*
2. *Both A and B are semisimple.*
3. *As an $A \otimes B$-module,*

$$V^{\otimes n} \simeq \bigoplus_{\lambda \vdash n} V_\lambda \otimes L_\lambda,$$

where zero summands are omitted, V_λ is the irreducible $\mathfrak{S}_n$-module indexed by λ, and, for an occurring summand,

$$L_\lambda = \mathrm{Hom}_{\mathfrak{S}_n}(V_\lambda, V^{\otimes n})$$

is a nonzero irreducible $\mathfrak{gl}(V)$-module.

Proof. The image A of $\mathbb{C}\mathfrak{S}_n$ is semisimple. Its centralizer is the Schur algebra B, by Theorem 13.2. The double centralizer theorem then gives the mutual centralizer statement, semisimplicity of B, and the displayed decomposition. Finally, B is the image of $\mathcal{U}(\mathfrak{g})$. A subspace of L_λ is $\mathfrak{g}$-stable if and only if it is $\mathcal{U}(\mathfrak{g})$-stable, and hence if and only if it is B-stable. Thus the irreducibility supplied by the double centralizer theorem is also irreducibility as a $\mathfrak{gl}(V)$-module. □

Remark 13.5. If $\dim V \geq n$, then the map

$$\mathbb{C}\mathfrak{S}_n \longrightarrow \mathrm{End}(V^{\otimes n})$$

is injective. In this range, every irreducible $\mathfrak{S}_n$-module V_λ occurs in $V^{\otimes n}$. For smaller $\dim V = d$, the summands with $\ell(\lambda) > d$ vanish.

One often states Schur-Weyl duality for the group $\mathrm{GL}(V)$. The group acts diagonally by

$$\rho_{\mathrm{GL}}\colon \mathrm{GL}(V) \longrightarrow \mathrm{End}(V^{\otimes n}), \qquad g \longmapsto g^{\otimes n}.$$

This action also commutes with $\mathfrak{S}_n$.

Lemma 13.6. *The linear span of $\rho_{\mathrm{GL}}(\mathrm{GL}(V))$ is the whole Schur algebra*

$$B = \mathrm{End}_{\mathbb{C}\mathfrak{S}_n}(V^{\otimes n}).$$

Proof. We have identified B with

$$\big(\mathrm{End}(V)^{\otimes n}\big)^{\mathfrak{S}_n}.$$

By Lemma 13.3(1), this space is linearly spanned by tensors $b^{\otimes n}$, where $b \in \mathrm{End}(V)$.

Let B' be the linear span of $g^{\otimes n}$ with $g \in \mathrm{GL}(V)$. For any $b \in \mathrm{End}(V)$, the operator $t\,\mathrm{Id} + b$ is invertible for all but finitely many $t \in \mathbb{C}$. Hence

$$(t\,\mathrm{Id} + b)^{\otimes n} \in B'$$

for infinitely many values of t. As a polynomial in t, this expression has coefficient $b^{\otimes n}$ at t^0. Since B' is a vector subspace, interpolation shows that every coefficient, in particular $b^{\otimes n}$, lies in B'. Therefore $B' = B$. □

Corollary 13.7. *As a representation of $\mathrm{GL}(V) \times \mathfrak{S}_n$,*

$$V^{\otimes n} \simeq \bigoplus_{\lambda \vdash n} L_\lambda \otimes V_\lambda,$$

where zero summands are omitted, V_λ is the irreducible $\mathfrak{S}_n$-module indexed by λ, and L_λ is a nonzero irreducible $\mathrm{GL}(V)$-module. If L_λ and L_μ are nonzero and $\lambda \neq \mu$, then $L_\lambda \not\simeq L_\mu$.

Proof. The displayed decomposition is the Schur–Weyl decomposition above, now with the diagonal $\mathrm{GL}(V)$-action. By the preceding lemma, a subspace of a multiplicity space is $\mathrm{GL}(V)$-stable if and only if it is B-stable. Likewise, a linear map between two multiplicity spaces intertwines $\mathrm{GL}(V)$ if and only if it intertwines their B-actions. The irreducibility and pairwise non-isomorphism therefore follow from the double centralizer theorem. □

For a finite-dimensional complex vector space X, write

$$L_\lambda(X) = \mathrm{Hom}_{\mathfrak{S}_n}(V_\lambda, X^{\otimes n}).$$

Thus $L_\lambda(X) = 0$ when $\ell(\lambda) > \dim X$.

Corollary 13.8 (Cauchy Decomposition)**.** *Let U and W be finite-dimensional complex vector spaces. As a $\mathrm{GL}(U) \times \mathrm{GL}(W)$-module,*

$$\mathrm{Sym}^n(U \otimes W) \simeq \bigoplus_{\lambda \vdash n} L_\lambda(U) \otimes L_\lambda(W),$$

where a summand is understood to be zero if $\ell(\lambda) > \dim U$ or $\ell(\lambda) > \dim W$.

Proof. Regrouping tensor factors gives an equivariant identification

$$(U \otimes W)^{\otimes n} \simeq U^{\otimes n} \otimes W^{\otimes n},$$

under which the permutation of the n copies of $U \otimes W$ becomes the diagonal action of $\mathfrak{S}_n$ on the right-hand side. Applying Schur–Weyl duality to both tensor powers gives

$$U^{\otimes n} \otimes W^{\otimes n} \simeq \bigoplus_{\lambda,\mu \vdash n} L_\lambda(U) \otimes L_\mu(W) \otimes V_\lambda \otimes V_\mu.$$

Every permutation is conjugate to its inverse, so $\chi_{V_\lambda^*}(w) = \chi_{V_\lambda}(w^{-1}) = \chi_{V_\lambda}(w)$; hence $V_\lambda^* \simeq V_\lambda$. Schur's lemma now gives

$$(V_\lambda \otimes V_\mu)^{\mathfrak{S}_n} \simeq \operatorname{Hom}_{\mathfrak{S}_n}(V_\lambda^*, V_\mu) \simeq \begin{cases} \mathbb{C}, & \lambda = \mu, \\ 0, & \lambda \neq \mu. \end{cases}$$

Taking diagonal $\mathfrak{S}_n$-invariants therefore yields the stated decomposition of $\operatorname{Sym}^n(U \otimes W)$. Here we use the characteristic-zero symmetrization isomorphism $\operatorname{Sym}^n(U \otimes W) \simeq ((U \otimes W)^{\otimes n})^{\mathfrak{S}_n}$. $\square$

Part IV

Continuous and Algebraic Representations

Chapter 14

Introduction to Lie Algebras

14.1 Definitions and Examples

Let k be a field of characteristic zero, usually $k = \mathbb{C}$ or $k = \mathbb{R}$. Lie algebras should be thought of as linearized symmetry objects: they remember the infinitesimal part of a group action.

Definition 14.1. A **Lie algebra** is a k-vector space $\mathfrak{g}$ equipped with a bilinear map

$$[-,-]\colon \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g},$$

called the **Lie bracket**, satisfying:

(1) **Antisymmetry**: $[x, y] = -[y, x]$ for all $x, y \in \mathfrak{g}$ (equivalently in characteristic zero, $[x, x] = 0$).

(2) **Jacobi identity**:
$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$$
for all $x, y, z \in \mathfrak{g}$.

Equivalently, the Jacobi identity may be written as

$$[x, [y, z]] = [[x, y], z] + [y, [x, z]].$$

Thus the operator $\mathrm{ad}(x)\colon y \mapsto [x, y]$ acts as a derivation of the bracket.

Example 14.2 (Basic examples)**.**

(1) **Abelian Lie algebras.** Any vector space V becomes a Lie algebra by setting $[x, y] = 0$ for all $x, y \in V$.

(2) **Associative algebras.** If A is an associative k-algebra, then A is a Lie algebra under the commutator bracket
$$[a, b] = ab - ba.$$
Associativity implies the Jacobi identity by direct expansion.

(3) **The general linear Lie algebra.** Taking $A = \mathrm{End}_k(V)$ gives the Lie algebra

$$\mathfrak{gl}(V) = \mathrm{End}_k(V), \qquad [X, Y] = XY - YX.$$

(4) **Derivations.** If A is an algebra, a linear map $D\colon A \to A$ is a **derivation** if

$$D(ab) = D(a)b + aD(b).$$

The space $\mathrm{Der}(A)$ of all derivations is a Lie algebra under the commutator bracket $[D, D'] = D \circ D' - D' \circ D$.

Verification for Derivations. Suppose $D, D' \in \mathrm{Der}(A)$. Expanding twice gives

$$D(D'(ab)) = DD'(a)b + D'(a)D(b) + D(a)D'(b) + aDD'(b),$$
$$D'(D(ab)) = D'D(a)b + D(a)D'(b) + D'(a)D(b) + aD'D(b).$$

Subtracting cancels the mixed terms, so

$$[D, D'](ab) = [D, D'](a)b + a[D, D'](b).$$

Hence $[D, D']$ is again a derivation. □

Remark 14.3. Over $\mathbb{R}$ or $\mathbb{C}$, Lie algebras arise from continuous symmetries by differentiation. If $g\colon (-\varepsilon, \varepsilon) \to \mathrm{Aut}(A)$ is a differentiable family of algebra automorphisms with $g(0) = \mathrm{id}$, then

$$D(a) = \left.\frac{d}{dt}\right|_{t=0} g(t)(a)$$

is a derivation. Conversely, when the exponential series converges, a derivation D gives a one-parameter family of automorphisms $\exp(tD)$. Formally, $\exp(tD)$ is a $k[[t]]$-algebra automorphism of $A[[t]]$.

Example 14.4 (Matrix Lie algebras)**.**

(1) The **special linear Lie algebra** is

$$\mathfrak{sl}_n = \{X \in \mathfrak{gl}_n \mid \mathrm{tr}(X) = 0\}.$$

It is closed under commutators because $\mathrm{tr}(XY - YX) = 0$.

(2) The Lie algebra $\mathfrak{sl}_2$ has basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

with relations

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

(3) The **Heisenberg Lie algebra** $\mathfrak{h}_3$ is the Lie algebra of strictly upper-triangular 3×3 matrices. It has a basis X, Y, C satisfying

$$[X, Y] = C, \qquad [X, C] = [Y, C] = 0.$$

(4) The **orthogonal Lie algebra** is

$$\mathfrak{so}_n = \{X \in \mathfrak{gl}_n \mid X^T + X = 0\}.$$

It is closed under commutators, so it is a Lie subalgebra of $\mathfrak{gl}_n$.

14.2 Subalgebras, Homomorphisms, and Representations

Definition 14.5. A **Lie subalgebra** of $\mathfrak{g}$ is a vector subspace $\mathfrak{a} \subset \mathfrak{g}$ such that $[\mathfrak{a}, \mathfrak{a}] \subset \mathfrak{a}$.

Definition 14.6. A **Lie algebra homomorphism** is a linear map $\varphi\colon \mathfrak{g}_1 \to \mathfrak{g}_2$ satisfying

$$\varphi([x, y]) = [\varphi(x), \varphi(y)].$$

Definition 14.7. A **representation** of a Lie algebra $\mathfrak{g}$ is a vector space V together with a Lie algebra homomorphism

$$\rho\colon \mathfrak{g} \longrightarrow \mathfrak{gl}(V).$$

Equivalently, each $x \in \mathfrak{g}$ acts as a linear operator $\rho(x)$ on V, and

$$\rho([x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x).$$

Example 14.8 (Important representations)**.**

(1) The **trivial representation** is defined by $\rho(x) = 0$ for all $x \in \mathfrak{g}$.

(2) The **adjoint representation** is the action of $\mathfrak{g}$ on itself by

$$\mathrm{ad}(x)(y) = [x, y].$$

The Jacobi identity gives

$$[\mathrm{ad}(x), \mathrm{ad}(y)] = \mathrm{ad}([x, y]),$$

so $\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is a Lie algebra homomorphism.

(3) For matrix Lie algebras such as $\mathfrak{gl}_n$, $\mathfrak{sl}_n$, and $\mathfrak{so}_n$, the natural action on column vectors in k^n is the **standard representation**.

14.3 The Universal Enveloping Algebra

The bracket on a Lie algebra is not an associative product. To use the tools of associative algebra, we build an associative algebra whose commutator recovers the Lie bracket.

Definition 14.9. The **universal enveloping algebra** $\mathcal{U}(\mathfrak{g})$ is the quotient

$$\mathcal{U}(\mathfrak{g}) = T^\bullet(\mathfrak{g}) \Big/ \big\langle x \otimes y - y \otimes x - [x, y] \mid x, y \in \mathfrak{g} \big\rangle,$$

where

$$T^\bullet(\mathfrak{g}) = \bigoplus_{n \geq 0} \mathfrak{g}^{\otimes n}$$

is the tensor algebra.

Example 14.10 (Enveloping Algebras)**.**

- If $\mathfrak{g}$ is abelian, then

$$\mathcal{U}(\mathfrak{g}) \simeq T^\bullet(\mathfrak{g}) / \langle x \otimes y - y \otimes x \rangle \simeq S^\bullet(\mathfrak{g}).$$

- For $\mathfrak{sl}_2$ over k,

$$\mathcal{U}(\mathfrak{sl}_2) = k\langle e, f, h\rangle \big/ \langle he - eh - 2e,\ hf - fh + 2f,\ ef - fe - h\rangle.$$

- For the Heisenberg algebra,

$$\mathcal{U}(\mathfrak{h}_3) = k\langle X, Y, C\rangle \big/ \langle XY - YX - C,\ XC - CX,\ YC - CY\rangle.$$

  The central quotient $\mathcal{U}(\mathfrak{h}_3)/\langle C - 1\rangle$ is the first Weyl algebra, the algebra generated by X and Y with relation $XY - YX = 1$.

Proposition 14.11. *The category of representations of $\mathfrak{g}$ is equivalent to the category of left $\mathcal{U}(\mathfrak{g})$-modules:*

$$\mathrm{Rep}(\mathfrak{g}) \simeq \mathcal{U}(\mathfrak{g})\text{-Mod}\,.$$

Here both categories contain representations on arbitrary vector spaces, and all $\mathcal{U}(\mathfrak{g})$-modules are unital. Restricting both sides to finite-dimensional objects gives the corresponding finite-dimensional equivalence.

Proof. Let $\rho\colon \mathfrak{g} \to \mathrm{End}(V)$ be a Lie algebra representation. By the universal property of the tensor algebra, ρ extends uniquely to an associative algebra homomorphism

$$\widetilde{\rho}\colon T^{\bullet}(\mathfrak{g}) \longrightarrow \mathrm{End}(V).$$

For every $x, y \in \mathfrak{g}$,

$$\widetilde{\rho}(x \otimes y - y \otimes x - [x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x) - \rho([x, y]) = 0.$$

Hence $\widetilde{\rho}$ factors through $\mathcal{U}(\mathfrak{g})$, giving a left $\mathcal{U}(\mathfrak{g})$-module structure on V.

Conversely, any left $\mathcal{U}(\mathfrak{g})$-module restricts along the canonical Lie algebra map $\mathfrak{g} \to \mathcal{U}(\mathfrak{g})$ to a representation of $\mathfrak{g}$. The defining relation in $\mathcal{U}(\mathfrak{g})$ exactly says that the commutator of the resulting operators is the action of the Lie bracket. These two constructions are inverse on objects and morphisms. □

Remark 14.12 (PBW Theorem)**.** We use without proof the following finite-dimensional form of the Poincaré–Birkhoff–Witt theorem; see [5]. If $\mathfrak{g}$ is finite-dimensional and we choose an ordered basis $x_1, \dots, x_m$ of $\mathfrak{g}$, the monomials

$$x_1^{a_1} \cdots x_m^{a_m}, \qquad a_i \geq 0,$$

form a basis of $\mathcal{U}(\mathfrak{g})$. In particular, the canonical map $\mathfrak{g} \to \mathcal{U}(\mathfrak{g})$ is injective, so the enveloping algebra contains the original Lie algebra faithfully through its canonical Lie map.

Chapter 15

Representation Theory of $\mathfrak{sl}_2$

15.1 The Highest Weight Vector

We classify the finite-dimensional irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$. Recall the standard basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

with relations

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

Let V be a finite-dimensional representation. Since we work over $\mathbb{C}$, the operator h has at least one eigenvalue.

Lemma 15.1. *Let λ be an eigenvalue of h whose real part is maximal among all eigenvalues of h on V. Then e annihilates the generalized λ-eigenspace of h.*

Proof. From $[h, e] = 2e$ we get

$$(h - (\lambda + 2))e = e(h - \lambda).$$

Hence, for every $N \geq 1$,

$$(h - (\lambda + 2))^N e = e(h - \lambda)^N.$$

If v is a generalized λ-eigenvector, then $(h - \lambda)^N v = 0$ for $N \gg 0$, and therefore

$$(h - (\lambda + 2))^N (ev) = 0.$$

Thus either $ev = 0$, or ev is a generalized eigenvector with eigenvalue $\lambda + 2$. The second possibility contradicts the maximality of $\mathrm{Re}(\lambda)$. □

In particular, the generalized λ-eigenspace contains an actual h-eigenvector killed by e.

Definition 15.2. A nonzero vector $v \in V$ satisfying

$$hv = \lambda v, \qquad ev = 0,$$

is called a **highest weight vector**. The scalar λ is its **highest weight**.

15.2 The Representation Chain

Starting from a highest weight vector, the operator f moves downward through the weights.

Lemma 15.3. *If $ew = 0$, then for every $k \geq 1$,*

$$e^k f^k w = k!\, h(h-1)\cdots(h-k+1)w.$$

Equivalently, with divided powers $e^{(k)} = e^k/k!$ and $f^{(k)} = f^k/k!$,

$$e^{(k)} f^{(k)} w = \binom{h}{k} w.$$

Proof. First compute

$$\begin{aligned}[e, f^k] &= \sum_{i=1}^{k} f^{k-i}[e, f]f^{i-1} \\ &= \sum_{i=1}^{k} f^{k-i} h f^{i-1}.\end{aligned}$$

Since $hf = f(h-2)$, the i-th summand is $f^{k-1}(h-2(i-1))$. Therefore

$$[e, f^k] = k f^{k-1}(h-k+1).$$

Applying this to a vector w with $ew = 0$ gives

$$e f^k w = k f^{k-1}(h-k+1)w.$$

Also $e\, p(h)w = p(h-2)ew = 0$ for every polynomial p. Induction on k now gives

$$e^k f^k w = k!\, h(h-1)\cdots(h-k+1)w.$$

Dividing by $(k!)^2$ gives the divided-power form. □

15.3 Classification Theorem

We can now classify the irreducible finite-dimensional representations.

Theorem 15.4. *The finite-dimensional irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$ are indexed by integers $N \geq 0$. The representation $V(N)$ has dimension $N+1$, and h acts diagonally with simple eigenvalues*

$$N,\ N-2,\ N-4,\ \ldots,\ -N.$$

Proof. First let V be finite-dimensional and irreducible. By the first lemma, choose a highest weight vector v with

$$hv = \lambda v, \qquad ev = 0.$$

Since $hf = f(h-2)$, the vector $f^k v$, if nonzero, has weight $\lambda - 2k$. These weights are distinct, so the nonzero vectors $v, fv, f^2v, \ldots$ are linearly independent. Finite dimensionality therefore forces the chain to stop: for some $N \geq 0$,

$$f^N v \neq 0, \qquad f^{N+1} v = 0.$$

Apply the commutator formula to the zero vector $f^{N+1}v$:

$$0 = ef^{N+1}v = (N+1)f^N(h-N)v = (N+1)(\lambda - N)f^N v.$$

Since $f^N v \neq 0$, we get $\lambda = N$. Thus the highest weight is a nonnegative integer, and the chain has exactly the weights

$$N,\ N-2,\ \ldots,\ -N.$$

The span

$$W = \mathrm{Span}_{\mathbb{C}}\{v, fv, \ldots, f^N v\}$$

is stable under h and f, and the formula for $ef^k v$ shows it is stable under e. Hence W is a nonzero subrepresentation of the irreducible module V, so $W = V$.

This proves uniqueness: any irreducible representation with highest weight N has a basis $v_i = f^i v$ for $0 \leq i \leq N$, and the action is forced:

$$hv_i = (N-2i)v_i, \qquad fv_i = v_{i+1}, \qquad ev_i = i(N-i+1)v_{i-1},$$

with the conventions $v_{-1} = v_{N+1} = 0$.

Conversely, for each $N \geq 0$, define a vector space with basis $v_0, \ldots, v_N$ and operators given by the displayed formulas. A direct check shows that they satisfy $[h,e] = 2e$, $[h,f] = -2f$, and $[e,f] = h$, hence define an $\mathfrak{sl}_2$-module. Let $0 \neq U$ be a submodule. Since h is diagonalizable in the displayed basis, the h-stable space U contains a nonzero weight vector; choose one of maximal weight. Every weight space is one-dimensional, so this vector is a scalar multiple of some v_i. If $i > 0$, then

$$ev_i = i(N-i+1)v_{i-1} \neq 0$$

lies in U and has larger weight, a contradiction. Hence $v_0 \in U$, and applying f gives every basis vector. Therefore this module is irreducible. It is the representation $V(N)$. □

Proposition 15.5 (Polynomial Model)**.** *For every $N \geq 0$, the module $V(N)$ is isomorphic to $\mathrm{Sym}^N(\mathbb{C}^2)$, realized as the space of homogeneous polynomials of degree N in variables x, y. The action is*

$$e = x\frac{\partial}{\partial y}, \qquad f = y\frac{\partial}{\partial x}, \qquad h = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

In this model the vectors

$$v_i = \frac{N!}{(N-i)!}x^{N-i}y^i, \qquad 0 \leq i \leq N,$$

satisfy the action formulas in the classification theorem.

Proof. The three differential operators satisfy $[h,e] = 2e$, $[h,f] = -2f$, and $[e,f] = h$. Direct differentiation gives

$$hv_i = (N-2i)v_i, \qquad fv_i = v_{i+1}, \qquad ev_i = i(N-i+1)v_{i-1},$$

with $v_{-1} = v_{N+1} = 0$. Hence this module has the same basis and action as $V(N)$. □

Chapter 16

Representations of General Linear Groups and $\mathrm{SL}_2(\mathbb{F}_q)$

16.1 Weyl Character Formula for $\mathrm{GL}(V)$

Let V be an N-dimensional complex vector space. Schur-Weyl duality gives the decomposition

$$V^{\otimes n} \simeq \bigoplus_{\lambda \vdash n} L_\lambda \otimes V_\lambda,$$

where V_λ is the irreducible $\mathfrak{S}_n$-module indexed by λ, and L_λ is either an irreducible polynomial $\mathrm{GL}(V)$-module or zero.

Let $g \in \mathrm{GL}(V)$ and $w \in \mathfrak{S}_n$. Suppose w has cycle type $\sigma = (\sigma_1, \ldots, \sigma_l)$. First suppose that g is diagonalizable, and choose an eigenbasis with eigenvalues $x_1, \ldots, x_N$. A basis tensor contributes to the trace of $g^{\otimes n} w$ precisely when its indices are constant along every cycle of w. Thus

$$\mathrm{Tr}_{V^{\otimes n}}(g^{\otimes n} w) = p_{\sigma_1} \cdots p_{\sigma_l} = p_\sigma(x_1, \ldots, x_N),$$

where $p_m = x_1^m + \cdots + x_N^m$. Both sides are polynomial functions of the matrix entries of g, since the right-hand side is also $\prod_j \mathrm{Tr}(g^{\sigma_j})$. Diagonalizable matrices are dense in $\mathrm{GL}_N(\mathbb{C})$, so the identity holds for every $g \in \mathrm{GL}(V)$.

The same trace computed from Schur-Weyl duality is

$$\mathrm{Tr}_{V^{\otimes n}}(g^{\otimes n} w) = \sum_{\lambda \vdash n} \mathrm{Tr}_{L_\lambda}(g)\, \chi^\lambda(w).$$

The Frobenius character formula says

$$p_\sigma = \sum_{\lambda \vdash n} \chi^\lambda(\sigma) s_\lambda.$$

Comparing the two expansions gives the character of L_λ.

Theorem 16.1 (Weyl Character Formula)**.** *Let $\lambda \vdash n$.*

(1) If $\ell(\lambda) > N$, then $L_\lambda = 0$.

(2) If $\ell(\lambda) \le N$, then

$$\mathrm{Tr}_{L_\lambda}(g) = s_\lambda(x_1, \dots, x_N) = \frac{\det\left(x_i^{\lambda_j + N - j}\right)_{1 \le i,j \le N}}{\det\left(x_i^{N-j}\right)_{1 \le i,j \le N}},$$

where $x_1, \dots, x_N$ are the eigenvalues of g. The determinant quotient is read first for pairwise distinct x_i and then by its polynomial continuation for arbitrary eigenvalues.

Corollary 16.2 (Dimension Formula)**.** *If $\ell(\lambda) \le N$, with $\lambda_i = 0$ for $i > \ell(\lambda)$, then*

$$\dim L_\lambda = \prod_{1 \le i < j \le N} \frac{\lambda_i - \lambda_j + j - i}{j - i}.$$

Proof. Set $g = I$ in the character formula, so that $\dim L_\lambda = s_\lambda(1, \dots, 1)$. Evaluating the bialternant formula for s_λ at $x_1 = \cdots = x_N = 1$ (equivalently, taking the Vandermonde limit) gives

$$s_\lambda(1^N) = \prod_{1 \le i < j \le N} \frac{\lambda_i - \lambda_j + j - i}{j - i},$$

which is the stated formula. □

16.2 Algebraic Representations of $\mathrm{GL}_N(\mathbb{C})$

Fix a basis and write $G = \mathrm{GL}_N(\mathbb{C})$. Its coordinate ring is

$$R = \mathbb{C}[G] = \mathbb{C}[x_{ij} \mid 1 \le i, j \le N][\det(x)^{-1}].$$

Definition 16.3. A finite-dimensional representation Y of G is **algebraic** if all of its matrix coefficients are regular functions on G, that is, elements of R.

The representations V, V^*, tensor powers, and the polynomial modules L_λ are algebraic. To include negative weights, let

$$\mu = (\mu_1 \ge \mu_2 \ge \cdots \ge \mu_N)$$

be a weakly decreasing sequence of integers. Choose an integer $r \ge \max\{0, -\mu_N\}$; then $\mu + (r^N)$ is a partition. Define

$$L_\mu = L_{\mu + (r^N)} \otimes (\wedge^N V^*)^{\otimes r}.$$

The determinant-twist identity

$$L_{\nu + (s^N)} \simeq L_\nu \otimes (\det)^s \qquad (s \ge 0),$$

for every partition ν with $\ell(\nu) \le N$, is obtained by adding s full columns in the Schur construction (and is reflected by $s_{\nu+(s^N)} = (x_1 \cdots x_N)^s s_\nu$), shows that the isomorphism class of this representation is independent of the chosen r. The set of all such μ is denoted by Λ^+.

Theorem 16.4 (Peter-Weyl Theorem for Algebraic Groups)**.**

(1) The category of finite-dimensional algebraic representations of G is semisimple.

(2) Under the $G \times G$ action

$$(g, h)\phi(x) = \phi(g^{-1}xh),$$

the coordinate ring decomposes as

$$R \simeq \bigoplus_{\mu \in \Lambda^+} L_\mu^* \otimes L_\mu.$$

Sketch of proof. As a $G \times G$-module,

$$\mathbb{C}[\mathrm{End}(V)] = \mathrm{Sym}^\bullet(V^* \otimes V).$$

Corollary 13.8, applied to V^* and V, and the natural identification $L_\lambda(V^*) \simeq L_\lambda(V)^*$ give

$$\mathrm{Sym}^n(V^* \otimes V) \simeq \bigoplus_{\lambda \vdash n} L_\lambda^* \otimes L_\lambda.$$

Passing from $\mathrm{End}(V)$ to $\mathrm{GL}(V)$ localizes by $\det^{-1}$. Multiplication by $\det^{-1}$ identifies the summand indexed by $\lambda + (1^N)$ with the determinant twist of the summand indexed by λ. The localization therefore produces exactly one summand $L_\mu^* \otimes L_\mu$ for each $\mu \in \Lambda^+$.

To deduce complete reducibility and the exhaustion of irreducibles, we use without proof the characteristic-zero structure theorem that GL_N is linearly reductive and that every irreducible algebraic representation occurs in its coordinate ring through its matrix coefficients; see [6]. This is the algebraic-group input not supplied by the preceding finite-group arguments. □

Corollary 16.5 (Classification of Algebraic Representations)**.** *Every finite-dimensional irreducible algebraic representation of $\mathrm{GL}_N(\mathbb{C})$ is isomorphic to L_μ for a unique $\mu \in \Lambda^+$. If $r \geq \max\{0, -\mu_N\}$, then*

$$\mathrm{Tr}_{L_\mu}(g) = (x_1 \cdots x_N)^{-r} s_{\mu+(r^N)}(x_1, \ldots, x_N),$$

where $x_1, \ldots, x_N$ are the eigenvalues of g; the expression is independent of r.

Proof. Each L_μ is a determinant twist of an irreducible polynomial module, so it is irreducible, and the displayed character formula follows from the Weyl character formula. Its independence of r is exactly the determinant-twist identity above. If $\mu \neq \nu$, choose one r for which both $\mu + (r^N)$ and $\nu + (r^N)$ are partitions. The corresponding Schur polynomials are distinct, so $L_\mu \not\simeq L_\nu$. Finally, every irreducible algebraic representation occurs through its matrix coefficients in the coordinate-ring decomposition, and hence is isomorphic to one of the L_μ. □

16.3 Structure of $\mathrm{SL}_2(\mathbb{F}_q)$

Now let p be an odd prime, $q = p^r$, and

$$G = \mathrm{SL}_2(\mathbb{F}_q).$$

Then $|G| = q(q^2 - 1)$ and the center is $Z = \{\pm I_2\}$. All representations of this finite group in what follows are finite-dimensional over $\mathbb{C}$; characters and their inner products are understood in this complex setting.

The group G acts transitively on $\mathbb{P}^1(\mathbb{F}_q) \simeq \mathbb{F}_q \cup \{\infty\}$, which has $q + 1$ points. The basic subgroups are:

- The **Borel subgroup** B is the stabilizer of $[1:0]$; it consists of upper triangular matrices and has order $q(q-1)$.
- The **split torus** is
$$T = \{d(a) = \mathrm{diag}(a, a^{-1}) \mid a \in \mathbb{F}_q^\times\},$$
so $|T| = q-1$.
- The **unipotent radical** is
$$U = \left\{u(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{F}_q\right\},$$
so $|U| = q$.

We have $B = T \ltimes U$. Let
$$s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$
This element represents the nontrivial Weyl group element and gives the Bruhat decomposition
$$G = B \sqcup BsB.$$

We use the following classical structure theorem without proof; see [1].

Theorem 16.6. *For $q > 3$, the group G is perfect: $[G, G] = G$. Moreover*
$$\mathrm{PSL}_2(\mathbb{F}_q) = G/Z$$
is simple.

If we temporarily drop the odd-q standing assumption, then for comparison, $\mathrm{PSL}_2(\mathbb{F}_2) \simeq \mathfrak{S}_3$ and $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$ are solvable, whereas $\mathrm{PSL}_2(\mathbb{F}_4) \simeq \mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$.

16.4 Conjugacy Classes and Harish-Chandra Induction

Besides the split torus T, the group G contains a **nonsplit torus** T'. It is obtained from the norm-one elements in $\mathbb{F}_{q^2}^\times$:
$$T' \simeq \{z \in \mathbb{F}_{q^2}^\times \mid \mathrm{Nm}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(z) = 1\}.$$
It is cyclic of order $q+1$.

For odd q, the $q+4$ conjugacy classes of G are distributed as follows:

- the central elements $\pm I_2$;
- $(q-3)/2$ split regular classes represented by $d(a)$ with $a \in \mathbb{F}_q^\times \setminus \{\pm 1\}$, where a and a^{-1} give the same class;
- $(q-1)/2$ nonsplit regular classes represented by elements of $T' \setminus \{\pm 1\}$, again modulo inversion;

- four noncentral repeated-eigenvalue classes. If ϵ is a fixed nonsquare in $\mathbb{F}_q^\times$, write

$$u_+ = u(1), \qquad u_- = u(\epsilon).$$

  The two classes represented by u_+, u_- are unipotent, while those represented by $-u_+, -u_-$ are negative-unipotent (and are not unipotent in odd characteristic).

The count is

$$2 + \frac{q-3}{2} + \frac{q-1}{2} + 4 = q + 4.$$

16.4.1 Harish-Chandra Series

Let $\alpha \in \mathrm{Irr}(T)$. Inflate α along $B \twoheadrightarrow B/U \simeq T$ (equivalently, make it trivial on U), and define

$$R(\alpha) = \mathrm{Ind}_B^G(\alpha).$$

Since $|G/B| = q+1$, we have $\dim R(\alpha) = q+1$.

Using Mackey theory and the Bruhat decomposition,

$$\langle R(\alpha), R(\beta)\rangle_G = \langle \alpha, \beta\rangle_T + \langle \alpha, {}^s\beta\rangle_T,$$

where

$${}^s\beta(t) = \beta(sts^{-1}) = \beta(t^{-1}) = \beta^{-1}(t).$$

The formula gives the principal series:

1. If $\alpha^2 \neq 1$, then

$$\langle R(\alpha), R(\alpha)\rangle_G = 1,$$

   so $R(\alpha)$ is irreducible of dimension $q+1$. More generally, the Mackey formula shows that two such representations have nonzero inner product exactly when their parameters differ by inversion. Hence $R(\alpha) \simeq R(\alpha^{-1})$, giving $(q-3)/2$ distinct irreducibles.

2. If $\alpha = 1$, then $R(1)$ is the permutation representation on $\mathbb{P}^1(\mathbb{F}_q)$. Frobenius reciprocity gives

$$\langle R(1), \mathbf{1}\rangle_G = 1,$$

   while the Mackey formula gives $\langle R(1), R(1)\rangle_G = 2$. Thus the complement of the constants is one irreducible representation, and

$$R(1) \simeq \mathbf{1} \oplus \mathrm{St},$$

   where St is the q-dimensional Steinberg representation.

3. If α_0 is the unique quadratic character of T, then $\langle R(\alpha_0), R(\alpha_0)\rangle_G = 2$. Semisimplicity therefore shows that it has two distinct irreducible constituents, each with multiplicity one. A calculation with the normalized intertwining operator, omitted here, shows that their dimensions are equal; see [1]. Thus

$$R(\alpha_0) \simeq R(\alpha_0)^+ \oplus R(\alpha_0)^-,$$

   with both summands irreducible of dimension $(q+1)/2$.

Thus the principal series accounts for $(q+5)/2$ irreducible representations. The nonsplit torus supplies the remaining irreducibles. We use without proof the following part of the character classification; see [1]. The discrete series consist of $(q-1)/2$ generic representations of dimension $q-1$, indexed by inverse pairs of characters θ of the nonsplit torus T' satisfying $\theta^2 \neq 1$, together with two exceptional constituents of dimension $(q-1)/2$ attached to its quadratic character. A construction is given by Deligne–Lusztig induction from T'. Their orthogonality and the class count above show that these are precisely the remaining $(q+3)/2$ irreducibles.

The number of representations just listed is

$$\frac{q+5}{2} + \frac{q+3}{2} = q+4,$$

in agreement with the conjugacy-class count. Their dimensions also satisfy

$$\begin{aligned} &\frac{q-3}{2}(q+1)^2 + 1 + q^2 + 2\left(\frac{q+1}{2}\right)^2 \\ &\qquad + \frac{q-1}{2}(q-1)^2 + 2\left(\frac{q-1}{2}\right)^2 = q(q^2-1) = |G|, \end{aligned}$$

as required by the decomposition of the regular representation.

Bibliography

[1] R. W. Carter, *Finite Groups of Lie Type: Conjugacy Classes and Complex Characters*, Wiley, 1985.

[2] C. W. Curtis and I. Reiner, *Representation Theory of Finite Groups and Associative Algebras*, Wiley, 1962.

[3] G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition, CRC Press, 2016.

[4] W. Fulton and J. Harris, *Representation Theory: A First Course*, Springer, 1991.

[5] J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Springer, 1972.

[6] J. C. Jantzen, *Representations of Algebraic Groups*, second edition, American Mathematical Society, 2003.

[7] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Oxford University Press, 1995.

[8] A. Okounkov and A. Vershik, "A new approach to representation theory of symmetric groups," *Selecta Mathematica* **2** (1996), 581–605.

[9] B. E. Sagan, *The Symmetric Group*, second edition, Springer, 2001.

[10] R. Schiffler, *Quiver Representations*, Springer, 2014.